

Bird of Prey: Practical Signature Combiners Preserving Strong Unforgeability

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Abstract Following the announcement of the first winners of the NIST post-quantum cryptography standardization process in 2022, cryptographic protocols are now undergoing migration to the newly standardized schemes. In most cases, this transition is realized through a hybrid approach, in which algorithms based on classical hardness assumptions, such as the discrete logarithm problem, are combined with post-quantum algorithms that rely on quantum-resistant assumptions, such as the Short Integer Solution (SIS) problem.

A combiner for signature schemes can be obtained by simply concatenating the signatures of both schemes. This construction preserves unforgeability of the underlying schemes; however, it does not extend to stronger notions, such as *strong unforgeability*. Several applications, including authenticated key exchange and secure messaging, inherently require strong unforgeability, yet no existing combiner is known to achieve this property.

This work introduces three practical combiners that preserve strong unforgeability and all BUFF (beyond unforgeability features) properties. Each combiner is tailored to a specific class of classical signature schemes capturing all broadly used schemes that are strongly unforgeable. Remarkably, all combiners can be instantiated with any post-quantum signature scheme in a black-box way making deployment practical and significantly less error prone. The proposed solutions are further highly efficient and have signatures that are at most the size of the (insecure) concatenation combiner. For instance, our most efficient combiner enables the combination of EdDSA with ML-DSA, yielding a signature size that is smaller than the sum of an individual EdDSA signature and an individual ML-DSA signature.

Additionally, we identify a novel signature property that we call random-message validity and show that it can be used to replace the BUFF transform with the more efficient Pornin-Stern transform. The notion may be of independent interest.

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1 Introduction

POST-QUANTUM SECURITY. Since the end of the first NIST post-quantum (PQ) competition [NIS16], cryptographic protocols are being migrated to use the newly standardized schemes. The selected algorithms have undergone a multi-year standardization process and the underlying assumptions have been known even longer. Nevertheless, they have yet to attain the same level of confidence or withstand the depth of cryptanalysis as classical hardness assumptions such as the discrete logarithm problem or RSA. For this reason, PQ schemes are mostly deployed in a hybrid manner, i.e. rather than replacing a classical scheme outright, it is augmented with a combiner that incorporates both the classical and the PQ scheme. The security of a combiner is expected to hold if the security of the classical *or* the PQ scheme holds. This approach offers protection both against the long-term threat posed by large-scale quantum computers, which could compromise classical assumptions, and against the current uncertainty due to the relatively limited cryptanalytic scrutiny that PQ assumptions have undergone. Recent years have witnessed substantial efforts aimed at both the academic evaluation of real-world post-quantum cryptographic schemes [BCNS15, PST20, BBCT22, Ste24, LSB24] and their integration into practical systems [Lan16, Lan18, KV19, WR19, KS24].

HYBRIDS/COMBINER. As noted earlier, a central aspect of the proposed adaptations is the use of *hybrid* or *combiner* approaches. This strategy has also received endorsement from several national security agencies. The French National Agency for the Security of Information Systems (ANSSI) recommends a hybrid adoption of PQC [ANS23], and the German Federal Office for Information Security (BSI) “*only recommends the hybrid use of quantum-safe methods in combination with classical methods*” [BSI24]. As already pointed out by [BH23], NIST will explicitly validate hybrid solutions if one of the components is approved by NIST. Using the term “dual signatures” for a hybrid approach they write [NIS25, FAQ]

Existing NIST standards and guidelines accommodate their use provided that at least one component digital signature algorithm is NIST-approved.

and

[...], NIST will accommodate the use of a hybrid key-establishment mode and dual signatures in FIPS 140 validation when suitably combined with a NIST-approved scheme.

This implies that compliance can be achieved by (suitably) combining one NIST-approved component with an arbitrary counterpart.

On the academic front, there has been growing interest in cryptographic combiner solutions [BCD⁺24, CHH⁺25, FH25, HR25, LL25, GRSV25, GHJ25]. In particular, key encapsulation mechanisms (KEMs) have received significant attention, with a number of recent works addressing this area [BCD⁺24, CHH⁺25, GHP18]. The recently proposed KEM combiner X-Wing [BCD⁺24] efficiently integrates the classical scheme X25519 with the post-quantum ML-KEM [MLK24] and identifies the properties required for a secure combination. This framework was later generalized in Starfighters [CHH⁺25], which captures a broader class of KEM instantiations. In contrast, developing a corresponding framework for digital signatures that captures all relevant security properties remains an open problem.

SIGNATURES. As part of the first post-quantum cryptography standardization competition, NIST selected three digital signature schemes. Dilithium [DKL⁺18] and Sphincs+ [BHK⁺19] have already been standardized as ML-DSA in FIPS 204 [MLD24] and as SLH-DSA in FIPS 205 [SLH24], respectively. Falcon [PFH⁺20] is going to be standardized as FN-DSA, although the official standard is still under development. A commonly considered threat model in the context of encryption is harvest-now-decrypt-later, i.e. adversaries may store classical ciphertexts today with the intention of decrypting them in the future, once large-scale quantum computers become available. This threat model does not directly apply to the basic use of digital signature schemes, which are primarily used for authentication and are only vulnerable to active adversaries who can participate in the communication

process. While the urgency of post-quantum migration is less acute for signature schemes than for encryption, it remains an important and time-sensitive challenge. Cryptographic migration is inherently complex and requires considerable time to implement effectively. For example, attacks on the collision-resistance of MD5 were published in the early 2000s [WFLY04] yet new system vulnerabilities continue to emerge due to its ongoing use [GHH⁺24]. On the other hand, for long-lived products, such as those in the automotive industry¹ or other hardware-driven products, security requirements projected 15 to 20 years into the future must be considered during today’s design and deployment phases. In industry, the migration process is already underway [Alg25, AWS25, MKTW25], highlighting the need for practical and standardized solutions as soon as possible.

STRONG UNFORGEABILITY. While existential unforgeability suffices for many use cases, certain applications require a signature scheme to satisfy a stronger security notion. *Strong unforgeability* ensures that an adversary cannot produce a new, valid signature on a given message – even if they have previously obtained a different valid signature for the same message. This is for example required in authenticated key exchange [BHJ⁺15, JKRS21], SSH [BDK⁺14], cryptocurrency applications [AEE⁺21, Kli17, TMM21], signcryption [ABF12, AJKL23, JMOR25], and as recently shown, in the group messaging protocol MLS [CGWZ25]. Fortunately, most classical signature schemes currently in use already satisfy the notion of strong unforgeability, including RSA [RSA78] (in their standardized variant RSASSA-PSS in PKCS#1 v2.2 [MKJR16]), BLS [BLS04], and EdDSA [BDL⁺12].² Among the NIST-selected signature schemes, two out of three have been shown to achieve strong unforgeability, namely ML-DSA [DKL⁺18, KLS18] and Falcon [GJK24].

1.1 Signature Combiners

Signature combiners have been explored in prior work [BHMS17, BH23, GKP⁺23], resulting in several efficient constructions. However, these approaches either fail to preserve strong unforgeability [BHMS17, GKP⁺23], or are not compliant with FIPS standards due to their elegant but complex design [BH23]. Additionally, there are recent standardization efforts [OGP⁺26, PWJ⁺25].

BLACK-BOX COMBINER. The strongest and most convenient class of combiners are black-box combiners. This means the combination of two schemes of the desired type in a black-box manner, i.e. the combiner interacts with them solely through their defined interfaces, without relying on any internal details. This black-box approach is particularly advantageous, as it yields more generic constructions and avoids making assumptions about which component of a scheme may become vulnerable in the future. For instance, the security of an underlying scheme may fail either due to an inherent flaw in its design or analysis, such as a flawed security proof, or because the hardness of an assumed underlying problem is later invalidated. In *any* case, a black-box combiner preserves its security guarantees as long as at least one of the constituent schemes remains secure.

When considering combiners in the context of post-quantum cryptography, i.e. the combination of a scheme based on a classical assumption with one based on a PQ assumption, the focus lies primarily on the underlying hardness assumptions, assuming the soundness of the respective security proofs (see [GHJ25] for further details). An alternative to this approach is the use of non-black-box combiners, which may exploit structural properties of the underlying schemes or interleave their internal components to achieve desired security guarantees. A common way to leverage the security of non-black-box combiners is to rely on statistical arguments wherever feasible [HR25, LL25, GRSV25, GHJ25]. Such arguments remain valid even in the presence of unbounded adversaries. For all computational assumptions, however, it is crucial to obtain an OR-property; that is, the security of the combiner should hold if one *or* the other assumption holds.

A NAÏVE COMBINER. For signature schemes, the simplest form of a combiner is the parallel or concatenation combiner:

$$\sigma = (\text{Sgn}_1(\text{sk}_1, m), \text{Sgn}_2(\text{sk}_2, m)).$$

¹ <https://prism.sustainability-directory.com/term/post-quantum-automotive-security>

² For EdDSA, it depends on the concrete implementation, see [BCJZ21] for more details.

The signature is accepted if both signature components verify. It is straightforward to show that this combiner achieves unforgeability as long as at least one of the underlying schemes is unforgeable.

However, as noted earlier, strong unforgeability of one underlying signature scheme does not imply that the combiner inherits this property. This occurs because an adversary controlling the compromised component can produce a fresh signature that results in a combined signature which is itself fresh. More formally, consider a scenario where the security of the combiner relies solely on the strong unforgeability of signature scheme Sig_1 , while Sig_2 may be entirely broken. An adversary can query the signing oracle on a message m obtaining a combined signature (σ_1, σ_2) . Since Sig_2 is not assumed to be strongly unforgeable, the adversary can generate a new signature σ'_2 on the same message and thus construct a new signature for the combiner: (σ_1, σ'_2) .

An alternative to the parallel combiner is a sequential combiner, previously proposed in [BHMS17]. In a sequential combiner, the message is first signed using one scheme, and the resulting signature (potentially along with the original message) is signed using the second scheme. This construction achieves strong unforgeability provided that the second scheme is strongly unforgeable.³ However, if security is to rely solely on the first scheme, the same limitations arise as with the parallel combiner described above. More generally, any black-box combiner seems to inherit this issue without additional assumptions because there needs to be a “last” signature that remains vulnerable to the described attack. With further assumptions on the signature schemes the problem can be circumvented; we later present a combiner following the sequential paradigm which is secure under the assumption that the final signature scheme is unique.

We emphasize that constructing strongly unforgeable black-box combiners is not impossible in general. For instance, generic transformations exist that upgrade plain unforgeability to strong unforgeability, for example based on chameleon hash functions [SPW07]. Hence, one could first construct a combiner achieving only plain unforgeability and then apply such a transformation to obtain strong unforgeability.⁴ Since our goal is to develop efficient solutions that are practical to implement, we need to avoid the additional overhead introduced by chameleon hash functions or other heavy cryptographic primitives. Under these constraints and without further assumptions constructing a black-box combiner that preserves strong unforgeability seems to be infeasible.

ADDITIONAL PROPERTIES. There are additional signature properties relevant in various application contexts, referred to as “BUFF” properties [CDF⁺21, DFH⁺24, DFHS24]. Similarly to strong unforgeability, these properties are not necessarily preserved by naïve combiners such as the parallel combiner. For example, exclusive ownership requires that it should be computationally infeasible to produce a signature that verifies under two distinct public keys. If one of the underlying signature schemes fails to satisfy exclusive ownership, then the concatenation combiner inherits this limitation. A property of particular interest is non-resignability which captures that, given a valid signature, it should be hard for an adversary to produce another valid signature on the same message under a different public key – without knowing the message itself. Since the original non-resignability property in [CDF⁺21] was not achievable, [DFH⁺24, DFHS24] introduced relaxed variants. They further showed that a salted variant [DFHS24] of the original BUFF transform and the original transform [DFH⁺24] satisfy the relaxed notions.

NON-SEPARABILITY. Non-separability was introduced as a specific property for signature combiners by [BHMS17] and later refined by [BH23]. Informally, this property captures that it should be hard to extract a valid signature for one of the underlying schemes from the combined signature. We provide a brief overview and discuss to what extent our schemes satisfy this property in Appendix A.

1.2 Our Approach

NON-BLACK-BOX. Due to the limitations of black-box combiners discussed before, our focus shifts to non-black-box combiners, i.e. constructions in which certain signature schemes are opened and combined at a lower level. This introduces several challenges, particularly when working with PQ schemes. On the one

³ This approach was utilized in [GKP⁺23].

⁴ In the case of chameleon hash functions, a suitable combiner for chameleon hash functions would also be required.

hand, PQ schemes are relatively new, and the standardized variants have typically undergone more rigorous evaluation than any modified versions. On the other hand, organizations are often required to use standardized cryptographic algorithms to meet regulatory requirements. As previously noted, a signature combiner is FIPS-compliant if one of the underlying schemes is NIST-approved. However, modifying such a scheme by altering or interacting with its internal components may violate standardization constraints, thereby jeopardizing FIPS compliance.

Second, PQ schemes are often much more complex than their classical counterparts. Modifying or “opening up” complex constructions and implementations is likely to introduce a wide range of issues and subtle bugs. As NIST-approved PQ standards are increasingly being adopted in practice, it is prudent to adhere to the official reference implementations and maintained libraries, which have been thoroughly reviewed and tested.

Third, departing from standardized schemes may give rise to intellectual property concerns. During the standardization process, patents on FIPS schemes were waived. However, implementing these schemes in a manner that differs, even slightly, from the approved standard may risk violating applicable patents.⁵

Fourth, if a construction relies on the internal structure or specific properties of concrete signature schemes, it becomes inherently less generic. This limits its general applicability and reduces its compatibility with potential future schemes. In particular, from the perspective of cryptographic agility, it is preferable to design systems and frameworks that are easily adaptable to new algorithms and standards.

Taking all of these limitations into account, our goal is to design combiners in which at least one component can be treated in a black-box manner. In practical instantiations, this will typically be the PQ component, due to both compliance constraints and the complexity of its implementation. This consideration leads to the following design criterion:

Design Criterion 1: Combiners should use one of the signature schemes in a black-box manner.

COMPLEXITY. Combiners should be as simple as possible. This applies both to the additional primitives used – ideally limited to lightweight components such as hash functions –and to the overall construction, which should operate at the highest possible level of abstraction.⁶ From a practical perspective, such simplicity facilitates reuse of existing codebases and reduces the likelihood of implementation errors, thereby supporting more secure and maintainable deployments.

Design Criterion 2: Combiners should be as abstract as possible and rely only on lightweight additional primitives.

EFFICIENCY. Given that size remains the primary bottleneck for PQ instantiations, our goal is to design combiners that are at least as compact as the generic combiner described earlier.⁷

Design Criterion 3: The size of combined signatures should be at most the sum of the signatures of the underlying schemes.

This leads us to the following research question.

“Can we construct combiners satisfying the three listed design criteria?”

⁵ This argument was brought to our attention by a Eurocrypt 2026 reviewer.

⁶ For example, using specific classes like signature schemes based on the Fiat-Shamir paradigm rather than concrete constructions.

⁷ The running time should also not be significantly higher than that of the underlying schemes.

1.3 Contributions

We present the BIRD-OF-PREY class containing three **strongly unforgeable** signature combiners that:

- make black-box use of any NIST-approved PQ signature resulting in FIPS compliance
- are classically instantiable with BLS, EDDSA, or RSA
- only need the two signature schemes and a hash function
- have compact signatures whose sizes are at most the sum of the underlying schemes
- preserve all BUFF properties

As expected for combiners, strong unforgeability is preserved if it holds for at least one of the underlying schemes. An overview can be found in Table 1 which also lists the additional requirements necessary to prove security. Unlike strong unforgeability, these additional requirements can be fulfilled unconditionally, i.e. either perfectly or statistically, and thus remain valid even if underlying computational assumptions are broken. The requirements are selected to ensure that our framework can accommodate any strongly unforgeable signature scheme in widespread use today. For each combiner category, the most representative example and the resulting signature size are listed in the final column. For the following overview of the constructions, we assume the first component to be classical and the second component to be the post-quantum instantiation.

Construction	Requirements		Size	Example Instantiations		
	Classical	PQ		Classical	PQ	Size (bytes)
BIRD-OF-PREY-1 (Figure 5)	unique	—	$ \sigma_1 + \sigma_2 $	BLS*	ML-DSA-44 FALCON-512	2 468 714
BIRD-OF-PREY-2 (Figure 6)	UR	MBS, RMV	$ \text{rsp}_1 + \sigma_2 $	EdDSA25519	ML-DSA-44 FALCON-512	2 452 698
BIRD-OF-PREY-3 (Figure 7)	salt-unique	—	$ \sigma_1 + \sigma_2 $	RSASSA-PSS [†]	ML-DSA-44 FALCON-512	2 676 922

Table 1. Overview of our constructions listing requirements for each component to preserve strong unforgeability. **UR** stands for unique responses for an identification scheme and **rsp** for its response, which requires the signature scheme to follow the Fiat-Shamir paradigm. **MBS** stands for message-bound security and **RMV** for random-message validity. The classical component refers to the first component in the constructions, the post-quantum component refers to the second component.

*BLS is instantiated with curve BLS12-381.

[†]RSA is instantiated with a 2048 bit key.

BIRD-OF-PREY-1. The first construction requires the classical signature scheme to be unique. A concrete instantiation meeting this requirement is BLS [BLS04]. Otherwise, both schemes are treated in a black-box manner.

BIRD-OF-PREY-2. The second construction applies to signature schemes following the Fiat-Shamir paradigm such as EDDSA [BDL⁺12]. These schemes are based on (canonical) identification (ID) schemes and we require these ID schemes to have unique responses (**UR**). Additionally, the combiner imposes two requirements on the PQ component: message-bound security (**MBS**), introduced in [BCJZ21], and random-message validity, which ensures that it is hard for an adversary to output a public key and a valid signature on a message which is randomly chosen after the adversary terminated. Both properties are unconditionally satisfied by ML-DSA and FALCON. Compared to a naïve combiner, this construction allows for improved compactness: the commitment/challenge of the classical component can be omitted. As a result, the final signature consists solely of the PQ signature and the response of the classical ID scheme.

BIRD-OF-PREY-3. The third construction applies (classical) signature schemes that are based on a salt and are unique given a fixed salt value, a property we call **salt-uniqueness**. This construction generalizes the

first one⁸ and encompasses, for example, all RSA variants. This includes (randomized) RSA-FDH [BR93], PSS [BR96], and the widely adopted RSASSA-PSS as specified in PKCS#1 v2.2 [MKJR16].

NEW PROPERTY AND OLD TRANSFORM. As introduced in the context of our second combiner, we identify a new property called random-message validity (**RMV**), which is strictly weaker than message-bound security. This property captures that it should be hard for an adversary to generate a public key and a valid signature that verifies for a randomly chosen message. From a theoretical perspective, **RMV** can serve as a useful intermediate property to enable other practically relevant guarantees, one example being our second combiner. In addition, we show that **RMV** can be leveraged to generically achieve non-resignability. As previously mentioned, [DFH⁺24] showed that the BUFF transform suffices to ensure non-resignability. The downside is that the transform comes at the cost of an increased signature size.

We go one step further and show that the third Pornin-Stern transform [PS05] is sufficient to achieve non-resignability, provided that the underlying signature scheme satisfies **RMV**. The Pornin-Stern transform does not increase the signature size. This approach has two additional key advantages. First, **RMV** is a significantly simpler property than non-resignability, making it easier to analyze and verify. Second, many widely used signature schemes naturally satisfy **RMV**, including hash-based signatures and those following the Fiat-Shamir or Full-Domain-Hash paradigms. A related result was shown in [DS24], though it relied on the stronger **MBS** assumption and addressed a weaker variant of non-resignability.

2 Preliminaries

We introduce some relevant notation and definitions used throughout the paper.

2.1 Notations

SETS AND ALGORITHMS. We write $s \leftarrow^{\$} \mathcal{S}$ to denote the uniform sampling of s from the finite set $\mathcal{S}$. For an integer n , we define $[n] := \{1, \dots, n\}$. For two sets A, B , we denote the set of all functions from A to B by $\{A \rightarrow B\}$. The empty string is denoted by ε . We use uppercase letters $\mathcal{A}, \mathcal{B}, \dots$ to denote algorithms. Unless otherwise stated, algorithms are probabilistic, and we write $(y_1, \dots) \leftarrow^{\$} \mathcal{A}(x_1, \dots)$ to denote that $\mathcal{A}$ returns $(y_1, \dots)$ when run on input $(x_1, \dots)$. We write $\mathcal{A}^{\mathcal{B}}$ to denote that $\mathcal{A}$ has oracle access to $\mathcal{B}$ during its execution. The support of a discrete random variable X is defined as $\text{supp}(X) := \{x \in \mathbb{R} \mid \Pr[X = x] > 0\}$. We sometimes simply write $x \in X$ as a shorthand for $x \in \text{supp}(X)$. We denote the running time of an algorithm $\mathcal{A}$ by $t_{\mathcal{A}}$. By “log” we denote the logarithm of base 2. We use **return** x to denote that an algorithm terminates and outputs x . Additionally, we use **output** x in sub algorithms, e.g. oracles, to denote that the higher level algorithm terminates and outputs x . For a distribution $\mathcal{D}$, $\mathcal{H}_{\infty}(\mathcal{D})$ is defined as its min-entropy.

SECURITY GAMES. We use standard code-based security games [BR06]. A *game* G is a probability experiment in which an adversary $\mathcal{A}$ interacts with an implicit challenger that answers oracle queries issued by $\mathcal{A}$. The game G has one *main procedure* and an arbitrary amount of additional *oracle procedures* which describe how these oracle queries are answered. We denote the (binary) output b of game G between a challenger and an adversary $\mathcal{A}$ as $\mathsf{G}(\mathcal{A}) \Rightarrow b$. $\mathcal{A}$ is said to *win* G if $\mathsf{G}(\mathcal{A}) \Rightarrow 1$, or shortly $\mathsf{G} \Rightarrow 1$. Unless otherwise stated, the randomness in the probability term $\Pr[\mathsf{G}(\mathcal{A}) \Rightarrow 1]$ is over all the random coins in game G and adversary $\mathcal{A}$. To provide a cleaner description and avoid repetitions, we sometimes refer to procedures of different games. To call the oracle procedure **Oracle** of game G on input x , we shortly write $\mathsf{G}.\text{Oracle}(x)$. If a game is aborted the output is 0. For our analysis we rely on the commonly used main difference lemma [BR06].

RANDOM ORACLES. We use the random oracle model [BR93] and let a scheme $\mathcal{S}$ specify a set $\mathcal{OS}$ of functions, called the oracle space. Hence, we also need to define primitives with respect to an oracle space. A security game samples a function $H \leftarrow^{\$} \mathcal{OS}$ at random and provides a random oracle RO to the adversary which on input x returns $H(x)$. Since the random oracle is always defined as described here, we do not necessarily define it for each security notion individually. If an algorithm $A(x, \dots)$ has access to random oracle H , we write $A[H](x, \dots)$. In case the oracle space is the empty set, we might ignore it.

⁸ The first construction can be viewed as a special case of the third where the salt space is empty.

2.2 Signatures

We recall the syntax and standard security notions of signatures.

Definition 1 (Signature Scheme). A *signature scheme* Sig is defined as a tuple $(\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$ with oracle space $\mathcal{OS}$ and the following three algorithms for any $H \in \mathcal{OS}$:

- $(\text{sk}, \text{pk}) \leftarrow^{\$} \text{Gen}[H]$: The probabilistic key generation algorithm returns a secret key sk and a corresponding public key pk .
- $\sigma \leftarrow^{\$} \text{Sgn}[H](\text{sk}, m)$: Given a secret key sk and a message m , the probabilistic signing algorithm Sgn returns a signature σ .
- $b \leftarrow \text{Ver}[H](\text{pk}, m, \sigma)$: Given a public key pk , a message m , and a signature σ , the deterministic⁹ verification algorithm Ver returns a bit $b \in \{0, 1\}$.

Sig has ε -correctness error if for all $H \in \mathcal{OS}$, $(\text{sk}, \text{pk}) \in \text{sup}(\text{Gen})$ and all m $\Pr[\text{Ver}[H](\text{pk}, m, \text{Sgn}[H](\text{sk}, m)) \neq 1] \leq \varepsilon$, where the probability is taken over the random choices of Sgn .

By derivePK , we denote a mapping from any sk to pk such that $(\text{sk}, \text{pk}) \in \text{Gen}$. Such a function exists in general if we assume the public key being part of the secret key. This requirement could be even relaxed by only relying on some public identifier/tag instead of the full public key¹⁰ but would need some changes in the concrete constructions and proofs.

Definition 2 (Signature Spreadness). For a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, we define its *spreadness* as

$$\gamma_{\text{Sig}} := \max_{\sigma^*, m} \Pr_{H \leftarrow^{\$} \mathcal{OS}} \left[\sigma^* = \sigma \mid \begin{array}{l} (\text{sk}, \text{pk}) \leftarrow^{\$} \text{Gen}[H] \\ \sigma \leftarrow^{\$} \text{Sgn}[H](\text{sk}, m) \end{array} \right].$$

In other words, signatures have a min-entropy of at least $-\log(\gamma_{\text{Sig}})$ bits.

Definition 3 ((Strong) Unforgeability). The notions of *(strong) existential unforgeability under chosen/no message attacks* are formalised via the games in Figure 1. We define the advantage functions of adversary $\mathcal{A}$ as

$$\begin{aligned} \text{Adv}_{\text{Sig}, \mathcal{A}}^{(Q_s, Q_{R0})\text{-UF-CMA}} &:= \Pr[(Q_s, Q_{R0})\text{-UF-CMA}_{\text{Sig}}(\mathcal{A}) \Rightarrow 1], \\ \text{Adv}_{\text{Sig}, \mathcal{A}}^{(Q_s, Q_{R0})\text{-SUF-CMA}} &:= \Pr[(Q_s, Q_{R0})\text{-SUF-CMA}_{\text{Sig}}(\mathcal{A}) \Rightarrow 1], \end{aligned}$$

Definition 4 (Uniqueness). A signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$ is called *unique* if given a public key $(\cdot, \text{pk}) \in \text{Gen}$ and a message m there exists exactly one signature σ such that $\text{Ver}(\text{pk}, m, \sigma) = 1$.

Definition 5 (Exclusive Ownership [BCJZ21]). For a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, we define *malicious strong universal exclusive ownership* against some adversary $\mathcal{A}$ via their advantage function

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{\text{EO}} := \Pr_{H \leftarrow^{\$} \mathcal{OS}} \left[\begin{array}{l} \text{Ver}[H](\text{pk}_1, m_1, \sigma) = 1, \\ \text{Ver}[H](\text{pk}_2, m_2, \sigma) = 1, \\ \text{pk}_1 \neq \text{pk}_2 \end{array} \mid (\text{pk}_1, \text{pk}_2, m_1, m_2, \sigma) \leftarrow^{\$} \mathcal{A}^{R0(\cdot)} \right].$$

Definition 6 (Message-bound Security [BCJZ21]). For a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, we define *message-bound security* against some adversary $\mathcal{A}$ via their advantage function

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{\text{MBS}} := \Pr_{H \leftarrow^{\$} \mathcal{OS}} \left[\begin{array}{l} \text{Ver}[H](\text{pk}, m_1, \sigma) = 1, \\ \text{Ver}[H](\text{pk}, m_2, \sigma) = 1, m_1 \neq m_2 \end{array} \mid (\text{pk}, m_1, m_2, \sigma) \leftarrow^{\$} \mathcal{A}^{R0(\cdot)} \right].$$

⁹ All results in this work also apply to a probabilistic verification procedure.

¹⁰ For instance, this is done in Hawk [BBD⁺23] which stores a hash of the public key.

Games (Q_s, Q_{R0}) -(S)UF-CMA _{Sig} ($\mathcal{A}$)	Oracle Sgn(m)
01 $H \xleftarrow{\$} \mathcal{OS}$	07 $\sigma \xleftarrow{\$} \text{Sgn}[H](sk, m)$
02 $\mathcal{Q} \leftarrow \emptyset$	08 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
03 $(sk, pk) \xleftarrow{\$} \text{Gen}[H]$	09 return σ
04 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), R0(\cdot)}(pk)$	
05 return $\text{Ver}[H](pk, m^*, \sigma^*) \wedge (m^*, \cdot) \notin \mathcal{Q}$	Oracle R0(x)
06 return $\text{Ver}[H](pk, m^*, \sigma^*) \wedge (m^*, \sigma^*) \notin \mathcal{Q}$	10 return $H(x)$
	// UF-CMA
	// SUF-CMA

Figure 1. Games defining **UF-CMA** and **SUF-CMA** for a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$ and adversary $\mathcal{A}$ making at most Q_s queries to **Sgn** and at most Q_{R0} queries to **R0**.

Definition 7 (Non-resignability [DFH⁺24]). The notion of *(strong) non-resignability* for a signature scheme Sig and an auxiliary function aux is formalised via the game in Figure 2. We define the advantage function of adversary $\mathcal{A}$ and $\mathcal{D}$ as

$$\text{Adv}_{\text{Sig}, \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} := \Pr[(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}_{\text{Sig}, \text{aux}}(\mathcal{A}, \mathcal{D}) \Rightarrow 1].$$

Game $(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}_{\text{Sig}, \text{aux}}(\mathcal{A}, \mathcal{D})$	Oracle R0(x)
01 $H \xleftarrow{\$} \mathcal{OS}$	07 return $H(x)$
02 $(sk, pk) \xleftarrow{\$} \text{Gen}[H]$	
03 $m \xleftarrow{\$} \mathcal{D}^{R0}(sk)$	
04 $\sigma \xleftarrow{\$} \text{Sgn}[H](sk, m)$	
05 $(pk^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{R0}(sk, \sigma, \text{aux}(sk, m))$	
06 return $pk \neq pk^* \wedge \text{Ver}[H](pk^*, m, \sigma^*)$	

Figure 2. Game defining **NR** for a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, an auxiliary function aux , an adversary $\mathcal{A}$, and an adversary $\mathcal{D}$ where $\mathcal{A}$ makes at most $Q_{\mathcal{A}}$ and $\mathcal{D}$ makes at most $Q_{\mathcal{D}}$ queries to **R0**.

2.3 Canonical Identification Schemes

Definition 8 (Identification Scheme). A *canonical identification scheme* ID is defined as a tuple $\text{ID} := (\mathcal{OS}, \text{Gen}, \text{Com}, \text{Rsp}, \text{Ver}, \text{ChlSet})$ of an oracle space $\mathcal{OS}$, a challenge set ChlSet , and the following algorithms for any $H \in \mathcal{OS}$:

- $(sk, pk) \xleftarrow{\$} \text{Gen}[H]$: The generation algorithm Gen returns a secret key sk and a corresponding public key pk .
- $(com, st) \xleftarrow{\$} \text{Com}[H](sk)$: Given a secret key sk , the commitment algorithm Com returns a commitment com and a state st .
- $rsp \xleftarrow{\$} \text{Rsp}[H](sk, com, chl, st)$: Given a secret key sk , a commitment com , a challenge $chl \in \text{ChlSet}$ and a state st , the response algorithm Rsp returns a response rsp .
- $b \leftarrow \text{Ver}[H](pk, com, chl, rsp)$: Given a public key pk , a commitment com , a challenge chl , and a response rsp , the deterministic verification algorithm Ver returns a bit $b \in \{0, 1\}$.

The correctness error δ_{ID} is defined as the smallest value such that for all $H \in \mathcal{OS}$ and $(sk, pk) \in \text{Gen}$ it holds that

$$\Pr \left[\text{Ver}[H](pk, com, chl, rsp) \neq 1 \mid \begin{array}{l} (com, st) \xleftarrow{\$} \text{Com}[H](sk) \\ rsp \xleftarrow{\$} \text{Rsp}[H](sk, com, chl, st) \end{array} \right] \leq \delta_{\text{ID}}.$$

By `derivePK`, we denote a mapping from `sk` to `pk` such that $(\text{sk}, \text{pk}) \in \text{Gen}$.

Definition 9 (Commitment Extractability). An identification scheme $\text{ID} := (\mathcal{OS}, \text{Gen}, \text{Com}, \text{Rsp}, \text{Ver}, \text{ChlSet})$ is called *commitment extractable* if there exists an algorithm `ExtCom` which takes as input a public key `pk`, a challenge `chl` $\in \text{ChlSet}$, and a response `rsp` and outputs a commitment `com` such that for all $H \in \mathcal{OS}$, $(\cdot, \text{pk}) \in \text{Gen}$, `chl` $\in \text{ChlSet}$, and `rsp` in the response space it holds $\text{Ver}[H](\text{pk}, \text{com}, \text{chl}, \text{rsp}) = 1$

Definition 10 ((Parallel) Impersonation under Passive Attacks [KMP16]). Let ID be an identification scheme. Consider the game **PIMP-PA** in Fig. 3. The advantage of an adversary $\mathcal{A}$ is defined as

$$\text{Adv}_{\text{ID}, \mathcal{A}}^{Q_{\text{chl}}\text{-PIMP-PA}} := \Pr[Q_{\text{chl}}\text{-PIMP-PA}_{\text{ID}}(\mathcal{A}) \Rightarrow 1].$$

Game $Q_{\text{chl}}\text{-PIMP-PA}_{\text{ID}}(\mathcal{A})$ <pre> 01 $H \xleftarrow{\\$} \mathcal{OS}$ 02 $(\text{sk}, \text{pk}) \xleftarrow{\\$} \text{Gen}[H]$ 03 $\text{cnt} \leftarrow 0$ 04 $\mathcal{L}_{\text{Chl}} \leftarrow \emptyset$ 05 $(\text{com}^*, \text{chl}^*, \text{rsp}^*) \xleftarrow{\\$} \mathcal{A}^{\text{Trans}, \text{Chl}(\cdot), \text{R0}(\cdot)}(\text{pk})$ 06 if $\text{cnt} > Q_{\text{chl}} \vee (\text{com}^*, \text{chl}^*) \notin \mathcal{L}_{\text{Chl}}$ 07 return 0 08 return $\text{Ver}[H](\text{pk}, \text{com}^*, \text{chl}^*, \text{rsp}^*)$ </pre> Game $\text{UR}_{\text{ID}}(\mathcal{A})$ <pre> 09 $H \xleftarrow{\\$} \mathcal{OS}$ 10 $(\text{sk}, \text{pk}) \xleftarrow{\\$} \text{Gen}[H]$ 11 $(\text{com}^*, \text{chl}^*, \text{rsp}_1^*, \text{rsp}_2^*) \xleftarrow{\\$} \mathcal{A}^{\text{Trans}, \text{R0}(\cdot)}(\text{pk})$ 12 if $\text{Ver}[H](\text{pk}, \text{com}^*, \text{chl}^*, \text{rsp}_1^*) = 1$ $\wedge \text{Ver}[H](\text{pk}, \text{com}^*, \text{chl}^*, \text{rsp}_2^*) = 1 \wedge \text{rsp}_1^* \neq \text{rsp}_2^*$ 13 return 1 14 return 0 </pre>	Oracle Trans <pre> 15 $(\text{com}, \text{st}) \xleftarrow{\\$} \text{Com}[H](\text{sk})$ 16 $\text{chl} \xleftarrow{\\$} \text{ChlSet}$ 17 $\text{rsp} \xleftarrow{\\$} \text{Rsp}[H](\text{sk}, \text{com}, \text{chl}, \text{st})$ 18 return $(\text{com}, \text{chl}, \text{rsp})$ </pre> Oracle $\text{Chl}(\text{com})$ <pre> 19 $\text{cnt} \leftarrow \text{cnt} + 1$ 20 $\text{chl} \xleftarrow{\\$} \text{ChlSet}$ 21 $\mathcal{L}_{\text{Chl}} \leftarrow \mathcal{L}_{\text{Chl}} \cup \{(\text{com}, \text{chl})\}$ 22 return chl </pre> Oracle $\text{R0}(x)$ <pre> 23 return $H(x)$ </pre>
--	--

Figure 3. Games defining **PIMP-PA** and **UR** for an ID scheme $\text{ID} = (\mathcal{OS}, \text{Gen}, \text{Com}, \text{Rsp}, \text{Ver}, \text{ChlSet})$.

Note that compared to standard definitions of unique responses, our definition includes a transcript oracle. Under honest-verifier zero-knowledge of the ID scheme, these notions are equivalent.

Definition 11 ((Computationally) Unique Responses). Let $\text{ID} = (\mathcal{OS}, \text{Gen}, \text{Com}, \text{Rsp}, \text{Ver}, \text{ChlSet})$ be an identification scheme. Consider the game in Figure 3. The advantage of an adversary $\mathcal{A}$ as

$$\text{Adv}_{\text{ID}, \mathcal{A}}^{\text{UR}} := \Pr[\text{UR}_{\text{ID}}(\mathcal{A}) \Rightarrow 1].$$

2.4 Hash Functions

Collision resistance is usually defined for hash functions or hash function families. Our abstraction of an oracle space allows us to give a more general notion which can be instantiated by common notions for collision resistance.

Definition 12 (Collision Resistance). For a set of oracles $\mathcal{OS}$, we define *collision resistance* against some adversary $\mathcal{A}$ via their advantage function

$$\text{Adv}_{\mathcal{OS}, \mathcal{A}}^{\text{CR}} := \Pr_{H \xleftarrow{\$} \mathcal{OS}} [H(x_1) = H(x_2), x_1 \neq x_2 \mid (x_1, x_2) \xleftarrow{\$} \mathcal{A}^{\text{R0}(\cdot)}].$$

Definition 13 (Hide-and-Seek [DFH⁺24]). For a set of oracles $\mathcal{OS}$, we define the *hide-and-seek* property against some adversaries $\mathcal{A}$ and $\mathcal{D}$ via their advantage function

$$\text{Adv}_{\mathcal{OS}, \mathcal{A}, \mathcal{D}}^{\text{HnS}} := \Pr_{\mathbf{H} \leftarrow \mathbb{S}_{\mathcal{OS}}} \left[x = x^* \mid \begin{array}{l} (x, z) \leftarrow \mathbb{S} \mathcal{D}^{\text{RO}(\cdot)} \\ x^* \leftarrow \mathbb{S} \mathcal{A}^{\text{RO}(\cdot)}(\mathbf{H}(x), z) \end{array} \right].$$

Note that this property is only meaningful for appropriate adversaries $\mathcal{D}$; in particular, for adversaries $\mathcal{D}$ with a sufficiently high min-entropy on m given z .

3 A New Property and Non-resignability

In this section, we identify an additional BUFF property which can be seen as a strictly weaker version of message-bound security, **MBS**. It will be useful to prove our combiners. Additionally, we show that this property allows to obtain non-resignability by not using the original BUFF transform [CDF⁺21] but the more efficient Pornin-Stern transform [PS05].¹¹

3.1 Random-message Validity

Definition 14 (Random-message Validity). For a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, we define *random-message validity* for some adversary $\mathcal{A}$ as

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{\mathcal{M}\text{-RMV}} := \Pr_{\mathbf{H} \leftarrow \mathbb{S}_{\mathcal{OS}}} \left[\text{Ver}[\mathbf{H}](\text{pk}, m \| x, \sigma) \mid \begin{array}{l} m \leftarrow \mathbb{S} \mathcal{M} \\ (\text{pk}, \sigma, x) \leftarrow \mathbb{S} \mathcal{A}^{\text{RO}(\cdot)} \end{array} \right].$$

The adversary's output x may also be the empty string. In this case, the challenger only checks the randomly chosen message. The hardness obviously relies on size of $\mathcal{M}$.¹²

This property seems natural but is not implied by unforgeability (see Appendix B). It is, however, implied by another BUFF property, **MBS**, but is strictly weaker (also Appendix B). Still most natural signature schemes fulfill the security notion: if the message is hashed as part of the signing procedure and if this hash is somehow also checked in the verification procedure, **RMV** can be reduced to the collision resistance of said hash function. This is for example the case for hash-based signature schemes, Fiat-Shamir-based signatures, or signatures following the Full-Domain-Hash paradigm, therefore capturing all current NIST standards. Further information on **RMV**, how it could be weakened and where it is needed can be found in Appendix B.

3.2 Non-resignability for Signature Schemes with RMV

The most subtle BUFF property is non-resignability. The first version was introduced in [CDF⁺21] together with the BUFF transform enabling this and other properties. It was later shown [DFHS24] that the BUFF transform actually does not fulfill the notion introduced in [CDF⁺21] and that the notion is in general nearly impossible to achieve. Based on these findings, the authors of [DFHS24] present a different notion of non-resignability which was later strengthened in [DFH⁺24]. They also show that the notion can be achieved by a salted BUFF transform [DFHS24] and even by the original BUFF transform [DFH⁺24].

In the remainder of this section, we show that when assuming **RMV** of the underlying signature scheme, one can even achieve non-resignability with Pornin-Stern transformation and does not have to rely on the less efficient BUFF transform. This is particularly interesting because the signature sizes do not grow and **RMV** is much easier to analyze for new schemes than the less handy non-resignability.

¹¹ A similar results was shown requiring (the stronger) **MBS** instead of **RMV** and for a weaker notion of non-resignability [DS24].

¹² One could also allow non-uniform distributions. Then the hardness is related to the min-entropy of such a distribution.

CONSTRUCTION. Both, the Pornin-Stern transform PS as well as the BUFF transform BUFF transform a signature scheme $\text{Sig} := (\mathcal{OS}_{\text{Sig}}, \text{Gen}, \text{Sgn}, \text{Ver})$ together with an output size $\lambda \in \mathbb{N}$ of a random oracle into a new signature scheme Sig' . It holds

$$\begin{aligned}\text{BUFF}[\text{Sig}, \lambda] &:= (\mathcal{OS} \times \mathcal{OS}_{\text{Sig}}, \text{Gen}', \text{Sgn}', \text{Ver}'), \\ \text{PS}[\text{Sig}, \lambda] &:= (\mathcal{OS} \times \mathcal{OS}_{\text{Sig}}, \text{Gen}', \text{Sgn}', \text{Ver}'),\end{aligned}$$

where $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ and the three algorithms are defined in Figure 4; the black code is for the PS transform and the black and blue code for the BUFF transform.

<u>Gen'[H, H_{Sig}]</u>	<u>Sgn'[H, H_{Sig}](sk, m)</u>
01 (sk, pk) $\xleftarrow{\$}$ Gen[H _{Sig}]	05 pk $\leftarrow$ derivePK(sk)
02 return (sk, pk)	06 m' $\leftarrow$ H(pk m)
<u>Ver'[H, H_{Sig}](pk, m, ($\sigma, \hat{m}$))</u>	07 $\sigma \xleftarrow{\$}$ Sgn[H _{Sig}](sk, m')
03 m' $\leftarrow$ H(pk m)	08 return ($\sigma, \textcolor{blue}{m'}$)
04 return Ver[H _{Sig}](pk, m', σ) $\wedge$ $\textcolor{blue}{m'} = \hat{m}$	

Figure 4. Pornin-Stern transform and BUFF transform (with additional blue code).

SECURITY. The following theorem can be seen as an analogue of the main result of [DFH⁺24] which introduced and showed how to use the **HnS** property. The first part of the proof is also taken from [DFH⁺24].

Theorem 1 (NR). For any adversaries $\mathcal{A}$ and $\mathcal{D}$ against the **NR** security of PS[Sig, λ] (Figure 4), there exist **HnS** adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ and an **RMV** adversary $\mathcal{C}$ against Sig with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}}$ and $t_{\mathcal{D}} = t_{\bar{\mathcal{D}}}$ such that

$$\text{Adv}_{\text{PS}[\text{Sig}, \lambda], \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}} + \text{Adv}_{\text{Sig}, \mathcal{C}}^{\{0, 1\}^\lambda\text{-RMV}}$$

and

$$\mathcal{H}_{\infty}^{(sk, pk) \xleftarrow{\$} \text{Gen}} (m \mid \text{RO}, sk, \text{aux}(sk, m)) \leq \mathcal{H}_{\infty}^{(x, z) \xleftarrow{\$} \bar{\mathcal{D}}^{\text{RO}}} (x \mid \text{RO}, z).$$

The proof can be found in Appendix B.2.

Remark 1. To interpret the theorem statement, note that **NR** as well as **HnS** are only meaningful if their distribution $\mathcal{D}/\bar{\mathcal{D}}$ has sufficient min-entropy. The theorem statement says that the min-entropy of **NR** is upper bounded by the min-entropy of **HnS**. Hence, a meaningful distribution property for the underlying component transfers to a meaningful property for the higher-level scheme.

4 Construction from Two Signature Schemes

4.1 The Scheme

Our simplest construction takes two signatures in a black-box manner. We require the first signature to be unique.¹³ The construction is detailed in Figure 5.

¹³ For the proof, it would also be sufficient to only require statistical uniqueness instead of perfect uniqueness.

We only make directly used random oracles explicit and assume random oracles for underlying schemes since they can be implemented straightforwardly by just forwarding calls. This results in oracle space

$$\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\} \times \mathcal{OS}_1 \times \mathcal{OS}_2,$$

where only the first oracle is made explicit in construction and proof.

<u>Gen[H]</u>	<u>Ver[H](pk, m, σ)</u>
01 $(sk_1, pk_1) \xleftarrow{\$} \text{Gen}_1$	10 $pk \rightarrow (pk_1, pk_2)$
02 $(sk_2, pk_2) \xleftarrow{\$} \text{Gen}_2$	11 $\sigma \rightarrow (\sigma_1, \sigma_2)$
03 return $((sk_1, sk_2), (pk_1, pk_2))$	12 $m' \leftarrow H(pk_1 pk_2 m)$
<u>Sgn[H](sk, m)</u>	13 if $\text{Ver}_1(pk_1, m' \sigma_2, \sigma_1)$
04 $sk \rightarrow (sk_1, sk_2)$	$\wedge \text{Ver}_2(pk, m', \sigma_2)$
05 $(pk_1, pk_2) \leftarrow (\text{derivePK}(sk_1), \text{derivePK}(sk_2))$	14 return 1
06 $m' \leftarrow H(pk_1 pk_2 m)$	15 return 0
07 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(sk_2, m')$	
08 $\sigma_1 \xleftarrow{\$} \text{Sgn}_1(sk_1, m' \sigma_2)$	
09 return (σ_1, σ_2)	

Figure 5. Construction $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda] = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$ from signature schemes $\text{Sig}_1 = (\mathcal{OS}_1, \text{Gen}_1, \text{Sgn}_1, \text{Ver}_1)$ and $\text{Sig}_2 = (\mathcal{OS}_2, \text{Gen}_2, \text{Sgn}_2, \text{Ver}_2)$.

4.2 Security

We can prove that the combiner is secure as long as one of the underlying schemes is secure. Note that for a unique signature scheme, unforgeability and strong unforgeability are equivalent which is why we only require Sig_1 to be (plain) unforgeable.

Theorem 2 (SUF-CMA). If Sig_1 is unique, then for any adversary $\mathcal{A}$ against the **SUF-CMA** security of $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$ (Figure 5), there exist a **CR** adversary $\mathcal{B}$ against $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$, a **UF-CMA** adversary $\mathcal{C}$ against Sig_1 and a **SUF-CMA** adversary $\mathcal{D}$ against Sig_2 with $t_{\mathcal{A}} \approx t_{\mathcal{B}} \approx t_{\mathcal{C}} \approx t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{(Q_s, Q_{ro})\text{-SUF-CMA}} \leq \min \left\{ \text{Adv}_{\text{Sig}_1, \mathcal{C}}^{(Q_s, Q_{ro})\text{-UF-CMA}}, \text{Adv}_{\text{Sig}_2, \mathcal{D}}^{(Q_s, Q_{ro})\text{-SUF-CMA}} \right\} + \text{Adv}_{\mathcal{OS}', \mathcal{B}}^{\text{CR}}.$$

The proof can be found in Appendix C.1.

Theorem 3 (EO). For any adversary $\mathcal{A}$ against the **EO** security of $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$ (Figure 5), there exist an **EO** adversary $\mathcal{B}_1$ against Sig_1 , an **MBS** adversary $\mathcal{C}_1$ against Sig_1 , an **EO** adversary $\mathcal{B}_2$ against Sig_2 , an **MBS** adversary $\mathcal{C}_2$ against Sig_2 , and an **CR** adversary $\mathcal{D}$ against $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}_1} = t_{\mathcal{C}_1} = t_{\mathcal{B}_2} = t_{\mathcal{C}_2} = t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{EO}} \leq \min \left\{ \text{Adv}_{\text{Sig}_1, \mathcal{B}_1}^{\text{EO}} + \text{Adv}_{\text{Sig}_1, \mathcal{C}_1}^{\text{MBS}}, \text{Adv}_{\text{Sig}_2, \mathcal{B}_2}^{\text{EO}} + \text{Adv}_{\text{Sig}_2, \mathcal{C}_2}^{\text{MBS}} \right\} + \text{Adv}_{\mathcal{OS}', \mathcal{D}}^{\text{CR}}.$$

The proof can be found in Appendix C.2.

Theorem 4 (MBS). For any adversary $\mathcal{A}$ against the **MBS** security of $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$ (Figure 5), there exist an **MBS** adversary $\mathcal{B}$ against Sig_1 , an **MBS** adversary $\mathcal{C}$ against Sig_2 , and an **CR** adversary $\mathcal{D}$ against $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}} = t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{MBS}} \leq \min \left\{ \text{Adv}_{\text{Sig}_1, \mathcal{B}}^{\text{MBS}}, \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\text{MBS}} \right\} + \text{Adv}_{\mathcal{OS}', \mathcal{D}}^{\text{CR}}.$$

The proof can be found in Appendix C.3.

Theorem 5 (NR). For any adversaries $\mathcal{A}$ and $\mathcal{D}$ against the **NR** security of $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$ (Figure 5), there exist **HnS** adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$, an **RMV** adversary $\mathcal{C}_1$ against Sig_1 , and an **RMV** adversary $\mathcal{C}_2$ against Sig_2 with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}_1} = t_{\mathcal{C}_2}$ and $t_{\mathcal{D}} = t_{\bar{\mathcal{D}}}$ such that

$$\text{Adv}_{\text{PS}[\text{Sig}, \lambda], \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}} + \min \left\{ \text{Adv}_{\text{Sig}_1, \mathcal{C}_1}^{\{0, 1\}^\lambda\text{-RMV}}, \text{Adv}_{\text{Sig}_2, \mathcal{C}_2}^{\{0, 1\}^\lambda\text{-RMV}} \right\}$$

and

$$\mathcal{H}_{\infty}^{(m \mid \text{R0}, \text{sk}, \text{aux}(\text{sk}, m))}_{(\text{sk}, \text{pk}) \xleftarrow{\$} \text{Gen}, m \xleftarrow{\$} \mathcal{D}^{\text{R0}}(\text{sk})} \leq \mathcal{H}_{\infty}^{(x \mid \text{R0}, z)}_{(x, z) \xleftarrow{\$} \bar{\mathcal{D}}^{\text{R0}}}$$

The proof follows by Theorem 1. Remark 1 interprets the theorem statement.

5 Construction from Identification and Signature Scheme

Since we think the construction in this section is the most interesting one, we try to give the most comprehensive information, i.e. precise theorem statements for **BUFF** properties and full **SUF-CMA** proof in the main body (we only defer some space consuming full rolled out reductions).

5.1 The Scheme

In Figure 6, we construct a signature scheme from an ID scheme which is commitment extractable and a signature scheme. As mentioned before, we only make directly used random oracles explicit and assume random oracles for underlying schemes. This results in oracle space

$$\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\} \times \{\{0, 1\}^* \rightarrow \text{ChlSet}\} \times \mathcal{OS}_1 \times \mathcal{OS}_2,$$

where only the first two oracles are made explicit in construction and proof. Further, **ChlSet** denotes the challenge set of ID and the scheme is parametrized over the output size of the first random oracle, λ . As can be seen in the theorem, this output space must be sufficiently large.

For concrete instantiations of the second component one could even optimize the construction without changing the second signature schemes. This is because a lot of signature schemes hash the message and use the result, e.g. Fiat-Shamir or Full-Domain-Hash.

5.2 Security

We can show that the scheme is strong unforgeable if the underlying signature is strongly unforgeable or the underlying ID scheme is secure against parallel impersonation attacks. In the second case, we additionally require the signature scheme to fulfill message-bound security and random-message validity. However, these properties can be fulfilled information-theoretically and hence do not contradict the combiner behavior we aim for. The same can hold for unique responses, which needs to be fulfilled unconditionally, for the ID scheme which can be statistical or even perfect.

Note that the theorem bound implicitly requires the first signature component (which is based on the ID scheme) to be strongly unforgeable. This is because **PIMP-PA** implies **UF-CMA** in the random oracle model and unique responses lift this to strong unforgeability [KMP16].

Theorem 6 (SUF-CMA). For any adversary $\mathcal{A}$, making at most Q_s signing queries and at most Q_{R0} random oracle queries, against the **SUF-CMA** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$ (Figure 6) in the random oracle model, there exist an **UR** adversary $\mathcal{B}$ against ID, a **SUF-CMA** adversary $\mathcal{C}$ against Sig_2 , an **MBS**

<u>Gen[H₁, H₂]</u>	<u>Sgn[H₁, H₂](sk, m)</u>
01 (sk _{ID} , pk _{ID}) $\leftarrow^{\$}$ Gen ₁	12 sk $\rightarrow$ (sk _{ID} , sk _{Sig})
02 (sk _{Sig} , pk _{Sig}) $\leftarrow^{\$}$ Gen ₂	13 (com, st) $\leftarrow^{\$}$ Com(sk _{ID})
03 return ((sk _{ID} , sk _{Sig}), (pk _{ID} , pk _{Sig}))	14 (pk _{ID} , pk _{Sig}) $\leftarrow$ (derivePK(sk _{ID}), derivePK(sk _{Sig}))
<u>Ver[H₁, H₂](pk, m, σ)</u>	15 m' $\leftarrow$ H ₁ (pk _{ID} pk _{Sig} m com)
04 pk $\rightarrow$ (pk _{ID} , pk _{Sig})	16 σ ₂ $\leftarrow^{\$}$ Sgn ₂ (sk _{Sig} , m')
05 σ $\rightarrow$ (rsp, σ ₂)	17 chl $\leftarrow$ H ₂ (σ ₂)
06 chl $\leftarrow$ H ₂ (σ ₂)	18 rsp $\leftarrow^{\$}$ Rsp(sk _{ID} , com, chl, st)
07 com $\leftarrow$ ExtCom(pk _{ID} , chl, rsp)	19 σ $\leftarrow$ (rsp, σ ₂)
08 m' $\leftarrow$ H ₁ (pk _{ID} pk _{Sig} m com)	20 return σ
09 if Ver ₂ (pk _{Sig} , m', σ ₂)	
10 return 1	
11 return 0	

Figure 6. Construction $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda] = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$ from an ID scheme $\text{ID} = (\mathcal{OS}_1, \text{Gen}_1, \text{Com}, \text{Rsp}, \text{Ver}_1, \text{ChlSet})$ which is commitment extractable, using ExtCom , and a signature scheme $\text{Sig}_2 = (\mathcal{OS}_2, \text{Gen}_2, \text{Sgn}_2, \text{Ver}_2)$.

adversary $\mathcal{D}$ against Sig_2 , an **RMV** adversary $\mathcal{E}$ against Sig_2 , and a **PIMP-PA** adversary $\mathcal{F}$ against ID with $t_{\mathcal{A}} \approx t_{\mathcal{B}} \approx t_{\mathcal{C}} \approx t_{\mathcal{D}} \approx t_{\mathcal{E}} \approx t_{\mathcal{F}}$ such that

$$\begin{aligned} \text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} &\leq \min \left\{ \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}, \text{Adv}_{\text{ID}, \mathcal{F}}^{(Q_{\text{RO}}+1)\text{-PIMP-PA}} \right. \\ &\quad \left. + \text{Adv}_{\text{Sig}_2, \mathcal{D}}^{\text{MBS}} + Q_{\text{RO}}^2 \cdot \text{Adv}_{\text{Sig}_2, \mathcal{E}}^{\{0,1\}^\lambda\text{-RMV}} \right\} \\ &\quad + \text{Adv}_{\text{ID}, \mathcal{B}}^{\text{UR}} + Q_s(Q_{\text{RO}} + Q_s)\gamma_{\text{Sig}_2} + \frac{Q_{\text{RO}} + Q_s}{|\text{ChlSet}|}, \end{aligned}$$

where ChlSet is the challenge set of ID.

The proof can be found in Appendix D.1.

By slightly adapting the construction, we could also rely on the exclusive ownership and message-bound security of the first component instead of the second component, i.e. achieving a **OR**-property. Since both properties can be fulfilled unconditionally¹⁴, we rely on the simpler construction and security bound. The same holds for Theorem 8.

Theorem 7 (EO). For any adversary $\mathcal{A}$ against the **EO** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$ (Figure 6), there exist a **EO** adversary $\mathcal{B}$ against Sig_2 , a **MBS** adversary $\mathcal{C}$ against Sig_2 , and a **CR** adversary $\mathcal{D}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}} = t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{EO}} \leq \text{Adv}_{\text{Sig}_2, \mathcal{B}}^{\text{EO}} + \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\text{MBS}} + \text{Adv}_{\mathcal{OS}, \mathcal{D}}^{\text{CR}}.$$

The proof can be found in Appendix D.2.

Theorem 8 (MBS). For any adversary $\mathcal{A}$ against the **MBS** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$ (Figure 6), there exist a **MBS** adversary $\mathcal{B}$ against Sig_2 and a **CR** adversary $\mathcal{C}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}}$ such that

$$\text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{MBS}} \leq \text{Adv}_{\text{Sig}_2, \mathcal{B}}^{\text{MBS}} + \text{Adv}_{\mathcal{OS}, \mathcal{C}}^{\text{CR}}.$$

The proof can be found in Appendix D.3.

¹⁴ Especially, this holds for all instantiations we are considering.

Theorem 9 (NR). For any adversaries $\mathcal{A}$ and $\mathcal{D}$ against the **NR** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda] := (\text{Gen}, \cdot, \cdot)$ (Figure 6), there exist **HnS** adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ and an **RMV** adversary $\mathcal{C}$ against Sig_2 with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}}$ and $t_{\mathcal{D}} = t_{\bar{\mathcal{D}}}$ such that

$$\text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}} + \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\{0, 1\}^\lambda\text{-RMV}}$$

and

$$\mathcal{H}_{\infty}^{(\text{sk}, \text{pk}) \xleftarrow{\text{Gen}}}_{m \xleftarrow{\mathcal{D}^{\text{R0}}}(\text{sk})} (m \mid \text{R0}, \text{sk}, \text{aux}(\text{sk}, m)) \leq \mathcal{H}_{\infty}^{(x, z) \xleftarrow{\bar{\mathcal{D}}^{\text{R0}}}} (x \mid \text{R0}, z).$$

The proof can be found in Appendix D.4. Remark 1 interprets the theorem.

6 Construction from a Salt-based Signature Scheme and a Signature Scheme

We first introduce another signature abstraction level named salt-based signature schemes.

Definition 15 (Salt-based Signature Scheme). A *salt-based signature scheme* SigS is defined as a tuple $(\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Sgn}_{\text{salt}}, \text{Ext}, \text{Ver})$ such that $(\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$ defines a signature scheme (where Gen additionally defines a salt space $\mathcal{S}$) and the remaining algorithms are defined as follows for any $H \in \mathcal{OS}$:

$\sigma \xleftarrow{\text{S}} \text{Sgn}_{\text{salt}}[H](\text{sk}, m, r)$: Given a secret key sk , a message m , and a salt $r \in \mathcal{S}$, the probabilistic salt-specific signing algorithm Sgn_{salt} returns a signature σ .
 $r \leftarrow \text{Ext}[H](\text{pk}, \sigma)$: Given a public key pk and a signature σ , the deterministic extraction algorithm Ext returns a salt r .

Further, it is required that distribution $\{\sigma : \sigma \xleftarrow{\text{S}} \text{Sgn}[H](\text{sk}, m)\}$ and $\{\sigma : r \xleftarrow{\text{S}} \mathcal{S}, \sigma \xleftarrow{\text{S}} \text{Sgn}_{\text{salt}}[H](\text{sk}, m, r)\}$ are equal for every $(\text{sk}, \cdot) \in \text{Gen}[H]$, m , and $H \in \mathcal{OS}$. For every $(\text{sk}, \cdot) \in \text{Gen}[H]$, $m, r \in \mathcal{S}$, $H \in \mathcal{OS}$, it must hold $r = \text{Ext}[H](\text{pk}, \text{Sgn}_{\text{salt}}(\text{sk}, m, r))$. The definition of the correctness error is the same as for signature schemes.

This abstraction can be instantiated by several salt-based signature schemes for which the signing process consists of sampling a uniformly random salt and then using Sgn_{salt} as a subroutine. Hence, the two distributions induced by Sgn and Sgn_{salt} are equal for such schemes. In cases in which the salt is part of the signature, the extraction algorithm is trivial. This is the case for signature schemes based on the Full-Domain-Hash (FDH) paradigm like RSA-FDH [BR93], probabilistic GPV [GPV08], and Falcon [PFH⁺20]. In other cases, the extraction is implicitly done in the verification procedure, e.g. signatures following the PSS design [BR96] like RSASSA-PSS [MKJR16].

Natural salt-based signature schemes are randomized and thus inherently not unique. However, some of them have a property which we call salt-uniqueness; it captures that the signature is unique for a fixed salt.

Definition 16 (Salt-uniqueness). A salt-based signature scheme $\text{SigS} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Sgn}_{\text{salt}}, \text{Ext}, \text{Ver})$ is called *salt-unique* if, for every $H \in \mathcal{OS}$, given a public key $(\cdot, \text{pk}) \in \text{Gen}[H]$, a message m , and a salt $r \in \mathcal{S}$ there exists exactly one signature σ such that $\text{Ver}[H](\text{pk}, m, \sigma) = 1$.

6.1 The Scheme

The scheme is defined with respect to an oracle space

$$\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\} \times \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\} \times \mathcal{OS}_1 \times \mathcal{OS}_2,$$

where $\mathcal{OS}_1$ and $\mathcal{OS}_2$ defines the oracle space of SigS and Sig respectively. As in Section 5, we do not make the random oracles underlying SigS and Sig explicit and assume that queries are simply forwarded. The construction is depicted in Figure 7.

$\text{Gen}[H_1, H_2]$	$\text{Sgn}[H_1, H_2](\text{sk}, m)$
01 $(\text{sk}_1, \text{pk}_1) \xleftarrow{\$} \text{SigS.Gen}$	10 $\text{sk} \rightarrow (\text{sk}_1, \text{sk}_2)$
02 $(\text{sk}_2, \text{pk}_2) \xleftarrow{\$} \text{Sig.Gen}$	11 $r \xleftarrow{\$} \{0, 1\}^\kappa$
03 return $((\text{sk}_1, \text{sk}_2), (\text{pk}_1, \text{pk}_2))$	12 $(\text{pk}_1, \text{pk}_2) \leftarrow (\text{derivePK}(\text{sk}_1), \text{derivePK}(\text{sk}_2))$
$\text{Ver}[H_1, H_2](\text{pk}, m, \sigma)$	13 $m' \leftarrow H_1(\text{pk}_1 \parallel \text{pk}_2 \parallel m)$
04 $\text{pk} \rightarrow (\text{pk}_1, \text{pk}_2)$	14 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(\text{sk}_2, m' \parallel r)$
05 $\sigma \rightarrow (\sigma_1, \sigma_2)$	15 $h \leftarrow H_2(m' \parallel \sigma_2 \parallel r)$
06 $r \leftarrow \text{Ext}(\text{pk}_1, \sigma_1)$	16 $\sigma_1 \xleftarrow{\$} \text{Sgn}_{\text{salt}}(\text{sk}_1, h, r)$
07 $m' \leftarrow H_1(\text{pk}_1 \parallel \text{pk}_2 \parallel m)$	17 return (σ_1, σ_2)
08 $h \leftarrow H_2(m' \parallel \sigma_2 \parallel r)$	
09 return $\text{Ver}_1(\text{pk}_1, h, \sigma_1) \wedge \text{Ver}_2(\text{pk}_2, m' \parallel r, \sigma_2)$	

Figure 7. Construction $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda] = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$ from a salt-based signature scheme $\text{SigS} = (\mathcal{OS}_1, \text{SigS.Gen}, \text{Sgn}_1, \text{Sgn}_{\text{salt}}, \text{Ext}, \text{Ver}_1)$ and a signature scheme $\text{Sig} = (\mathcal{OS}_2, \text{Sig.Gen}, \text{Sgn}_2, \text{Ver}_2)$.

6.2 Security

Theorem 10 ($(\text{SUF}_1 \vee \text{SUF}_2) \wedge \text{Sig}_1 \text{ salt-unique} \Rightarrow \text{SUF}$). If Sig_1 is salt-unique, then for any adversary $\mathcal{A}$, making at most Q_s signing queries and Q_{RO} random oracle queries, against the **SUF-CMA** security of $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda]$ (Figure 7) in the random oracle model, there exist an **SUF-CMA** adversary $\mathcal{B}$ against Sig_2 and an **SUF-CMA** adversary $\mathcal{C}$ against Sig_1 with $t_{\mathcal{A}} \approx t_{\mathcal{B}} \approx t_{\mathcal{C}}$ such that

$$\text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} \leq \min \left\{ \text{Adv}_{\text{Sig}, \mathcal{B}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}, \text{Adv}_{\text{Sig}, \mathcal{C}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} + Q_{\text{RO}} \cdot (\gamma_{\text{Sig}} 2^{-\kappa} + 2^{-\lambda+1}) \right\}.$$

The proof can be found in Appendix E.1.

Theorem 11 (EO). For any adversary $\mathcal{A}$ against the **EO** security of $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda]$ (Figure 7), there exist an **EO** adversary $\mathcal{B}$ against Sig , an **MBS** adversary $\mathcal{C}$ against Sig , and an **CR** adversary $\mathcal{D}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}} = t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \mathcal{A}}^{\text{EO}} \leq \text{Adv}_{\text{Sig}, \mathcal{B}}^{\text{EO}} + \text{Adv}_{\text{Sig}, \mathcal{C}}^{\text{MBS}} + \text{Adv}_{\mathcal{OS}, \mathcal{D}}^{\text{CR}}.$$

Proof. The proof can be done analogously to the proof of Theorem 7. ■

Theorem 12 (MBS). For any adversary $\mathcal{A}$ against the **MBS** security of $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda]$ (Figure 7), there exist an **MBS** adversary $\mathcal{B}$ against Sig and an **CR** adversary $\mathcal{C}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}}$ such that

$$\text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \mathcal{A}}^{\text{MBS}} \leq \text{Adv}_{\text{Sig}, \mathcal{B}}^{\text{MBS}} + \text{Adv}_{\mathcal{OS}, \mathcal{D}}^{\text{CR}}.$$

Proof. The proof can be done analogously to the proof of Theorem 8. ■

Theorem 13 (NR). For any adversaries $\mathcal{A}$ and $\mathcal{D}$ against the **NR** security of $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda] := (\text{Gen}, \cdot, \cdot)$ (Figure 7), there exist **HnS** adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$, an **RMV** adversary $\mathcal{C}$ against Sig , and an **RMV** adversary $\mathcal{E}$ against SigS with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}} = t_{\mathcal{E}}$ and $t_{\mathcal{D}} = t_{\bar{\mathcal{D}}}$ such that

$$\text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}} + \min \left\{ \text{Adv}_{\text{Sig}, \mathcal{C}}^{\{0, 1\}^\lambda\text{-RMV}}, \text{Adv}_{\text{SigS}, \mathcal{E}}^{\{0, 1\}^\lambda\text{-RMV}} + \frac{Q_{\mathcal{A}}}{2^\lambda} \right\}.$$

and

$$\mathcal{H}_\infty \left(m \mid \text{RO}, \text{sk}, \text{aux}(\text{sk}, m) \right) \leq \mathcal{H}_\infty \left(x \mid \text{RO}, z \right)_{\substack{(\text{sk}, \text{pk}) \xleftarrow{\$} \text{Gen} \\ m \xleftarrow{\$} \mathcal{D}^{\text{RO}}(\text{sk})}}_{(x, z) \xleftarrow{\$} \bar{\mathcal{D}}^{\text{RO}}}$$

The proof can be found in Appendix E.2. Remark 1 interprets the theorem.

7 Instantiation and Concrete Security

In this section, we describe how our constructions can be instantiated. We again assume that the first component will be classical and the second component will be post-quantum. We start with discussing candidates for the PQ component that can be plugged into each of our constructions as a second component. Then, we show how each of the constructions can be instantiated by a specific class of classical signature schemes and that the respective instantiations achieve all requirements needed for our security bounds. An overview can be found in Table 1.

7.1 Post-Quantum Schemes

From the NIST winners, ML-DSA and FALCON (standardized as FN-DSA), are proven to be strongly unforgeable [DKL⁺18, KLS18, GJK24]. Besides strong unforgeability, the required properties are message-bound security (**MBS**), random-message validity (**RMV**) and signature spreadness (denoted γ) which are all needed for BIRD-OF-PREY-2. To preserve exclusive ownership (and message-bound security) we also require the two properties from the PQ scheme.

ML-DSA. Message-bound security and exclusive ownership was shown in [CDF⁺21] and signature spreadness was shown in [KLS18]. Random-message validity holds since in ML-DSA the message is hashed to obtain the challenge. For a random message, a signature only verifies if there is a collision in the hash.

FALCON. Message-bound security was shown in [CDF⁺21] and exclusive ownership follows with similar arguments as in ML-DSA if the public key is included in the hash. This is not the case in the round three submission of FALCON but considered to be included in the final FIPS standard. Random-message validity holds since FALCON follows the full-domain hash paradigm and the message is hashed. Hence, **RMV** reduces to a collision in the hash function. It remains to show that FALCON signatures have a sufficient min-entropy. The signature includes a uniformly chosen salt of 320 bits resulting in $\gamma_{\text{FALCON}} \leq 2^{-320}$. This is already sufficient even though there is more entropy involved in the preimage sampling step.

7.2 Instantiating our Constructions

BIRD-OF-PREY-1. This construction can be instantiated with BLS since it is unique. Since **EO** and **MBS** are information theoretically fulfilled by the PQ components, we do not require any further properties from the classical component.

BIRD-OF-PREY-2. This construction can be instantiated using Schnorr signatures [Sch91]. The most prominent and widely used example is EdDSA [BDL⁺12]. Parallel impersonation security is implied by the signature's (no-message) unforgeability [KMP16]. Unforgeability of EdDSA was shown in [BCJZ21] and the responses are perfectly unique fulfilling **UR** without any loss. Further, the challenge set is sufficiently large (512 bits) compensating the statistical term of Theorem 6.

BIRD-OF-PREY-3. This construction can be instantiated using any RSA-based signature scheme. The most prominent example is RSASSA-PSS as used in PKCS#1 v2.2 [MKJR16]. According to [MKJR16], the salt size is usually 0 (making the scheme deterministic) or the size of the range of the hash function which is usually sufficient on their own to compensate the statistical term in Theorem 10. In case the deterministic version is chosen (salt size being equal to 0), we can rely on the min-entropy of signatures of the PQ component for the security proof. As discussed in the beginning of the section, all PQ candidates achieve a sufficiently large spreadness.

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References

- [ABF12] Afonso Arriaga, Manuel Barbosa, and Pooya Farshim. On the joint security of signature and encryption schemes under randomness reuse: Efficiency and security amplification. In Feng Bao, Pierangela Samarati, and Jianying Zhou, editors, *ACNS 12: 10th International Conference on Applied Cryptography and Network Security*, volume 7341 of *Lecture Notes in Computer Science*, pages 206–223, Singapore, June 26–29, 2012. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-642-31284-7_13. (Cited on page 4.)
- [AEE⁺21] Lukas Aumayr, Oguzhan Ersoy, Andreas Erwig, Sebastian Faust, Kristina Hostáková, Matteo Maffei, Pedro Moreno-Sanchez, and Siavash Riahi. Generalized channels from limited blockchain scripts and adaptor signatures. In Mehdi Tibouchi and Huaxiong Wang, editors, *Advances in Cryptology – ASIACRYPT 2021, Part II*, volume 13091 of *Lecture Notes in Computer Science*, pages 635–664, Singapore, December 6–10, 2021. Springer, Cham, Switzerland. doi:10.1007/978-3-030-92075-3_22. (Cited on page 4.)
- [AJKL23] Joël Alwen, Jonas Janneck, Eike Kiltz, and Benjamin Lipp. The pre-shared key modes of HPKE. In Jian Guo and Ron Steinfeld, editors, *Advances in Cryptology – ASIACRYPT 2023, Part VI*, volume 14443 of *Lecture Notes in Computer Science*, pages 329–360, Guangzhou, China, December 4–8, 2023. Springer, Singapore, Singapore. doi:10.1007/978-981-99-8736-8_11. (Cited on page 4.)
- [Alg25] Algorand. Leading on post-quantum technology, 2025. URL: <https://algorand.co/technology/post-quantum>. (Cited on page 4.)
- [ANS23] ANSSI. Anssi views on the post-quantum cryptography transition (2023 follow up), 2023. URL: https://cyber.gouv.fr/sites/default/files/document/follow_up_position_paper_on_post_quantum_cryptography.pdf. (Cited on page 3.)
- [AWS25] AWS. Aws kms adds support for post-quantum ml-dsa digital signatures, 2025. URL: <https://aws.amazon.com/about-aws/whats-new/2025/06/aws-kms-post-quantum-ml-dsa-digital-signatures/>. (Cited on page 4.)
- [BBCT22] Daniel J. Bernstein, Billy Bob Brumley, Ming-Shing Chen, and Nicola Tuveri. OpenSSLNTRU: Faster post-quantum TLS key exchange. In Kevin R. B. Butler and Kurt Thomas, editors, *USENIX Security 2022: 31st USENIX Security Symposium*, pages 845–862, Boston, MA, USA, August 10–12, 2022. USENIX Association. URL: <https://www.usenix.org/conference/usenixsecurity22/presentation/bernstein>. (Cited on page 3.)
- [BBD⁺23] Joppe W. Bos, Olivier Bronchain, Léo Ducas, Serge Fehr, Yu-Hsuan Huang, Thomas Pornin, Eamonn W. Postlethwaite, Thomas Prest, Ludo N. Pulles, and Wessel van Woerden. HAWK. Technical report, National Institute of Standards and Technology, 2023. available at <https://csrc.nist.gov/Projects/pqc-dig-sig/round-1-additional-signatures>. (Cited on page 9.)
- [BCD⁺24] Manuel Barbosa, Deirdre Connolly, João Diogo Duarte, Aaron Kaiser, Peter Schwabe, Karoline Varner, and Bas Westerbaan. X-Wing. *IACR Communications in Cryptology (CiC)*, 1(1):21, 2024. doi:10.62056/a3qj89n4e. (Cited on page 3.)
- [BCJZ21] Jacqueline Brendel, Cas Cremers, Dennis Jackson, and Mang Zhao. The provable security of Ed25519: Theory and practice. In *2021 IEEE Symposium on Security and Privacy*, pages 1659–1676, San Francisco, CA, USA, May 24–27, 2021. IEEE Computer Society Press. doi:10.1109/SP40001.2021.00042. (Cited on pages 4, 7, 9, and 19.)
- [BCNS15] Joppe W. Bos, Craig Costello, Michael Naehrig, and Douglas Stebila. Post-quantum key exchange for the TLS protocol from the ring learning with errors problem. In *2015 IEEE Symposium on Security and Privacy*, pages 553–570, San Jose, CA, USA, May 17–21, 2015. IEEE Computer Society Press. doi:10.1109/SP.2015.40. (Cited on page 3.)
- [BDK⁺14] Florian Bergsma, Benjamin Dowling, Florian Kohlar, Jörg Schwenk, and Douglas Stebila. Multi-ciphersuite security of the Secure Shell (SSH) protocol. In Gail-Joon Ahn, Moti Yung, and Ninghui Li, editors, *ACM CCS 2014: 21st Conference on Computer and Communications Security*, pages 369–381, Scottsdale, AZ, USA, November 3–7, 2014. ACM Press. doi:10.1145/2660267.2660286. (Cited on page 4.)
- [BDL⁺12] Daniel J. Bernstein, Niels Duif, Tanja Lange, Peter Schwabe, and Bo-Yin Yang. High-speed high-security signatures. *Journal of Cryptographic Engineering*, 2(2):77–89, September 2012. doi:10.1007/s13389-012-0027-1. (Cited on pages 4, 7, and 19.)
- [BH23] Nina Bindel and Britta Hale. A note on hybrid signature schemes. Cryptology ePrint Archive, Report 2023/423, 2023. URL: <https://eprint.iacr.org/2023/423>. (Cited on pages 3, 4, 5, and 25.)

- [BHJ⁺15] Christoph Bader, Dennis Hofheinz, Tibor Jager, Eike Kiltz, and Yong Li. Tightly-secure authenticated key exchange. In Yevgeniy Dodis and Jesper Buus Nielsen, editors, *TCC 2015: 12th Theory of Cryptography Conference, Part I*, volume 9014 of *Lecture Notes in Computer Science*, pages 629–658, Warsaw, Poland, March 23–25, 2015. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-662-46494-6_26. (Cited on page 4.)
- [BHK⁺19] Daniel J Bernstein, Andreas Hülsing, Stefan Kölbl, Ruben Niederhagen, Joost Rijneveld, and Peter Schwabe. The sphincs+ signature framework. In *Proceedings of the 2019 ACM SIGSAC conference on computer and communications security*, pages 2129–2146, 2019. (Cited on page 3.)
- [BHMS17] Nina Bindel, Udyani Herath, Matthew McKague, and Douglas Stebila. Transitioning to a quantum-resistant public key infrastructure. In Tanja Lange and Tsuyoshi Takagi, editors, *Post-Quantum Cryptography - 8th International Workshop, PQCrypto 2017*, pages 384–405, Utrecht, The Netherlands, June 26–28, 2017. Springer, Cham, Switzerland. doi:10.1007/978-3-319-59879-6_22. (Cited on pages 4, 5, and 25.)
- [BLS04] Dan Boneh, Ben Lynn, and Hovav Shacham. Short signatures from the Weil pairing. *Journal of Cryptology*, 17(4):297–319, September 2004. doi:10.1007/s00145-004-0314-9. (Cited on pages 4 and 7.)
- [BN06] Mihir Bellare and Gregory Neven. Multi-signatures in the plain public-key model and a general forking lemma. In Ari Juels, Rebecca N. Wright, and Sabrina De Capitani di Vimercati, editors, *ACM CCS 2006: 13th Conference on Computer and Communications Security*, pages 390–399, Alexandria, Virginia, USA, October 30 – November 3, 2006. ACM Press. doi:10.1145/1180405.1180453. (Cited on page 27.)
- [BR93] Mihir Bellare and Phillip Rogaway. Random oracles are practical: A paradigm for designing efficient protocols. In Dorothy E. Denning, Raymond Pyle, Ravi Ganesan, Ravi S. Sandhu, and Victoria Ashby, editors, *ACM CCS 93: 1st Conference on Computer and Communications Security*, pages 62–73, Fairfax, Virginia, USA, November 3–5, 1993. ACM Press. doi:10.1145/168588.168596. (Cited on pages 8 and 17.)
- [BR96] Mihir Bellare and Phillip Rogaway. The exact security of digital signatures: How to sign with RSA and Rabin. In Ueli M. Maurer, editor, *Advances in Cryptology – EUROCRYPT’96*, volume 1070 of *Lecture Notes in Computer Science*, pages 399–416, Saragossa, Spain, May 12–16, 1996. Springer Berlin Heidelberg, Germany. doi:10.1007/3-540-68339-9_34. (Cited on pages 8 and 17.)
- [BR06] Mihir Bellare and Phillip Rogaway. The security of triple encryption and a framework for code-based game-playing proofs. In Serge Vaudenay, editor, *Advances in Cryptology – EUROCRYPT 2006*, volume 4004 of *Lecture Notes in Computer Science*, pages 409–426, St. Petersburg, Russia, May 28 – June 1, 2006. Springer Berlin Heidelberg, Germany. doi:10.1007/11761679_25. (Cited on page 8.)
- [BSI24] BSI. Cryptographic mechanisms: Recommendations and key lengths - bsi tr-02102-1, 2024. URL: <https://www.bsi.bund.de/SharedDocs/Downloads/EN/BSI/Publications/TechGuidelines/TG02102/BSI-TR-02102-1.pdf>. (Cited on page 3.)
- [CDF⁺21] Cas Cremers, Samed Düzl , Rune Fiedler, Marc Fischlin, and Christian Janson. BUFFing signature schemes beyond unforgeability and the case of post-quantum signatures. In *2021 IEEE Symposium on Security and Privacy*, pages 1696–1714, San Francisco, CA, USA, May 24–27, 2021. IEEE Computer Society Press. doi:10.1109/SP40001.2021.00093. (Cited on pages 5, 12, and 19.)
- [CGWZ25] Cas Cremers, Esra G nsay, Vera Wesselkamp, and Mang Zhao. ETK: External-operations TreeKEM and the security of MLS in RFC 9420. Cryptology ePrint Archive, Report 2025/229, 2025. URL: <https://eprint.iacr.org/2025/229>. (Cited on page 4.)
- [CHH⁺25] Deirdre Connolly, Kathrin H velmanns, Andreas H lsing, Stavros Kousidis, and Matthias Meijers. Starfighters—on the general applicability of X-wing. Cryptology ePrint Archive, Report 2025/1397, 2025. URL: <https://eprint.iacr.org/2025/1397>. (Cited on page 3.)
- [DFH⁺24] Jelle Don, Serge Fehr, Yu-Hsuan Huang, Jyun-Jie Liao, and Patrick Struck. Hide-and-seek and the non-resignability of the BUFF transform. In Elette Boyle and Mohammad Mahmoody, editors, *TCC 2024: 22nd Theory of Cryptography Conference, Part III*, volume 15366 of *Lecture Notes in Computer Science*, pages 347–370, Milan, Italy, December 2–6, 2024. Springer, Cham, Switzerland. doi:10.1007/978-3-031-78020-2_12. (Cited on pages 5, 8, 10, 12, and 13.)
- [DFHS24] Jelle Don, Serge Fehr, Yu-Hsuan Huang, and Patrick Struck. On the (in)security of the BUFF transform. In Leonid Reyzin and Douglas Stebila, editors, *Advances in Cryptology – CRYPTO 2024, Part I*, volume 14920 of *Lecture Notes in Computer Science*, pages 246–275, Santa Barbara, CA, USA, August 18–22, 2024. Springer, Cham, Switzerland. doi:10.1007/978-3-031-68376-3_8. (Cited on pages 5 and 12.)

- [DKL⁺18] Léo Ducas, Eike Kiltz, Tancrede Lepoint, Vadim Lyubashevsky, Peter Schwabe, Gregor Seiler, and Damien Stehlé. Crystals-dilithium: A lattice-based digital signature scheme. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, pages 238–268, 2018. (Cited on pages 3, 4, and 19.)
- [DS24] Samed Düzlül and Patrick Struck. The role of message-bound signatures for the beyond UnForgeability features and weak keys. In Nicky Mouha and Nick Nikiforakis, editors, *ISC 2024: 27th International Conference on Information Security, Part II*, volume 15258 of *Lecture Notes in Computer Science*, pages 61–80, Arlington, VA, USA, October 23–25, 2024. Springer, Cham, Switzerland. doi:10.1007/978-3-031-75764-8_4. (Cited on pages 8 and 12.)
- [FH25] Sebastian Faller and Julia Hesse. How to (not) combine oblivious pseudorandom functions. Cryptology ePrint Archive, Paper 2025/1084, 2025. URL: <https://eprint.iacr.org/2025/1084>. (Cited on page 3.)
- [GHH⁺24] Sharon Goldberg, Miro Haller, Nadia Heninger, Mike Milano, Dan Shumow, Marc Stevens, and Adam Suhl. RADIUS/UDP considered harmful. In Davide Balzarotti and Wenyuan Xu, editors, *USENIX Security 2024: 33rd USENIX Security Symposium*, Philadelphia, PA, USA, August 14–16, 2024. USENIX Association. URL: <https://www.usenix.org/conference/usenixsecurity24/presentation/goldberg>. (Cited on page 4.)
- [GHJ25] Phillip Gajland, Vincent Hwang, and Jonas Janneck. Shadowfax: A deniability-preserving AKEM combiner. Cryptology ePrint Archive, Report 2025/154, 2025. URL: <https://eprint.iacr.org/2025/154>. (Cited on pages 3 and 4.)
- [GHP18] Federico Giacon, Felix Heuer, and Bertram Poettering. KEM combiners. In Michel Abdalla and Ricardo Dahab, editors, *PKC 2018: 21st International Conference on Theory and Practice of Public Key Cryptography, Part I*, volume 10769 of *Lecture Notes in Computer Science*, pages 190–218, Rio de Janeiro, Brazil, March 25–29, 2018. Springer, Cham, Switzerland. doi:10.1007/978-3-319-76578-5_7. (Cited on page 3.)
- [GJK24] Phillip Gajland, Jonas Janneck, and Eike Kiltz. A closer look at falcon. Cryptology ePrint Archive, Report 2024/1769, 2024. URL: <https://eprint.iacr.org/2024/1769>. (Cited on pages 4 and 19.)
- [GKP⁺23] Diana Ghinea, Fabian Kaczmarczyk, Jennifer Pullman, Julien Cretin, Stefan Kölbl, Rafael Misoczki, Jean-Michel Picod, Luca Invernizzi, and Elie Bursztein. Hybrid post-quantum signatures in hardware security keys. In *International Conference on Applied Cryptography and Network Security*, pages 480–499. Springer, 2023. (Cited on pages 4 and 5.)
- [GPV08] Craig Gentry, Chris Peikert, and Vinod Vaikuntanathan. Trapdoors for hard lattices and new cryptographic constructions. In Richard E. Ladner and Cynthia Dwork, editors, *40th Annual ACM Symposium on Theory of Computing*, pages 197–206, Victoria, BC, Canada, May 17–20, 2008. ACM Press. doi:10.1145/1374376.1374407. (Cited on page 17.)
- [GRSV25] Felix Günther, Michael Rosenberg, Douglas Stebila, and Shannon Veitch. Hybrid obfuscated key exchange and KEMs. Cryptology ePrint Archive, Report 2025/408, 2025. URL: <https://eprint.iacr.org/2025/408>. (Cited on pages 3 and 4.)
- [HR25] Julia Hesse and Michael Rosenberg. PAKE combiners and efficient post-quantum instantiations. In *Advances in Cryptology – EUROCRYPT 2025, Part II*, Lecture Notes in Computer Science, pages 395–420. Springer, Cham, Switzerland, June 2025. doi:10.1007/978-3-031-91124-8_14. (Cited on pages 3 and 4.)
- [JKRS21] Tibor Jager, Eike Kiltz, Doreen Riepel, and Sven Schäge. Tightly-secure authenticated key exchange, revisited. In Anne Canteaut and François-Xavier Standaert, editors, *Advances in Cryptology – EUROCRYPT 2021, Part I*, volume 12696 of *Lecture Notes in Computer Science*, pages 117–146, Zagreb, Croatia, October 17–21, 2021. Springer, Cham, Switzerland. doi:10.1007/978-3-030-77870-5_5. (Cited on page 4.)
- [JMOR25] Jonas Janneck, Jonas Meers, Massimo Ostuzzi, and Doreen Riepel. Snake mackerel: An isogeny-based AKEM leveraging randomness reuse. Cryptology ePrint Archive, Report 2025/1474, 2025. URL: <https://eprint.iacr.org/2025/1474>. (Cited on page 4.)
- [Kli17] E. Klitzke. Bitcoin transaction malleability, 2017. URL: <https://eklitzke.org/bitcoin-transaction-malleability>. (Cited on page 4.)
- [KLS18] Eike Kiltz, Vadim Lyubashevsky, and Christian Schaffner. A concrete treatment of Fiat-Shamir signatures in the quantum random-oracle model. In Jesper Buus Nielsen and Vincent Rijmen, editors, *Advances in Cryptology – EUROCRYPT 2018, Part III*, volume 10822 of *Lecture Notes in Computer Science*, pages 552–586, Tel Aviv, Israel, April 29 – May 3, 2018. Springer, Cham, Switzerland. doi:10.1007/978-3-319-78372-7_18. (Cited on pages 4 and 19.)
- [KMP16] Eike Kiltz, Daniel Masny, and Jiaxin Pan. Optimal security proofs for signatures from identification schemes. In Matthew Robshaw and Jonathan Katz, editors, *Advances in Cryptology – CRYPTO 2016*,

- Part II*, volume 9815 of *Lecture Notes in Computer Science*, pages 33–61, Santa Barbara, CA, USA, August 14–18, 2016. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-662-53008-5_2. (Cited on pages 11, 15, and 19.)
- [KS24] Ehren Kret and Rolfe Schmidt. The pqxdh key agreement protocol, 2024. URL: <https://signal.org/docs/specifications/pqxdh/pqxdh.pdf>. (Cited on page 3.)
- [KV19] Kris Kwiatkowski and Luke Valenta. The TLS post-quantum experiment. Post on the Cloudflare blog, 2019. <https://blog.cloudflare.com/the-tls-post-quantum-experiment/>. (Cited on page 3.)
- [Lan16] Adam Langley. CECQP1 results. Blog post, 2016. <https://www.imperialviolet.org/2016/11/28/cecpq1.html>. (Cited on page 3.)
- [Lan18] Adam Langley. CECQP2. Blog post, 2018. <https://www.imperialviolet.org/2018/12/12/cecpq2.html>. (Cited on page 3.)
- [LL25] You Lyu and Shengli Liu. Hybrid password authentication key exchange in the UC framework. In *Advances in Cryptology – EUROCRYPT 2025, Part II*, Lecture Notes in Computer Science, pages 421–450. Springer, Cham, Switzerland, June 2025. doi:10.1007/978-3-031-91124-8_15. (Cited on pages 3 and 4.)
- [LSB24] Felix Linker, Ralf Sasse, and David Basin. A formal analysis of apple’s iMessage PQ3 protocol. Cryptology ePrint Archive, Paper 2024/1395, 2024. URL: <https://eprint.iacr.org/2024/1395>. (Cited on page 3.)
- [MKJR16] Kathleen Moriarty, Burt Kaliski, Jakob Jonsson, and Andreas Rusch. PKCS #1: RSA Cryptography Specifications Version 2.2. RFC 8017, November 2016. URL: <https://www.rfc-editor.org/info/rfc8017>, doi:10.17487/RFC8017. (Cited on pages 4, 8, 17, and 19.)
- [MKTW25] Jake Massimo, Panos Kampanakis, Sean Turner, and Bas Westerbaan. Internet X.509 Public Key Infrastructure - Algorithm Identifiers for the Module-Lattice-Based Digital Signature Algorithm (ML-DSA). Internet-Draft draft-ietf-lamps-dilithium-certificates-12, Internet Engineering Task Force, June 2025. Work in Progress. URL: <https://datatracker.ietf.org/doc/draft-ietf-lamps-dilithium-certificates-12/>. (Cited on page 4.)
- [MLD24] Module-lattice-based digital signature standard. National Institute of Standards and Technology NIST FIPS PUB 204, U.S. Department of Commerce, August 2024. URL: <http://dx.doi.org/10.6028/NIST.FIPS.204>, doi:10.6028/nist.fips.204. (Cited on page 3.)
- [MLK24] Module-lattice-based key-encapsulation mechanism standard. National Institute of Standards and Technology NIST FIPS PUB 203, U.S. Department of Commerce, August 2024. URL: <http://dx.doi.org/10.6028/NIST.FIPS.203>, doi:10.6028/nist.fips.203. (Cited on page 3.)
- [NIS16] NIST. Submission requirements and evaluation criteria for the post-quantum cryptography standardization process, 2016. <https://csrc.nist.gov/CSRC/media/Projects/Post-Quantum-Cryptography/documents/call-for-proposals-final-dec-2016.pdf>. (Cited on page 3.)
- [NIS25] NIST. Post-quantum cryptography pqc, 2025. URL: <https://csrc.nist.gov/projects/post-quantum-cryptography>. (Cited on page 3.)
- [OGP⁺26] Mike Ounsworth, John Gray, Massimiliano Pala, Jan Klaußner, and Scott Fluhrer. Composite ML-DSA for use in X.509 Public Key Infrastructure. Internet-Draft draft-ietf-lamps-pq-composite-sigs-14, Internet Engineering Task Force, January 2026. Work in Progress. URL: <https://datatracker.ietf.org/doc/draft-ietf-lamps-pq-composite-sigs-14/>. (Cited on page 4.)
- [PFH⁺20] Thomas Prest, Pierre-Alain Fouque, Jeffrey Hoffstein, Paul Kirchner, Vadim Lyubashevsky, Thomas Pornin, Thomas Ricosset, Gregor Seiler, William Whyte, and Zhenfei Zhang. Falcon. Submission to the NIST Post-Quantum Cryptography Standardization Project, 2020. URL: <https://falcon-sign.info/>. (Cited on pages 3 and 17.)
- [PS05] Thomas Pornin and Julien P. Stern. Digital signatures do not guarantee exclusive ownership. In John Ioannidis, Angelos Keromytis, and Moti Yung, editors, *ACNS 05: 3rd International Conference on Applied Cryptography and Network Security*, volume 3531 of *Lecture Notes in Computer Science*, pages 138–150, New York, NY, USA, June 7–10, 2005. Springer Berlin Heidelberg, Germany. doi:10.1007/11496137_10. (Cited on pages 8 and 12.)
- [PST20] Christian Paquin, Douglas Stebila, and Goutam Tamvada. Benchmarking post-quantum cryptography in TLS. In Jintai Ding and Jean-Pierre Tillich, editors, *Post-Quantum Cryptography - 11th International Conference, PQCrypto 2020*, pages 72–91, Paris, France, April 15–17, 2020. Springer, Cham, Switzerland. doi:10.1007/978-3-030-44223-1_5. (Cited on page 3.)

- [PWJ⁺25] Lucas Prabel, Guilin WANG, Jonas Janneck, Tirumaleswar Reddy.K, and John Preuß Mattsson. Hybrid Digital Signatures with Strong Unforgeability. Internet-Draft draft-prabel-cfrg-suf-hybrid-sigs-00, Internet Engineering Task Force, October 2025. Work in Progress. URL: <https://datatracker.ietf.org/doc/draft-prabel-cfrg-suf-hybrid-sigs/00/>. (Cited on page 4.)
- [RSA78] Ronald L Rivest, Adi Shamir, and Leonard Adleman. A method for obtaining digital signatures and public-key cryptosystems. *Communications of the ACM*, 21(2):120–126, 1978. (Cited on page 4.)
- [Sch91] Claus-Peter Schnorr. Efficient signature generation by smart cards. *Journal of Cryptology*, 4(3):161–174, January 1991. doi:10.1007/BF00196725. (Cited on page 19.)
- [SLH24] Stateless hash-based digital signature standard. National Institute of Standards and Technology NIST FIPS PUB 205, U.S. Department of Commerce, August 2024. URL: <http://dx.doi.org/10.6028/NIST.FIPS.205>, doi:10.6028/nist.fips.205. (Cited on page 3.)
- [SPW07] Ron Steinfeld, Josef Pieprzyk, and Huaxiong Wang. How to strengthen any weakly unforgeable signature into a strongly unforgeable signature. In *Cryptographers’ Track at the RSA Conference*, pages 357–371. Springer, 2007. (Cited on page 5.)
- [Ste24] Douglas Stebila. Security analysis of the iMessage PQ3 protocol. Cryptology ePrint Archive, Report 2024/357, 2024. URL: <https://eprint.iacr.org/2024/357>. (Cited on page 3.)
- [TMM21] Erkan Tairi, Pedro Moreno-Sanchez, and Matteo Maffei. Post-quantum adaptor signature for privacy-preserving off-chain payments. In Nikita Borisov and Claudia Díaz, editors, *FC 2021: 25th International Conference on Financial Cryptography and Data Security, Part II*, volume 12675 of *Lecture Notes in Computer Science*, pages 131–150, Virtual Event, March 1–5, 2021. Springer Berlin Heidelberg, Germany. doi:10.1007/978-3-662-64331-0_7. (Cited on page 4.)
- [WFLY04] Xiaoyun Wang, Dengguo Feng, Xuejia Lai, and Hongbo Yu. Collisions for hash functions MD4, MD5, HAVAL-128 and RIPEMD. Cryptology ePrint Archive, Report 2004/199, 2004. URL: <https://eprint.iacr.org/2004/199>. (Cited on page 4.)
- [WR19] Bas Westerbaan and Cefan Daniel Rubin. Defending against future threats: Cloudflare goes post-quantum. Post on the Cloudflare blog, 2019. <https://blog.cloudflare.com/post-quantum-for-all/>. (Cited on page 3.)

Supplementary Material

A Non-Separability

Non-separability was introduced in [BHMS17]. It describes a property specific to signature combiners and says that it should be hard given a combined signature to separate a valid signature of one of the underlying schemes. The authors define non-separability with respect to a class of valid messages for the security game and define a recognizer algorithm identifying signatures from said class. However, this security notion can be fulfilled by simply adding unique identifiers to messages before signing indicating that the signature originates from a combined signature scheme. To circumvent this, the notion is further strengthened in [BH23] to allow adversaries to output any message/signature pair which is valid. In the following we formalize the description of strong non-separability as textually described in [BH23] and compare it with the weaker version of [BHMS17].

For the purpose of the following combiner-specific notions, we use the following notation. We are considering a combiner C instantiated by two signature schemes Sig_1 and Sig_2 and potentially further primitives denoted by A . We will write $C[\text{Sig}_1, \text{Sig}_2, A]$ or just C if it is clear from context. Since a signature combiner is itself a signature scheme, we use the common syntax, e.g. $C.\text{Sgn}$ to denote its signing algorithm. Further, we assume that there also exists an algorithm ExtPK associated to C which on input a public key of C and an index $\tau \in \{1, 2\}$ extracts a valid public key of Sig_τ .¹⁵

Definition 17 (Non-Separability). For a signature combiner $C[\text{Sig}_1, \text{Sig}_2, A]$, *weak/strong non-separability* for $\text{Sig}_\tau, \tau \in \{1, 2\}$ is defined via the games in Figure 8. Weak non-separability is further defined with respect to a recognizer algorithm R . The advantage functions of an adversary $\mathcal{A}$ are defined as

$$\begin{aligned} \text{Adv}_{C[\text{Sig}_1, \text{Sig}_2, A], R, \mathcal{A}}^{\tau\text{-wNS}} &:= \Pr[\tau\text{-wNS}_{C[\text{Sig}_1, \text{Sig}_2, A], R}(\mathcal{A}) \Rightarrow 1], \\ \text{Adv}_{C[\text{Sig}_1, \text{Sig}_2, A], \mathcal{A}}^{\tau\text{-sNS}} &:= \Pr[\tau\text{-sNS}_{C[\text{Sig}_1, \text{Sig}_2, A]}(\mathcal{A}) \Rightarrow 1]. \end{aligned}$$

Games $\tau\text{-wNS}_{C[\text{Sig}_1, \text{Sig}_2, A], R}(\mathcal{A}) / \tau\text{-sNS}_{C[\text{Sig}_1, \text{Sig}_2, A]}(\mathcal{A})$	Oracle $\text{Sgn}(m)$
01 $(\text{sk}, \text{pk}) \xleftarrow{\$} C.\text{Gen}$	07 return $C.\text{Sgn}(\text{sk}, m)$
02 $\text{pk}_\tau \leftarrow C.\text{ExtPK}(\text{pk}, \tau)$	
03 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}}(\text{pk})$	
04 if $R(m^*) = 1$	// wNS
05 return 0	// wNS
06 return $\text{Sig}_\tau.\text{Ver}(\text{pk}_\tau, m^*, \sigma^*)$	

Figure 8. Games defining wNS and sNS for a signature combiner $C[\text{Sig}_1, \text{Sig}_2, A]$.

It is easy to see that if signature scheme Sig_τ is not forgeable the combiner cannot be τ -non-separable.

A.1 Non-separability of our Constructions

As mentioned before, weak non-separability is not an inherent since it can be solved by pre- or appending a combiner identifier to the messages. For strong non-separability, we distinguish between the classical and the PQ component. One of our requirements was to use the PQ component black-box for practical and compliance reasons. The (reasonable) black-box use of one signature components always leads to the

¹⁵ All reasonable signature combiners include both public keys in their combined key.

combiner non-separable for that component because the verification algorithm (which is publicly executable) must at some point extract the signature of that component which means that an adversary can separate it as well. Hence, our constructions do not achieve strong non-separability for the PQ component.

For the classical component, however, a non-separability property makes much more sense since the classical part of a combiner might be susceptible to downgrading attacks. For BIRD-OF-PREY-1, the black-box use of the classical component leads to not achieving the notion as well. The same holds for BIRD-OF-PREY-3 where the classical signature is not completely used in a black-box way but the verification is still with respect to publicly computable hash value and there we also do not achieve strong non-separability. For BIRD-OF-PREY-2 we can hope for more because the both components are much more intertwined. In particular, an adversary against strong non-separability has the following problem. If they want to reuse a transcript they need to adjust the message and redefine it as $\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m$ which works. This makes m' the output of the random oracle which, for a verification of the classical component on itself, is the challenge of the transcript that is verified. However, for the combiner the challenge of the ID transcript is not m' but the output of $H_2(\sigma_2)$ where σ_2 is a signature on m' and not m' itself. Therefore, the classical component only verifies if there is a “collision” in the two hash functions.

B Additional Material for Section 3

B.1 Random-message Validity

Some of our proofs actually require a weaker version of **RMV**. For completeness, we define the weaker notion, state the hierarchy between them and show that even the weakest version is not implied by **UF-CMA**.

We start by restating the version used throughout the paper.

Definition 14 (Random-message Validity). For a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, we define *random-message validity* for some adversary $\mathcal{A}$ as

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{\mathcal{M}\text{-RMV}} := \Pr_{\mathbf{H} \leftarrow \mathcal{OS}} \left[\text{Ver}[\mathbf{H}](\text{pk}, m \| x, \sigma) \mid \begin{array}{l} m \leftarrow \mathcal{M} \\ (\text{pk}, \sigma, x) \leftarrow \mathcal{A}^{\text{RO}(\cdot)} \end{array} \right].$$

The additional appendix x is only needed in the proof of Theorem 13. Everywhere else a “plain” version would be sufficient, denoted as plain random-message validity.

Definition 18 (Plain Random-message Validity). For a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, we define *plain random-message validity* for some adversary $\mathcal{A}$ as

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{\mathcal{M}\text{-pRMV}} := \Pr_{\mathbf{H} \leftarrow \mathcal{OS}} \left[\text{Ver}[\mathbf{H}](\text{pk}, m, \sigma) \mid \begin{array}{l} m \leftarrow \mathcal{M} \\ (\text{pk}, \sigma) \leftarrow \mathcal{A}^{\text{RO}(\cdot)} \end{array} \right].$$

For all non-resignability proofs, it is crucial that the adversary is allowed to choose the signing key themselves. For the proof of Theorem 6, it would be sufficient if the key pair is chosen by the challenger and given to the adversary. We denote this notion by weak random-message validity.

Definition 19 (Weak Random-message Validity). For a signature scheme $\text{Sig} = (\mathcal{OS}, \text{Gen}, \text{Sgn}, \text{Ver})$, we define *weak random-message validity* for some adversary $\mathcal{A}$ as

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{\mathcal{M}\text{-wRMV}} := \Pr_{\mathbf{H} \leftarrow \mathcal{OS}} \left[\text{Ver}[\mathbf{H}](\text{pk}, m, \sigma) \mid \begin{array}{l} (\text{sk}, \text{pk}) \leftarrow \text{Gen}[\mathbf{H}] \\ m \leftarrow \mathcal{M} \\ \sigma \leftarrow \mathcal{A}^{\text{RO}(\cdot)}(\text{sk}, \text{pk}) \end{array} \right].$$

It is obvious that the following chain of implications hold:

$$\text{RMV} \Rightarrow \text{pRMV} \Rightarrow \text{wRMV}.$$

The following two lemmata show that **RMV** is strictly weaker than **MBS**. Especially, note that the reduction to **MBS** is not tight.

Lemma 1 (MBS $\Rightarrow$ RMV). For any **RMV** adversary $\mathcal{A}$ against a signature scheme Sig , there exists an **MBS** adversary $\mathcal{B}$ against Sig with $t_{\mathcal{A}} = t_{\mathcal{B}}$ such that

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{\mathcal{M}\text{-RMV}} \leq \sqrt{\text{Adv}_{\text{Sig}, \mathcal{B}}^{\text{MBS}}} + \frac{1}{|\mathcal{M}|}.$$

Proof. We construct adversary $\mathcal{B}$ in Figure 9. The bound directly follows by the General Forking Lemma [BN06]. ■

Lemma 2 (RMV $\not\Rightarrow$ MBS). There exists a signature scheme Sig which is **RMV** but not **MBS**.

Proof. Let $\text{Sig}' := (\mathcal{OS}', \text{Gen}', \text{Sgn}', \text{Ver}')$ be a signature scheme which is **RMV** and let $\text{Sig} := (\mathcal{OS}', \text{Gen}', \text{Sgn}', \text{Ver})$ where the verification algorithm is defined in Figure 10. Sig is also **RMV** secure since $m = 0$ is only chosen with probability $1/|\mathcal{M}|$. However, there is a simple adversary $\mathcal{A}$ constructed in Figure 10 against **MBS** of Sig with winning probability equal to the correctness of Sig . ■

<u>$\mathcal{B}$</u>	
01	$(pk, \sigma, x) \xleftarrow{\$} \mathcal{A}$
02	$m_1 \xleftarrow{\$} \mathcal{M}$
03	$m_2 \xleftarrow{\$} \mathcal{M}$
04	return $(pk, m_1 \ x, m_2 \ x, \sigma)$

Figure 9. Adversary $\mathcal{B}$ against MBS simulating the game for $\mathcal{A}$.

<u>$\text{Ver}(pk, m, \sigma)$</u>	<u>$\mathcal{A}$</u>
01 if $m = 0$	04 $(sk, pk) \xleftarrow{\$} \text{Gen}$
02 return 1	05 $m_1 \leftarrow 0$
03 return $\text{Ver}'(pk, m, \sigma)$	06 $m_1 \leftarrow 1$
	07 $\sigma \xleftarrow{\$} \text{Sgn}(sk, m_1)$
	08 return (pk, m_0, m_1, σ)

Figure 10. Algorithm Ver for construction Sig and adversary $\mathcal{A}$ against MBS of Sig .

Remark 2. Note that the signature scheme Sig constructed in the proof of Lemma 2

We also show that none of the notions is implied by unforgeability.

Lemma 3 (UF-CMA $\not\Rightarrow$ wRMV). There exists a signature scheme Sig which is **UF-CMA** but not **wRMV**.

Proof. Let $\text{Sig}' := (\mathcal{OS}', \text{Gen}', \text{Sgn}', \text{Ver}')$ be a signature scheme which is **UF-CMA** and let $\text{Sig} := (\mathcal{OS}', \text{Gen}', \text{Sgn}', \text{Ver})$ where the verification algorithm is defined in Figure 11. Sig is also **UF-CMA** secure; more precisely, for any adversary $\mathcal{A}$ there exists an adversary $\mathcal{B}$ such that

$$\text{Adv}_{\text{Sig}, \mathcal{A}}^{(Q_s, Q_{\text{RD}})\text{-UF-CMA}} \leq \text{Adv}_{\text{Sig}', \mathcal{B}}^{(Q_s, Q_{\text{RD}})\text{-UF-CMA}}.$$

There is a simple adversary $\mathcal{A}$ against **RMV** of Sig :

<u>$\text{Ver}(pk, m, \sigma)$</u>
01 if $pk = \text{derivePK}(\sigma)$
02 return 1
03 return $\text{Ver}'(pk, m, \sigma)$

Figure 11. Algorithm Ver for construction Sig .

$$sk \xleftarrow{\$} \mathcal{A}^{\text{R0}}(sk, pk)$$

which has winning probability 1. ■

B.2 Proof of Theorem 1

Theorem 1 (NR). For any adversaries $\mathcal{A}$ and $\mathcal{D}$ against the **NR** security of $\text{PS}[\text{Sig}, \lambda]$ (Figure 4), there exist **HnS** adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ and an **RMV** adversary $\mathcal{C}$ against Sig with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}}$ and $t_{\mathcal{D}} = t_{\bar{\mathcal{D}}}$ such that

$$\text{Adv}_{\text{PS}[\text{Sig}, \lambda], \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}} + \text{Adv}_{\text{Sig}, \mathcal{C}}^{\{0, 1\}^\lambda\text{-RMV}}$$

and

$$\mathcal{H}_{\infty}^{(m \mid \text{RO}, \text{sk}, \text{aux}(\text{sk}, m))}_{(\text{sk}, \text{pk}) \leftarrow^{\mathbb{S}} \text{Gen}, m \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{RO}}(\text{sk})} \leq \mathcal{H}_{\infty}^{(x \mid \text{RO}, z)}_{(x, z) \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{RO}}}$$

Proof. In Figure 12, we present a sequence of games.

Game $\mathbf{G}_0$. We start with the **NR** game for **PS**:

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] = \text{Adv}_{\text{PS}[\text{Sig}, \lambda], \mathcal{A}, \mathcal{D}}^{\text{NR}}.$$

Games $\mathbf{G}_0 - \mathbf{G}_2$	Oracle $\text{RO}(x)$
01 $(\text{H}, \text{H}_{\text{Sig}}) \leftarrow^{\mathbb{S}} \mathcal{OS}$	12 return $\text{H}(x)$
02 $(\text{sk}, \text{pk}) \leftarrow^{\mathbb{S}} \text{Gen}[\text{H}_{\text{Sig}}]$	Oracle $\text{RO}'(x)$
03 $m^* \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{RO}(\cdot)}(\text{sk})$	13 $x \rightarrow \dots \ m$
04 $m' \leftarrow \text{H}(\text{pk} \ m^*)$	14 if $m = m^*$ $\quad \parallel \mathbf{G}_1 - \mathbf{G}_2$
05 $\sigma \leftarrow^{\mathbb{S}} \text{Sgn}[\text{H}_{\text{Sig}}](\text{sk}, m')$	15 abort $\quad \parallel \mathbf{G}_1 - \mathbf{G}_2$
06 $(\text{pk}^*, \sigma^*) \leftarrow^{\mathbb{S}} \mathcal{A}^{\text{RO}'(\cdot)}(\text{sk}, \sigma, \text{aux}(\text{sk}, m^*))$	16 return $\text{H}(x)$
07 if $\text{pk} = \text{pk}^*$	
08 return 0	
09 $m' \leftarrow \text{H}(\text{pk}^* \ m^*)$	
10 $m' \leftarrow^{\mathbb{S}} \{0, 1\}^{\lambda}$ $\quad \parallel \mathbf{G}_2$	
11 return $\text{Ver}[\text{H}](\text{pk}^*, m', \sigma^*)$	

Figure 12. Games $\mathbf{G}_0 - \mathbf{G}_2$ for the proof of Theorem 1.

Game $\mathbf{G}_1$. This is the same game as the previous one except that it aborts in the random oracle if $\mathcal{A}$ queries the random oracle on the challenge message m^* .

Claim 1: There exists an adversary $\mathcal{B}$ against **HnS** such that

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] - \Pr[\mathbf{G}_1^{\mathbf{A}} \Rightarrow 1] \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-HnS}}.$$

Proof. We prove the claim by a sequence of hybrids over the random oracle queries to RO' . The original game $\mathbf{G}_0$ does not abort in the random oracle and the i -th hybrid aborts if there is a random oracle query on m^* within the first i queries to RO' . The i -th reduction is denoted by $\mathcal{B}_i$ and formally constructed in Figure 13. The reduction is an adversary against **HnS** and returns a solution in the i -th query to RO' . We further need to define an appropriate adversary $\mathcal{D}$ which is also given in Figure 13. Note that the min-entropy of $\mathcal{D}$ equals the min-entropy of $\mathcal{D}$:

$$\begin{aligned} \mathcal{H}_{\infty}^{(x \mid \text{RO}, z)}_{(x, z) \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{RO}}} &= \mathcal{H}_{\infty}^{((\text{pk}, m) \mid \text{RO}, \text{sk}, \text{aux}(\text{sk}, m))}_{(\text{sk}, \text{pk}) \leftarrow^{\mathbb{S}} \text{Gen}, m \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{RO}}(\text{sk})} \\ &\geq \mathcal{H}_{\infty}^{(m \mid \text{RO}, \text{sk}, \text{aux}(\text{sk}, m))}_{(\text{sk}, \text{pk}) \leftarrow^{\mathbb{S}} \text{Gen}, m \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{RO}}(\text{sk})}. \end{aligned}$$

The last inequality holds since the min-entropy can only decrease when considering less random variables.¹⁶ ■

¹⁶ In the case of the public key being uniquely determined by the secret key, we could obtain an inequality. This does not gain any benefits over a lower bound though (see Remark 1).

Adversary $\mathcal{B}_i^{\bar{R}0}(y, z)$ 01 $\text{cnt} \leftarrow 0$ 02 $z \rightarrow (\text{sk}, a)$ 03 $\text{pk} \leftarrow \text{derivePK}(\text{sk})$ 04 $\sigma \xleftarrow{\$} \text{Sgn}(\text{sk}_{\text{Sig}}, y)$ 05 $(\text{pk}^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\bar{R}0'(\cdot)}(\text{sk}, \sigma, a)$ 06 return $\perp$ Adversary $\bar{\mathcal{D}}^{\bar{R}0}$ 07 $(\text{sk}, \text{pk}) \xleftarrow{\$} \text{Gen}$ 08 $m \xleftarrow{\$} \mathcal{D}^{\bar{R}0(\cdot)}(\text{sk})$ 09 $x \leftarrow (\text{pk} \ m)$ 10 $z \leftarrow (\text{sk}, \text{aux}(\text{sk}, m))$ 11 return (x, z)	Oracle $\bar{R}0(x)$ 12 return $\bar{R}0(x)$ Oracle $\bar{R}0'(x)$ 13 $\text{cnt} \leftarrow \text{cnt} + 1$ 14 $x \rightarrow \dots \ m$ 15 if $\text{cnt} = i$ 16 output m // terminate 17 return $\bar{R}0(x)$
--	---

Figure 13. Adversaries $\mathcal{B}_i$ and $\bar{\mathcal{D}}$ against **HnS** simulating the i -th hybrid between $\mathcal{G}_0/\mathcal{G}_1$ for adversaries $\mathcal{A}$ and $\mathcal{D}$.

Game $\mathcal{G}_2$. This is the same game as the previous one except that it replaces the output of H in the verification of $\mathcal{A}$'s signature by a uniformly random value from H 's output space (Line 10).

Claim 2: It holds that

$$\Pr[\mathcal{G}_1^{\mathcal{A}} \Rightarrow 1] = \Pr[\mathcal{G}_2^{\mathcal{A}} \Rightarrow 1].$$

Proof. Due to the changes in the previous game, $\mathcal{A}$ never queries random oracle $\bar{R}0'$ with the correct m^* . In contrast, $\mathcal{D}$ could have queried their random oracle $\bar{R}0$ on the correct values, i.e. the public key pk^* and the message m^* . However, the information $\mathcal{A}$ receives is independent of the output of such a query because $\mathcal{A}$ obtains sk which is independently generated and not chosen by $\mathcal{D}$, the signature σ which does not involve any additional information from $\mathcal{D}$ except for the message, and the auxiliary information which can only include information about sk and the message m^* itself. Note that the signature that $\mathcal{A}$ receives is based on a public key which must be different from the public key $\mathcal{A}$ outputs which means that the signature cannot contain information of the random oracle query on $\text{pk}^* \| m^*$. Since the query output is independent from $\mathcal{A}$'s view, reprogramming the random oracle is indistinguishable. ■

Final reduction. The final game can be reduced to random-message validity.

Claim 3: There exists an adversary $\mathcal{C}$ against **RMV** of **Sig** such that

$$\Pr[\mathcal{G}_2^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}, \mathcal{C}}^{\{0,1\}^\lambda\text{-RMV}}.$$

Proof. Reduction $\mathcal{C}$ can simulate $\mathcal{G}_2$ for adversary $\mathcal{A}$ as is. When $\mathcal{A}$ outputs a public key and a signature, $\mathcal{C}$ can forward it to their own game. Since the message m' is uniform due to the changes in the previous game, $\mathcal{C}$ wins their **RMV** game if $\mathcal{A}$'s signature verifies. The reduction is formally depicted in Figure 14. ■

The time of adversary $\mathcal{B}$ and $\mathcal{C}$ is approximately the running time of $\mathcal{A}$ and the one of $\bar{\mathcal{D}}$ is approximately the running time of $\mathcal{D}$ concluding the proof. ■

<u>Adversary $\mathcal{C}^{\bar{R}\bar{O}}$</u>	<u>Oracle $\bar{R}\bar{O}(x)$</u>
01 $(\cdot, H_{\text{Sig}}) \xleftarrow{\$} \mathcal{OS}$	10 return $\bar{R}\bar{O}(x)$
02 $(\text{sk}, \text{pk}) \xleftarrow{\$} \text{Gen}[H_{\text{Sig}}]$	<u>Oracle $\bar{R}\bar{O}'(x)$</u>
03 $m^* \xleftarrow{\$} \mathcal{D}^{\bar{R}\bar{O}(\cdot)}(\text{sk})$	11 $x \rightarrow \dots \ m$
04 $m' \leftarrow H(\text{pk} \ m^*)$	12 if $m = m^*$
05 $\sigma \xleftarrow{\$} \text{Sgn}[H_{\text{Sig}}](\text{sk}, m')$	13 abort
06 $(\text{pk}^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\bar{R}\bar{O}'(\cdot)}(\text{sk}, \sigma, \text{aux}(\text{sk}, m^*))$	14 return $\bar{R}\bar{O}(x)$
07 if $\text{pk} = \text{pk}^*$	
08 return 0	
09 return $(\text{pk}^*, \sigma^*, \varepsilon)$	

Figure 14. Adversary $\mathcal{C}$ against **RMV** simulating $\mathcal{G}_2$ for $\mathcal{A}$ and $\mathcal{D}$.

C Proofs of Section 4

C.1 Proof of Theorem 2

Theorem 2 (SUF-CMA). If Sig_1 is unique, then for any adversary $\mathcal{A}$ against the **SUF-CMA** security of $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$ (Figure 5), there exist a **CR** adversary $\mathcal{B}$ against $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$, a **UF-CMA** adversary $\mathcal{C}$ against Sig_1 and a **SUF-CMA** adversary $\mathcal{D}$ against Sig_2 with $t_{\mathcal{A}} \approx t_{\mathcal{B}} \approx t_{\mathcal{C}} \approx t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} \leq \min \left\{ \text{Adv}_{\text{Sig}_1, \mathcal{C}}^{(Q_s, Q_{\text{RO}})\text{-UF-CMA}}, \text{Adv}_{\text{Sig}_2, \mathcal{D}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} \right\} + \text{Adv}_{\mathcal{OS}', \mathcal{B}}^{\text{CR}}.$$

Proof. We proceed with a sequence of games depicted in Figure 15.

Game $\mathbf{G}_0$. This is the (Q_s, Q_{RO}) -**SUF-CMA** game for $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$, hence it holds that

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] = \text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}.$$

As noted in the description of the construction, we just forward random oracle queries to random oracle queries of the underlying schemes and hence do not write down the random oracle explicitly.

Games $\mathbf{G}_0 - \mathbf{G}_2$	Oracle $\text{Sgn}(m)$
01 $(H, \cdot, \cdot) \xleftarrow{\$} \mathcal{OS}$	16 $m' \leftarrow \text{RO}(\text{pk}_1 \ \text{pk}_2 \ m)$
02 $\mathcal{Q} \leftarrow \emptyset$	17 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(\text{sk}_2, m')$
03 $(\text{sk}_1, \text{pk}_1) \xleftarrow{\$} \text{Gen}_1$	18 $\sigma_1 \xleftarrow{\$} \text{Sgn}_1(\text{sk}_1, m' \ \sigma_2)$
04 $(\text{sk}_2, \text{pk}_2) \xleftarrow{\$} \text{Gen}_2$	19 $\sigma \leftarrow (\sigma_1, \sigma_2)$
05 $\text{pk} \leftarrow (\text{pk}_1, \text{pk}_2)$	20 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
06 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}(\cdot)}(\text{pk})$	21 return σ
07 if $(m^*, \sigma^*) \in \mathcal{Q}$	Oracle $\text{RO}(x)$
08 return 0	22 if $\exists x' \in \mathcal{H} : x' \neq x \wedge H(x') = H(x)$ $\parallel \mathbf{G}_1 - \mathbf{G}_2$
09 $\sigma^* \rightarrow (\sigma_1^*, \sigma_2^*)$	23 abort $\parallel \mathbf{G}_1 - \mathbf{G}_2$
10 $m' \leftarrow \text{RO}(\text{pk}_1 \ \text{pk}_2 \ m^*)$	24 $\mathcal{H} \leftarrow \mathcal{H} \cup \{x\}$ $\parallel \mathbf{G}_1 - \mathbf{G}_2$
11 if $(m^*, (\cdot, \sigma_2^*)) \in \mathcal{Q}$ $\parallel \mathbf{G}_2$	25 return $H(x)$
12 return 0 $\parallel \mathbf{G}_2$	
13 if $\text{Ver}_1(\text{pk}_1, m' \ \sigma_2^*, \sigma_1) \wedge \text{Ver}_2(\text{pk}_2, m', \sigma_2^*)$	
14 return 1	
15 return 0	

Figure 15. Games $\mathbf{G}_0 - \mathbf{G}_2$ for the proof of Theorem 2.

Game $\mathbf{G}_1$. This is the same as the previous game except that it aborts if there is a collision in $\mathcal{H}$.

Claim 4: There exists an adversary $\mathcal{B}$ against **CR** such that

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] - \Pr[\mathbf{G}_1^{\mathbf{A}} \Rightarrow 1] \leq \text{Adv}_{\mathcal{OS}', \mathcal{B}}^{\text{CR}},$$

with $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$.

Proof. Adversary $\mathcal{B}$ can simulate the complete game using their own random oracle. As soon as a collision occurs (and the new abort would trigger) they can abort and win their game. ■

Game G_2 . This is the same as the previous game except that it returns 0 if for the challenge message m^* and the second part of the forgery, σ_2^* , there was a previous signing query which contains both the values.

Claim 5: It holds that

$$\Pr[G_1^A \Rightarrow 1] = \Pr[G_2^A \Rightarrow 1].$$

Proof. Assume there exists a signing query with input m^* and output (σ_1', σ_2^*) . If $\sigma_1' = \sigma_1^*$, the game already returns in Line 08 and the winning probability after the change is the same. If $\sigma_1' \neq \sigma_1^*$, the verification of σ_1^* must fail because Sig_1 is unique and σ_1' is a valid signature for $H(\text{pk}_1 \parallel \text{pk}_2 \parallel m^*)$ and there are no collisions in R0 which means that the message must be the same. ■

Reduction to UF-CMA of Sig_1 . We can reduce G_2 to the unforgeability of Sig_1 .

Claim 6: There exists an adversary $\mathcal{C}$ against **UF-CMA** such that

$$\Pr[G_2^A \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_1, \mathcal{C}}^{(Q_s, Q_{\text{R0}})\text{-UF-CMA}}.$$

Proof. Reduction $\mathcal{C}$ is formalized in Figure 16. If $\mathcal{A}$ wins the game, $\mathcal{C}$'s winning conditions are also fulfilled. The validity of the signature is checked by $\mathcal{C}$ before output and the message $m' \parallel \sigma_2^*$ was never queried to oracle $\text{Sgn}_{\mathcal{C}}$ due to the check in Line 10 and the absence of collisions in R0 . ■

<u>Adversary $\mathcal{C}^{\text{Sgn}_{\mathcal{C}}}(\text{pk}_1)$</u>	<u>Oracle $\text{Sgn}(m)$</u>
01 $(H, \cdot, \cdot) \xleftarrow{\$} \mathcal{OS}$	15 $m' \leftarrow \text{R0}(\text{pk}_1 \parallel \text{pk}_2 \parallel m)$
02 $\mathcal{Q} \leftarrow \emptyset$	16 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(\text{sk}_2, m')$
03 $(\text{sk}_2, \text{pk}_2) \xleftarrow{\$} \text{Gen}_2$	17 $\sigma_1 \xleftarrow{\$} \text{Sgn}_{\mathcal{C}}(m' \parallel \sigma_2)$ // Sgn oracle
04 $\text{pk} \leftarrow (\text{pk}_1, \text{pk}_2)$	18 $\sigma \leftarrow (\sigma_1, \sigma_2)$
05 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{R0}(\cdot)}(\text{pk})$	19 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
06 if $(m^*, \sigma^*) \in \mathcal{Q}$	20 return σ
07 return 0	
08 $\sigma^* \rightarrow (\sigma_1^*, \sigma_2^*)$	<u>Oracle $\text{R0}(x)$</u>
09 $m' \leftarrow \text{R0}(\text{pk}_1 \parallel \text{pk}_2 \parallel m^*)$	21 if $\exists x' \in \mathcal{H} : x' \neq x \wedge H(x') = H(x)$
10 if $(m^*, (\cdot, \sigma_2^*)) \in \mathcal{Q}$	22 abort
11 return 0	23 $\mathcal{H} \leftarrow \mathcal{H} \cup \{x\}$
12 if $\text{Ver}_1(\text{pk}_1, m' \parallel \sigma_2^*, \sigma_1^*) \wedge \text{Ver}_2(\text{pk}, m', \sigma_2^*)$	24 return $H(x)$
13 return $(m' \parallel \sigma_2^*, \sigma_1^*)$ // win	
14 return 0	

Figure 16. Adversary $\mathcal{C}$ against **UF-CMA** of Sig_1 having access to oracle $\text{Sgn}_{\mathcal{C}}$ simulating G_2 for adversary $\mathcal{A}$.

*Reduction to **SUF-CMA** of Sig_2 .* We can reduce G_2 to the strong unforgeability of Sig_2 .

Claim 7: There exists an adversary $\mathcal{D}$ against **SUF-CMA** such that

$$\Pr[G_2^A \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_2, \mathcal{D}}^{(Q_s, Q_{\text{R0}})\text{-SUF-CMA}}.$$

Proof. Reduction $\mathcal{D}$ is formalized in Figure 17. If $\mathcal{A}$ wins the game, $\mathcal{D}$'s winning conditions are also fulfilled. The validity of the signature is checked by $\mathcal{D}$ before output and the tuple (m', σ_2^*) does not correspond to any previous query to $\text{Sgn}_{\mathcal{D}}$ due to the check in Line 10 and the absence of collision in R0 . ■

The running times of $\mathcal{B}, \mathcal{C}$, and $\mathcal{D}$ are approximately the same as for $\mathcal{A}$ which concludes the proof. ■

Adversary $\mathcal{D}^{\text{Sgn}\mathcal{D}}(\text{pk}_2)$	Oracle $\text{Sgn}(m)$
01 $(H, \cdot, \cdot) \xleftarrow{\$} \mathcal{OS}$	15 $m' \leftarrow \text{RO}(\text{pk}_1 \ \text{pk}_2 \ m)$
02 $\mathcal{Q} \leftarrow \emptyset$	16 $\sigma_2 \xleftarrow{\$} \text{Sgn}_{\mathcal{D}}(m')$ // Sgn oracle
03 $(\text{sk}_1, \text{pk}_1) \xleftarrow{\$} \text{Gen}_1$	17 $\sigma_1 \xleftarrow{\$} \text{Sgn}_1(\text{sk}_1, m' \ \sigma_2)$
04 $\text{pk} \leftarrow (\text{pk}_1, \text{pk}_2)$	18 $\sigma \leftarrow (\sigma_1, \sigma_2)$
05 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}(\cdot)}(\text{pk})$	19 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
06 if $(m^*, \sigma^*) \in \mathcal{Q}$	20 return σ
07 return 0	
08 $\sigma^* \rightarrow (\sigma_1^*, \sigma_2^*)$	Oracle $\text{RO}(x)$
09 $m' \leftarrow \text{RO}(\text{pk}_1 \ \text{pk}_2 \ m^*)$	21 if $\exists x' \in \mathcal{H} : x' \neq x \wedge H(x') = H(x)$
10 if $(m^*, (\cdot, \sigma_2^*)) \in \mathcal{Q}$	22 abort
11 return 0	23 $\mathcal{H} \leftarrow \mathcal{H} \cup \{x\}$
12 if $\text{Ver}_1(\text{pk}_1, m' \ \sigma_2^*, \sigma_1^*) \wedge \text{Ver}_2(\text{pk}, m', \sigma_2^*)$	24 return $H(x)$
13 return (m', σ_2^*) // win	
14 return 0	

Figure 17. Adversary $\mathcal{D}$ against **SUF-CMA** of Sig_2 having access to oracle $\text{Sgn}_{\mathcal{D}}$ simulating $\mathbf{G}_2$ for adversary $\mathcal{A}$.

C.2 Proof of Theorem 3

Theorem 3 (EO). For any adversary $\mathcal{A}$ against the **EO** security of $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$ (Figure 5), there exist an **EO** adversary $\mathcal{B}_1$ against Sig_1 , an **MBS** adversary $\mathcal{C}_1$ against Sig_1 , an **EO** adversary $\mathcal{B}_2$ against Sig_2 , an **MBS** adversary $\mathcal{C}_2$ against Sig_2 , and an **CR** adversary $\mathcal{D}$ against $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}_1} = t_{\mathcal{C}_1} = t_{\mathcal{B}_2} = t_{\mathcal{C}_2} = t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{EO}} \leq \min \left\{ \text{Adv}_{\text{Sig}_1, \mathcal{B}_1}^{\text{EO}}, \text{Adv}_{\text{Sig}_1, \mathcal{C}_1}^{\text{MBS}}, \text{Adv}_{\text{Sig}_2, \mathcal{B}_2}^{\text{EO}}, \text{Adv}_{\text{Sig}_2, \mathcal{C}_2}^{\text{MBS}} \right\} + \text{Adv}_{\mathcal{OS}', \mathcal{D}}^{\text{CR}}.$$

Proof. In Figure 18, we present a sequence of games.

Game $\mathbf{G}_0$. We start with the **EO** game for $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$:

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] = \text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{EO}}.$$

Game $\mathbf{G}_1$. This is the same game as the previous one except that it aborts if all signatures are valid and the signature keys pk_1 and $\hat{\text{pk}}_1$ are different.

Claim 8: There exists an adversary $\mathcal{B}$ against **EO** such that

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] - \Pr[\mathbf{G}_1^{\mathbf{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_1, \mathcal{B}}^{\text{EO}}.$$

Proof. The reduction is analogous to the proof of Theorem 7. ■

Game $\mathbf{G}_2$. This is the same game as the previous one except that it aborts if all signatures are valid and the messages m'_1 and m'_2 are different.

Claim 9: There exists an adversary $\mathcal{C}$ against **MBS** such that

$$\Pr[\mathbf{G}_1^{\mathbf{A}} \Rightarrow 1] - \Pr[\mathbf{G}_2^{\mathbf{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_1, \mathcal{C}}^{\text{MBS}}.$$

Proof. The reduction is analogous to the proof of Theorem 7. ■

Games $G_0 - G_2$	
01	$(H, \cdot, \cdot) \xleftarrow{\$} \mathcal{OS}$
02	$(pk, \hat{pk}, m_1, m_2, (\sigma_1, \sigma_2)) \xleftarrow{\$} \mathcal{A}^{RO(\cdot)}$
03	$pk \rightarrow (pk_1, pk_2)$
04	$\hat{pk} \rightarrow (\hat{pk}_1, \hat{pk}_2)$
05	$m'_1 \leftarrow RO(pk_1 \ pk_2 \ m_1)$
06	$m'_2 \leftarrow RO(\hat{pk}_1 \ \hat{pk}_2 \ m_2)$
07	if $Ver_1(pk_1, m'_1 \ \sigma_2, \sigma_1) \wedge Ver_1(\hat{pk}_1, m'_2 \ \sigma_2, \sigma_2)$
08	if $Ver_2(pk_2, m'_1, \sigma_2) \wedge Ver_2(\hat{pk}_2, m'_2, \sigma_2)$
09	if $pk_1 \neq \hat{pk}_1$ // $G_1 - G_2$
10	abort // $G_1 - G_2$
11	if $m'_1 \neq m'_2$ // G_2
12	abort // G_2
13	return $pk \neq \hat{pk}$
14	return 0

Figure 18. Games $G_0 - G_2$ for the proof of Theorem 7.

Final reduction. Claim 10: There exists an adversary $\mathcal{D}$ against **CR** such that

$$\Pr[G_2^A \Rightarrow 1] \leq \text{Adv}_{\mathcal{OS}', \mathcal{D}}^{\text{CR}},$$

with $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$

Proof. If adversary $\mathcal{A}$ it must hold $m'_1 = m'_2$ because otherwise the game aborts. It must further hold $pk \neq \hat{pk}$ which implies a collision in $\mathcal{OS}'$ and $\mathcal{D}$ wins their game. ■

Note that we can do the exactly same arguments with Sig_2 instead of Sig_1 resulting in the theorem bound. ■

C.3 Proof of Theorem 4

Theorem 4 (MBS). For any adversary $\mathcal{A}$ against the **MBS** security of $\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda]$ (Figure 5), there exist an **MBS** adversary $\mathcal{B}$ against Sig_1 , an **MBS** adversary $\mathcal{C}$ against Sig_2 , and an **CR** adversary $\mathcal{D}$ against $\mathcal{OS}' := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}} = t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-1}[\text{Sig}_1, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{MBS}} \leq \min \left\{ \text{Adv}_{\text{Sig}_1, \mathcal{B}}^{\text{MBS}}, \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\text{MBS}} \right\} + \text{Adv}_{\mathcal{OS}', \mathcal{D}}^{\text{CR}}.$$

Proof. Since the message m which is signed by the signature combiner is part of the message that is signed by Sig_1 and Sig_2 , the message-bound security can directly reduced to **MBS** of Sig_1 or Sig_2 by simply forwarding the correctly constructed messages. ■

D Proofs of Section 5

D.1 Proof of Theorem 6

Theorem 6 (SUF-CMA). For any adversary $\mathcal{A}$, making at most Q_s signing queries and at most Q_{RO} random oracle queries, against the **SUF-CMA** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$ (Figure 6) in the random oracle model, there exist an **UR** adversary $\mathcal{B}$ against ID , a **SUF-CMA** adversary $\mathcal{C}$ against Sig_2 , an **MBS** adversary $\mathcal{D}$ against Sig_2 , an **RMV** adversary $\mathcal{E}$ against Sig_2 , and a **PIMP-PA** adversary $\mathcal{F}$ against ID with $t_{\mathcal{A}} \approx t_{\mathcal{B}} \approx t_{\mathcal{C}} \approx t_{\mathcal{D}} \approx t_{\mathcal{E}} \approx t_{\mathcal{F}}$ such that

$$\begin{aligned} \text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} &\leq \min \left\{ \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}, \text{Adv}_{\text{ID}, \mathcal{F}}^{(Q_{\text{RO}}+1)\text{-PIMP-PA}} \right. \\ &\quad \left. + \text{Adv}_{\text{Sig}_2, \mathcal{D}}^{\text{MBS}} + Q_{\text{RO}}^2 \cdot \text{Adv}_{\text{Sig}_2, \mathcal{E}}^{\{0,1\}^\lambda\text{-RMV}} \right\} \\ &\quad + \text{Adv}_{\text{ID}, \mathcal{B}}^{\text{UR}} + Q_s(Q_{\text{RO}} + Q_s)\gamma_{\text{Sig}_2} + \frac{Q_{\text{RO}} + Q_s}{|\text{ChlSet}|}, \end{aligned}$$

where ChlSet is the challenge set of ID .

Proof. We proceed with a sequence of games depicted in Figure 19.

Game $\mathcal{G}_0$. This is the unforgeability game for $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$ where the random functions H_1 and H_2 are defined via lazy sampling. We also use an additional list $\mathcal{L}_{DQ}$ to mark (direct) queries to the random oracle RO_2 which does not influence the winning probability. By definition we have

$$\Pr[\mathcal{G}_0^{\mathcal{A}} \Rightarrow 1] = \text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}.$$

Game $\mathcal{G}_1$. This is the same game as the previous one except that it aborts in the signing oracle if σ_2 was already queried to the random oracle RO_2 before. The depiction is simplified by using a different oracle RO' which is only called from the signing oracle.

Claim 11: It holds that

$$\Pr[\mathcal{G}_0^{\mathcal{A}} \Rightarrow 1] - \Pr[\mathcal{G}_1^{\mathcal{A}} \Rightarrow 1] \leq Q_s(Q_{\text{RO}} + Q_s)\gamma_{\text{Sig}_2}.$$

Proof. For each query to **Sgn**, the oracle computes a fresh signature σ_2 . The probability that the signing algorithm outputs a specific σ_2 can be upper bounded by γ_{Sig_2} by definition. Further, the list of random oracle queries $\mathcal{L}_{H_2}$ contains at most $Q_{\text{RO}} + Q_s$ elements and the signing oracle is called at most Q_s times which results in the claimed bound. ■

Game $\mathcal{G}_2$. This is the same game as the previous one except that it aborts if there was a signing query for the challenge message m^* resulting in the same signature σ_2^* as the valid forgery but having a different response.

Claim 12: There exists an adversary $\mathcal{B}$ against **UR** such that

$$\Pr[\mathcal{G}_1^{\mathcal{A}} \Rightarrow 1] - \Pr[\mathcal{G}_2^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{ID}, \mathcal{B}}^{\text{UR}}.$$

Proof. The reduction $\mathcal{B}$ is formalized in Figure 20. The signing oracle can be simulated via the transcript oracle of $\mathcal{B}$. Transcripts $(\text{com}^*, \text{chl}^*, \text{rsp}^*)$ and $(\text{com}^*, \text{chl}^*, \text{rsp}')$ must verify due to the definition of algorithm ExtCom . Further, we require $\text{rsp}^* \neq \text{rsp}'$ which implies that if the new abort condition triggers, adversary $\mathcal{B}$ wins their game. ■

Adversary $\mathcal{B}^{\text{Trans}}(\text{pk}_{\text{ID}})$	Oracle $\text{Sgn}(m)$
01 $\mathcal{Q}, \mathcal{L}_{\text{H}_1}[], \mathcal{L}_{\text{H}_2}[] \leftarrow \emptyset$	17 $(\text{com}, \text{chl}, \text{rsp}) \xleftarrow{\$} \text{Trans} \quad // \text{trans oracle}$
02 $i \leftarrow 0$	18 $m' \leftarrow \text{H}(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m \parallel \text{com})$
03 $(\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}}) \xleftarrow{\$} \text{Gen}_2$	19 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(\text{sk}_{\text{Sig}}, m')$
04 $\text{pk} \leftarrow (\text{pk}_{\text{ID}}, \text{pk}_{\text{Sig}})$	20 if $\mathcal{L}_{\text{H}_2}[\sigma_2] = \perp$
05 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{pk})$	21 $\mathcal{L}_{\text{H}_2}[\sigma_2] \leftarrow \text{chl} \quad // \text{program RO}$
06 if $(m^*, \sigma^*) \in \mathcal{Q}$	22 else
07 return $\perp$	23 abort
08 $\sigma^* \rightarrow (\text{rsp}^*, \sigma_2^*)$	24 $\sigma \leftarrow (\text{rsp}, \sigma_2)$
09 $\text{chl}^* \leftarrow \text{RO}_2(\sigma_2^*)$	25 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
10 $\text{com}^* \leftarrow \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}^*)$	26 return σ
11 $m' \leftarrow \text{RO}_1(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^*)$	<u>Oracle $\text{RO}_1(x)$</u>
12 if $\text{Ver}(\text{pk}_{\text{Sig}}, m', \sigma_2^*)$	27 return $\text{G}_1.\text{RO}_1(x)$
13 if $\exists (m^*, (\text{rsp}', \sigma_2^*)) \in \mathcal{Q} :$	<u>Oracle $\text{RO}_2(x)$</u>
$\text{com}^* = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}') \wedge \text{rsp}^* \neq \text{rsp}'$	28 return $\text{G}_1.\text{RO}_2(x)$
14 return $(\text{com}^*, \text{chl}^*, \text{rsp}^*, \text{rsp}')$ $// \text{win}$	
15 return $\perp$	
16 return $\perp$	

Figure 20. Adversary $\mathcal{B}$ against **UR** of ID having access to oracle **Trans** simulating G_1/G_2 for adversary $\mathcal{A}$.

Assume that it is not fresh, i.e. there was a query $\text{Sgn}_c(m')$ in the signing oracle outputting σ_2^* . In such a query, the commitment must be the same as for the forgery because otherwise the game would abort due to the changes introduced in G_3 . The challenge for such a query must also be the same as the forgery challenge, chl^* , because it is the output of the random oracle with the same input σ_2^* . Hence, the response of such a query must be different since adversary $\mathcal{A}$ passed a triviality check in Line 06. However, in this case the game would have aborted in Line 07 already (note that for a forgery the same m' implies the same m^* due to the abort introduced in G_3) leading to a contradiction. ■

Game G_4 . This is the same game as the previous one except that it aborts if there exists a signing query such that the corresponding signature σ_2 and the challenge are the same as for the forgery but the message or the commitment is different.

Claim 15: There exists an adversary $\mathcal{D}$ against **MBS** such that

$$\Pr[\text{G}_3^{\text{A}} \Rightarrow 1] - \Pr[\text{G}_4^{\text{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_2, \mathcal{D}}^{\text{MBS}}.$$

Proof. Reduction $\mathcal{D}$ is formally constructed in Figure 21. As soon as the newly introduced abort condition is met, adversary $\mathcal{D}$ has two different messages such that σ_2^* is valid for these messages. Since $\mathcal{D}$ can freely choose the key they output, they can simulate the entire game as done by a normal challenger. ■

Game G_5 . This is the same game as the previous one except that it aborts in random oracle RO_1 (on a fresh input) if there was a previous query for which the input was a signature that verifies for the output that was chosen in the current RO query (see Line 34).

Claim 16: There exists an adversary $\mathcal{E}$ against **RMV** such that

$$\Pr[\text{G}_4^{\text{A}} \Rightarrow 1] - \Pr[\text{G}_5^{\text{A}} \Rightarrow 1] \leq Q_{\text{RO}}^2 \cdot \text{Adv}_{\text{Sig}_2, \mathcal{E}}^{\{0,1\}^{\wedge} \text{-RMV}}.$$

Proof. We proceed with a sequence of hybrids iterating over the random oracle queries to RO_1 , denoted as i_1 , and the random oracle queries to RO_2 , denoted as i_2 . By game $\text{G}_{4.(i_1, i_2)}$, we denote G_4 where the game

Adversary $\mathcal{D}$	Oracle $\text{Sgn}(m)$
01 $\mathcal{Q}, \mathcal{L}_{H_1}[], \mathcal{L}_{H_2}[] \leftarrow \emptyset$	23 return $G_3.\text{Sgn}(m)$
02 $(\text{sk}_{\text{ID}}, \text{pk}_{\text{ID}}) \xleftarrow{\$} \text{Gen}_1$	<u>Oracle $\text{RO}_1(x)$</u>
03 $(\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}}) \xleftarrow{\$} \text{Gen}_2$	24 return $G_3.\text{RO}_1(x)$
04 $\text{pk} \leftarrow (\text{pk}_{\text{ID}}, \text{pk}_{\text{Sig}})$	<u>Oracle $\text{RO}_2(x)$</u>
05 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{pk})$	25 return $G_3.\text{RO}_2(x)$
06 if $(m^*, \sigma^*) \in \mathcal{Q}$	<u>Oracle $\text{RO}'(\sigma_2)$</u>
07 return 0	26 return $G_3.\text{RO}'(\sigma_2)$
08 $\sigma^* \rightarrow (\text{rsp}^*, \sigma_2^*)$	
09 $\text{chl}^* \leftarrow \text{RO}_2(\sigma_2^*)$	
10 $\text{com}^* \leftarrow \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}^*)$	
11 $m' \leftarrow \text{RO}_1(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^*)$	
12 if $\exists x \neq \text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^* : \mathcal{L}_{H_1}[x] = m'$	
13 abort	
14 if $\text{Ver}(\text{pk}_{\text{Sig}}, m', \sigma_2^*)$	
15 if $\exists (m^*, (\text{rsp}', \sigma_2^*)) \in \mathcal{Q} :$	
$\text{com}^* = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}') \wedge \text{rsp}^* \neq \text{rsp}'$	
16 abort	
17 if $\exists (m, (\text{rsp}, \sigma_2^*)) \in \mathcal{Q} :$	
$\text{com} = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp})$	
$\wedge (m, \text{com}) \neq (m^*, \text{com}^*)$	
18 $m_1 \leftarrow \mathcal{L}_{H_1}[\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m \parallel \text{com}]$	
19 $m_2 \leftarrow m'$	
20 return $(\text{pk}_{\text{Sig}}, m_1, m_2, \sigma_2^*)$ // win	
21 return 1	
22 return 0	

Figure 21. Adversary $\mathcal{D}$ against MBS security of Sig_2 simulating G_3/G_4 for adversary $\mathcal{A}$.

additionally aborts in random oracle RO_1 if the condition introduced in G_5 is met, the current query number to RO_1 is less than i_1 , and the query number in which σ_2 was queried to RO_2/RO' is less than i_2 . By definition, we have $G_{4.(1,1)} = G_4$ and $G_{4.(Q_{\text{RO}}+1, Q_{\text{RO}}+1)} = G_5$. In Figure 23, we construct adversary $\mathcal{E}_{i_1^*, i_2^*}$ such that

$$\Pr[G_{4.(i_1^*, i_2^*-1)}^A \Rightarrow 1] - \Pr[G_{4.(i_1^*, i_2^*)}^A \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_2, \mathcal{E}_{i_1^*, i_2^*}}^{\{0,1\}^\lambda\text{-RMV}}$$

and

$$\Pr[G_{4.(i_1^*-1, i_2^*)}^A \Rightarrow 1] - \Pr[G_{4.(i_1^*, i_2^*)}^A \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_2, \mathcal{E}_{i_1^*, i_2^*}}^{\{0,1\}^\lambda\text{-RMV}}.$$

Aggregating over $i_1^*, i_2^* \in [Q_{\text{RO}}]$, we obtain the claimed bound. $\blacksquare$

Game G_6 . This is the same game as the previous one except that it aborts if the random oracle query corresponding to the forgery was not a direct query, i.e. $\sigma_2^* \notin \mathcal{L}_{DQ}$. Since we distinguish the oracles, $\mathcal{L}_{DQ}$ only contains direct queries to random oracle RO_2 and no implicit queries via the signing oracle.

Claim 17: It holds that

$$\Pr[G_5^A \Rightarrow 1] = \Pr[G_6^A \Rightarrow 1].$$

Proof. We want to show that the abort introduced in this game never occurs and by $\Pr[\text{abort}]$ we denote the probability that it occurs. By definition, we have

$$\Pr[\text{abort}] = \Pr[\sigma_2^* \notin \mathcal{L}_{DQ}],$$

Adversary $\mathcal{F}^{\text{Trans}, \text{Chl}}(\text{pk}_{\text{ID}})$	Oracle $\text{Sgn}(m)$
01 $\mathcal{Q}, \mathcal{L}_{H_1}[], \mathcal{L}_{H_2}[], \mathcal{L}_{DQ}[] \leftarrow \emptyset$	22 $(\text{com}, \text{chl}, \text{rsp}) \xleftarrow{\$} \text{Trans}$ // trans oracle
02 $(\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}}) \xleftarrow{\$} \text{Gen}_2$	23 $m' \leftarrow \text{RO}_1(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m \parallel \text{com})$
03 $\text{pk} \leftarrow (\text{pk}_{\text{ID}}, \text{pk}_{\text{Sig}})$	24 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(\text{sk}_{\text{Sig}}, m')$
04 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{pk})$	25 if $\mathcal{L}_{H_2}[\sigma_2] = \perp$
05 if $(m^*, \sigma^*) \in \mathcal{Q}$	26 $\mathcal{L}_{H_2}[\sigma_2] \leftarrow \text{chl}$ // program RO
06 return $\perp$	27 else
07 $\sigma^* \rightarrow (\text{rsp}^*, \sigma_2^*)$	28 abort
08 $\text{chl}^* \leftarrow \text{RO}_2(\sigma_2^*)$	29 $\sigma \leftarrow (\text{rsp}, \sigma_2)$
09 $\text{com}^* \leftarrow \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}^*)$	30 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
10 $m' \leftarrow \text{RO}_1(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^*)$	31 return σ
11 if $\exists x \neq \text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^* : \mathcal{L}_{H_1}[x] = m'$	Oracle $\text{RO}_1(x)$
12 abort	32 return $\text{G}_6.\text{RO}_1(x)$
13 if $\text{Ver}(\text{pk}_{\text{Sig}}, m', \sigma_2^*)$	Oracle $\text{RO}_2(x)$
14 if $\exists (m^*, (\text{rsp}', \sigma_2^*)) \in \mathcal{Q} :$	33 if $\mathcal{L}_{H_2}[x] = \perp$
$\text{com}^* = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}') \wedge \text{rsp}^* \neq \text{rsp}'$	34 $h \xleftarrow{\$} \text{ChlSet}$
15 abort	35 if $\exists x' \in \mathcal{L}_{H_1} : \text{Ver}(\text{pk}_{\text{Sig}}, \mathcal{L}_{H_1}[x'], x)$
16 if $\exists (m, (\text{rsp}, \sigma_2^*)) \in \mathcal{Q} :$	36 $x' \rightarrow \dots \parallel \text{com}$
$\text{com} = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp})$	37 $h \xleftarrow{\$} \text{Chl}(\text{com})$ // embed challenge
$\wedge (m, \text{com}) \neq (m^*, \text{com}^*)$	38 $\mathcal{L}_{DQ}[x] \leftarrow 1$
17 abort	39 $\mathcal{L}_{H_2}[x] \leftarrow h$
18 if $\sigma_2^* \notin \mathcal{L}_{DQ}$	40 return $\mathcal{L}_{H_2}[x]$
19 abort	
20 return $(\text{com}^*, \text{chl}^*, \text{rsp}^*)$ // win	
21 return $\perp$	

Figure 22. Adversary $\mathcal{F}$ against **PIMP-PA** security of ID having access to oracles **Trans** and **Chl** simulating G_6 for adversary $\mathcal{A}$.

For the winning probability of the two games to be equal it is sufficient to show:

$\Pr[\sigma_2^* \notin \mathcal{L}_{DQ}] = 0$: Let us assume $\Pr[\sigma_2^* \notin \mathcal{L}_{DQ}] \neq 0$, i.e. the RO query for the forgery challenge chl^* was issued in the signing oracle. Then, this particular query to the signing oracle output a signature where σ_2 , and chl must be the same as for the forgery (σ_2 is an input and chl is an output of said RO query). The message and the commitment must also be the same; otherwise the game would abort in Line 18 due to the changes introduced in G_4 . Finally, since the game of adversary $\mathcal{A}$ did not abort due to a trivial forgery (Line 06), rsp of the signing query must be different from the forgery one. However, this would lead to an abort in Line 16 such that we can conclude that the new abort is never reached. ■

Final Reduction. For basing the security on the ID scheme, we can reduce the last game to its **PIMP-PA** security.

Claim 18: There exists an adversary $\mathcal{F}$ against **PIMP-PA** such that

$$\Pr[\text{G}_6^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{ID}, \mathcal{F}}^{(\text{QH}+1)\text{-PIMP-PA}}.$$

Proof. The reduction $\mathcal{F}$ is formalized in Figure 22. The signing oracle can be simulated using the transcript oracle and programming the random oracle on the output challenge. Further, the reduction embeds an output of their challenge oracle in each of the random oracle queries to RO_2 . Due to the abort introduced in G_6 , adversary $\mathcal{A}$ can only win the game with a forgery corresponding to a direct RO query to RO_2 . The main issue for the reduction is to handle their own challenge oracle. In particular, the commitment of the ID scheme is input into a query to RO_1 . The output of this query is then signed and the signature is input of a query to RO_2 . The output of the RO_2 query needs to be the challenge such the reduction can win their **PIMP-PA**

Adversary $\mathcal{E}_{i_1^*, i_2^*}$	Oracle $\text{Sgn}(m)$
01 $i_1, i_2 \leftarrow 0$	22 return $\text{G}_4.\text{Sgn}(m)$
02 $\mathcal{Q}, \mathcal{L}_{H_1}[], \mathcal{L}_{H_2}[], \mathcal{L}_2[] \leftarrow \emptyset$	Oracle $\text{R0}_1(x)$
03 $(\text{sk}_{\text{ID}}, \text{pk}_{\text{ID}}) \xleftarrow{\$} \text{Gen}_1$	23 if $\mathcal{L}_{H_1}[x] = \perp$
04 $(\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}}) \xleftarrow{\$} \text{Gen}_2$	24 $i_1 \leftarrow i_1 + 1$
05 $\text{pk} \leftarrow (\text{pk}_{\text{ID}}, \text{pk}_{\text{Sig}})$	25 $\mathcal{L}_{H_1}[x] \xleftarrow{\$} \{0, 1\}^\lambda$
06 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{R0}_1(\cdot), \text{R0}_2(\cdot)}(\text{pk})$	26 if $i_1 < i_1^* \wedge (\exists \sigma_2 \in \mathcal{L}_{H_2}, i_2 :$
07 if $(m^*, \sigma^*) \in \mathcal{Q}$	$\text{Ver}(\text{pk}_{\text{Sig}}, \mathcal{L}_{H_1}[x], \sigma_2) \wedge \mathcal{L}[i_2] = \sigma_2 \wedge i_2 < i_2^*)$
08 return 0	27 abort
09 $\sigma^* \rightarrow (\text{rsp}^*, \sigma_2^*)$	28 if $i_1 = i_1^*$
10 $\text{chl}^* \leftarrow \text{R0}_2(\sigma_2^*)$	29 output $(\text{pk}_{\text{Sig}}, \mathcal{L}_2[i_2^*], \epsilon)$ // output game
11 $\text{com}^* \leftarrow \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}^*)$	30 return $\mathcal{L}_{H_1}[x]$
12 $m' \leftarrow \text{R0}_1(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^*)$	Oracle $\text{R0}_2(x)$
13 if $\exists x \neq \text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^* : \mathcal{L}_{H_1}[x] = m'$	31 if $\mathcal{L}_{H_2}[x] = \perp$
14 abort	32 $i_2 \leftarrow i_2 + 1$
15 if $\text{Ver}(\text{pk}_{\text{Sig}}, m', \sigma_2^*)$	33 $\mathcal{L}_2[x] \leftarrow i_2$
16 if $\exists (m^*, (\text{rsp}', \sigma_2^*)) \in \mathcal{Q} :$	34 $\mathcal{L}_{H_2}[x] \xleftarrow{\$} \text{ChlSet}$
$\text{com}^* = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}') \wedge \text{rsp}^* \neq \text{rsp}'$	35 return $\mathcal{L}_{H_2}[x]$
17 abort	Oracle $\text{R0}'(\sigma_2)$
18 if $\exists (m, (\text{rsp}, \sigma_2^*)) \in \mathcal{Q} :$	36 if $\mathcal{L}_{H_2}[\sigma_2] = \perp$
$\text{com} = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp})$	37 $i_2 \leftarrow i_2 + 1$
$\wedge (m, \text{com}) \neq (m^*, \text{com}^*)$	38 $\mathcal{L}_2[i_2] \leftarrow \sigma_2$
19 abort	39 $\mathcal{L}_{H_2}[\sigma_2] \xleftarrow{\$} \text{ChlSet}$
20 return 1	40 else
21 return 0	41 abort
	42 return $\mathcal{L}_{H_2}[\sigma_2]$

Figure 23. Adversary $\mathcal{E}_{i_1^*, i_2^*}$ against **RMV** security of Sig_2 simulating the game of adversary $\mathcal{A}$ used in the proof between G_4 and G_5 .

game. To this end, $\mathcal{F}$ checks for each (direct) query to R0_2 if there was a previous query to R0_1 , such that the current input is a valid signature of the output of the previous query to R0_1 (Line 35). If this is the case, the commitment is extracted from the previous query and input to $\mathcal{F}$'s challenge oracle such that the output challenge can be embedded as the RO output. In this way, we can make sure that every challenge is connected with the correct commitment. Due to the changes in G_5 , the game aborts whenever a freshly chosen R0_1 output is valid for a previously queried signature to R0_2 . Hence, if the game does not abort and the forged signature is valid the random oracles must have been queried in the correct order. This allows the reduction to embed a challenge for every possible forgery meeting their winning condition which requires the commitment/challenge pair to originate from a challenge query. Additionally, the output transcript is valid due to the definition of ExtCom . ■

The running times of the reductions are approximately the same as for $\mathcal{A}$. Collecting the bounds yields the theorem statement. ■

D.2 Proof of Theorem 7

Theorem 7 (EO). For any adversary $\mathcal{A}$ against the **EO** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$ (Figure 6), there exist a **EO** adversary $\mathcal{B}$ against Sig_2 , a **MBS** adversary $\mathcal{C}$ against Sig_2 , and a **CR** adversary $\mathcal{D}$ against

Adversary $\mathcal{C}^{\text{Sgn}_C}(\text{pk}_{\text{Sig}})$	Oracle $\text{Sgn}(m)$
01 $\mathcal{Q}, \mathcal{L}_{H_1}[], \mathcal{L}_{H_2}[] \leftarrow \emptyset$	18 $(\text{com}, \text{st}) \xleftarrow{\$} \text{Com}(\text{sk}_{\text{ID}})$
02 $(\text{sk}_{\text{ID}}, \text{pk}_{\text{ID}}) \xleftarrow{\$} \text{Gen}_1$	19 $m' \leftarrow \text{RO}_1(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m \parallel \text{com})$
03 $\text{pk} \leftarrow (\text{pk}_{\text{ID}}, \text{pk}_{\text{Sig}})$	20 $\sigma_2 \xleftarrow{\$} \text{Sgn}_C(m') \quad // \text{sign oracle}$
04 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{pk})$	21 $\text{chl} \leftarrow \text{RO}'(\sigma_2)$
05 if $(m^*, \sigma^*) \in \mathcal{Q}$	22 $\text{rsp} \xleftarrow{\$} \text{Rsp}(\text{sk}_{\text{ID}}, \text{com}, \text{chl}, \text{st})$
06 return $\perp$	23 $\sigma \leftarrow (\text{rsp}, \sigma_2)$
07 $\sigma^* \rightarrow (\text{rsp}^*, \sigma_2^*)$	24 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
08 $\text{chl}^* \leftarrow \text{RO}_2(\sigma_2^*)$	25 return σ
09 $\text{com}^* \leftarrow \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}^*)$	Oracle $\text{RO}_1(x)$
10 $m' \leftarrow \text{RO}_1(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^*)$	26 return $\text{G}_3.\text{RO}_1(x)$
11 if $\exists x \neq \text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m^* \parallel \text{com}^* : \mathcal{L}_{H_1}[x] = m'$	Oracle $\text{RO}_2(x)$
12 abort	27 return $\text{G}_3.\text{RO}_2(x)$
13 if $\text{Ver}(\text{pk}_{\text{Sig}}, m', \sigma_2^*)$	Oracle $\text{RO}'(\sigma_2)$
14 if $\exists (m^*, (\text{rsp}', \sigma_2^*)) \in \mathcal{Q} :$	28 return $\text{G}_3.\text{RO}'(\sigma_2)$
$\text{com}^* = \text{ExtCom}(\text{pk}_{\text{ID}}, \text{chl}^*, \text{rsp}') \wedge \text{rsp}^* \neq \text{rsp}'$	
15 abort	
16 return (m', σ_2^*) // win	
17 return $\perp$	

Figure 24. Adversary $\mathcal{C}$ against **SUF-CMA** of Sig_2 having access to oracle Sgn_C simulating G_3 for adversary $\mathcal{A}$.

$\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}} = t_{\mathcal{D}}$ such that

$$\text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{EO}} \leq \text{Adv}_{\text{Sig}_2, \mathcal{B}}^{\text{EO}} + \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\text{MBS}} + \text{Adv}_{\mathcal{OS}, \mathcal{D}}^{\text{CR}}.$$

Proof. In Figure 25, we present a sequence of games.

Game G_0 . We start with the **EO** game for $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$:

$$\Pr[\text{G}_0^{\mathcal{A}} \Rightarrow 1] = \text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{EO}}.$$

Game G_1 . This is the same game as the previous one except that it aborts if σ_2 is valid and the signature keys $\text{pk}_{\text{Sig}, 1}$ and $\text{pk}_{\text{Sig}, 2}$ are different.

Claim 19: There exists an adversary $\mathcal{B}$ against **EO** such that

$$\Pr[\text{G}_0^{\mathcal{A}} \Rightarrow 1] - \Pr[\text{G}_1^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_2, \mathcal{B}}^{\text{EO}}.$$

Proof. Reduction $\mathcal{B}$ is depicted in Figure 26. The winning conditions of $\mathcal{B}$ are checked before they output a solution. Hence, $\mathcal{B}$ wins if the abort statement is reached. ■

Game G_2 . This is the same game as the previous one except that it aborts if σ_2 is valid and the message m'_1 and m'_2 are different.

Claim 20: There exists an adversary $\mathcal{C}$ against **MBS** such that

$$\Pr[\text{G}_1^{\mathcal{A}} \Rightarrow 1] - \Pr[\text{G}_2^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\text{MBS}}.$$

Proof. Reduction $\mathcal{C}$ is depicted in Figure 26. The winning conditions of $\mathcal{C}$ are checked before they output a solution. Hence, $\mathcal{C}$ wins if the abort statement is reached. ■

<u>Adversary $\mathcal{C}$</u>	<u>Oracle $\text{RO}_1(x)$</u>
01 $H_1 \xleftarrow{\$} \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$	19 return $H_1(x)$
02 $H_2 \xleftarrow{\$} \{\{0, 1\}^* \rightarrow \text{ChlSet}\}$	<u>Oracle $\text{RO}_2(x)$</u>
03 $(pk_1, pk_2, m_1, m_2, \sigma) \xleftarrow{\$} \mathcal{A}^{\text{RO}_1(\cdot), \text{RO}_2(\cdot)}$	20 return $H_2(x)$
04 $pk_1 \rightarrow (pk_{ID,1}, pk_{Sig,1})$	
05 $pk_2 \rightarrow (pk_{ID,2}, pk_{Sig,2})$	
06 $\sigma \rightarrow (rsp, \sigma_2)$	
07 $chl \leftarrow \text{RO}_2(\sigma_2)$	
08 $com_1 \leftarrow \text{ExtCom}(pk_{ID,1}, chl, rsp)$	
09 $com_2 \leftarrow \text{ExtCom}(pk_{ID,2}, chl, rsp)$	
10 $m'_1 \leftarrow \text{RO}_1(pk_1 \ m_1 \ com_1)$	
11 $m'_2 \leftarrow \text{RO}_1(pk_2 \ m_2 \ com_2)$	
12 if $\text{Ver}_2(pk_{Sig,1}, m'_1, \sigma_2) \wedge \text{Ver}_2(pk_{Sig,2}, m'_2, \sigma_2)$	
13 if $pk_{Sig,1} \neq pk_{Sig,2}$	
14 abort	
15 if $m'_1 \neq m'_2$	
16 return $(pk_{Sig,1}, m'_1, m'_2, \sigma_2)$ // win	
17 return 0	
18 return 0	

Figure 27. Adversary $\mathcal{C}$ against **MBS** of Sig_2 simulating G_1/G_2 for adversary $\mathcal{A}$.

Proof. We give a reduction in Figure 28. If adversary $\mathcal{A}$ wins G_2 , m'_1 and m'_2 must be the same due to the changes in G_2 . Since one of the winning conditions of $\mathcal{A}$ is $pk_1 \neq pk_2$ the input to RO_1 must be different. Hence, $\mathcal{D}$'s output constitutes a valid solution. ■

<u>Adversary $\mathcal{D}^{\text{RO}_D}$</u>	<u>Oracle $\text{RO}_1(x)$</u>
01 $H_2 \xleftarrow{\$} \{\{0, 1\}^* \rightarrow \text{ChlSet}\}$	16 return $\text{RO}_D(x)$
02 $(pk_1, pk_2, m_1, m_2, \sigma) \xleftarrow{\$} \mathcal{A}^{\text{RO}_1(\cdot), \text{RO}_2(\cdot)}$	<u>Oracle $\text{RO}_2(x)$</u>
03 $\sigma \rightarrow (rsp, \sigma_2)$	17 return $H_2(x)$
04 $chl \leftarrow \text{RO}_2(\sigma_2)$	
05 $com_1 \leftarrow \text{ExtCom}(pk_{ID,1}, chl, rsp)$	
06 $com_2 \leftarrow \text{ExtCom}(pk_{ID,2}, chl, rsp)$	
07 $m'_1 \leftarrow \text{RO}_1(pk_1 \ m_1 \ com_1)$	
08 $m'_2 \leftarrow \text{RO}_1(pk_2 \ m_2 \ com_2)$	
09 if $\text{Ver}_2(pk_{Sig,1}, m'_1, \sigma_2) \wedge \text{Ver}_2(pk_{Sig,2}, m'_2, \sigma_2)$	
10 if $pk_{Sig,1} \neq pk_{Sig,2}$	
11 abort	
12 if $m'_1 \neq m'_2$	
13 abort	
14 return (m'_1, m'_2) // output	
15 return 0	

Figure 28. Adversary $\mathcal{D}$ against **CR** of $\mathcal{OS}$ simulating G_2 for adversary $\mathcal{A}$.

The running times of $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ are approximately the same as for $\mathcal{A}$ which concludes the proof. $\blacksquare$

D.3 Proof of Theorem 8

Theorem 8 (MBS). For any adversary $\mathcal{A}$ against the **MBS** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$ (Figure 6), there exist a **MBS** adversary $\mathcal{B}$ against Sig_2 and a **CR** adversary $\mathcal{C}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}}$ such that

$$\text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}}^{\text{MBS}} \leq \text{Adv}_{\text{Sig}_2, \mathcal{B}}^{\text{MBS}} + \text{Adv}_{\mathcal{OS}, \mathcal{C}}^{\text{CR}}.$$

Proof. The theorem can be proved analogously to Theorem 7 except that the first step is not needed since adversary $\mathcal{A}$ only outputs one public key. $\blacksquare$

D.4 Proof of Theorem 9

Theorem 9 (NR). For any adversaries $\mathcal{A}$ and $\mathcal{D}$ against the **NR** security of $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda] := (\text{Gen}, \cdot, \cdot)$ (Figure 6), there exist **HnS** adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$ and an **RMV** adversary $\mathcal{C}$ against Sig_2 with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}}$ and $t_{\mathcal{D}} = t_{\bar{\mathcal{D}}}$ such that

$$\text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}} + \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\{0, 1\}^\lambda\text{-RMV}}$$

and

$$\mathcal{H}_{\infty}^{(m \mid \text{R0}, \text{sk}, \text{aux}(\text{sk}, m))}_{(\text{sk}, \text{pk}) \leftarrow^{\mathbb{S}} \text{Gen}, m \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{R0}}(\text{sk})} \leq \mathcal{H}_{\infty}^{(x \mid \text{R0}, z)}_{(x, z) \leftarrow^{\mathbb{S}} \bar{\mathcal{D}}^{\text{R0}}}$$

Proof. In Figure 29, we present a sequence of games.

Game $\mathbf{G}_0$. We start with the **NR** game for $\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda]$:

$$\Pr[\mathbf{G}_0^{\mathcal{A}} \Rightarrow 1] = \text{Adv}_{\text{BoP-2}[\text{ID}, \text{Sig}_2, \lambda], \mathcal{A}, \mathcal{D}}^{\text{NR}}.$$

Game $\mathbf{G}_1$. This is the same game as the previous one except that it aborts in the first random oracle if adversary $\mathcal{A}$ queries it on message m^* . To ease the depiction, we denote it using a different oracle but using the same underlying function.

Claim 22: There exist adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against **HnS** such that

$$\mathcal{H}_{\infty}^{(m \mid \text{R0}, \text{sk}, \text{aux}(\text{sk}, m))}_{(\text{sk}, \text{pk}) \leftarrow^{\mathbb{S}} \text{Gen}, m \leftarrow^{\mathbb{S}} \mathcal{D}^{\text{R0}}(\text{sk})} \leq \mathcal{H}_{\infty}^{(x \mid \bar{\text{R0}}, z)}_{(x, z) \leftarrow^{\mathbb{S}} \bar{\mathcal{D}}^{\text{R0}}}$$

and

$$\Pr[\mathbf{G}_0^{\mathcal{A}} \Rightarrow 1] - \Pr[\mathbf{G}_1^{\mathcal{A}} \Rightarrow 1] \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}},$$

with $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$.

Proof. We prove the claim by a sequence of hybrids over the random oracle queries to $\text{R0}'_1$. The original game $\mathbf{G}_0$ does not abort in the random oracle and the i -th hybrid aborts if there is a random oracle query on m^* within the first i queries to $\text{R0}'_1$. The i -th reduction is denoted by $\mathcal{B}_i$ and formally constructed in Figure 30. The reduction is an adversary against **HnS** and returns a solution in the i -th query to $\text{R0}'_1$. We further need

Games $G_0 - G_2$	Oracle $RO_1(x)$
01 $(H_1, H_2, \cdot, \cdot) \xleftarrow{\$} OS$	22 return $H_1(x)$
02 $(sk_{ID}, pk_{ID}) \xleftarrow{\$} Gen_1$	Oracle $RO_2(x)$
03 $(sk_{Sig}, pk_{Sig}) \xleftarrow{\$} Gen_2$	23 return $H_2(x)$
04 $sk \leftarrow (sk_{ID}, sk_{Sig})$	Oracle $RO'_1(x)$
05 $m^* \xleftarrow{\$} \mathcal{D}^{RO_1(\cdot), RO_2(\cdot)}(sk)$	24 $x \rightarrow \dots \ m\ com$
06 $(com, st) \xleftarrow{\$} Com(sk_{ID})$	25 if $m = m^*$ // $G_1 - G_2$
07 $m' \leftarrow H_1(pk_{ID} \ pk_{Sig} \ m^* \ com)$	26 abort // $G_1 - G_2$
08 $\sigma_2 \xleftarrow{\$} Sgn_2(sk_{Sig}, m')$	27 return $H_1(x)$
09 $chl \leftarrow H_2(\sigma_2)$	
10 $rsp \xleftarrow{\$} Rsp(sk_{ID}, com, chl, st)$	
11 $\sigma \leftarrow (rsp, \sigma_2)$	
12 $(pk^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{RO'_1(\cdot), RO_2(\cdot)}(sk, \sigma, aux(sk, m^*))$	
13 if $(pk_{ID}, pk_{Sig}) = pk^*$	
14 return 0	
15 $pk^* \rightarrow (pk_{ID}^*, pk_{Sig}^*)$	
16 $\sigma^* \rightarrow (rsp, \sigma_2)$	
17 $chl \leftarrow H_2(\sigma_2)$	
18 $com \leftarrow ExtCom(pk_{ID}, chl, rsp)$	
19 $m' \leftarrow H_1(pk_{ID}^* \ pk_{Sig}^* \ m^* \ com)$	
20 $m' \xleftarrow{\$} \{0, 1\}^\lambda$ // G_2	
21 return $Ver_2(pk_{Sig}^*, m', \sigma_2)$	

Figure 29. Games $G_0 - G_2$ for the proof of Theorem 9.

to define an appropriate adversary $\bar{\mathcal{D}}$ which is also given in Figure 30. Note that the min-entropy of $\bar{\mathcal{D}}$ is not smaller than the min-entropy of $\mathcal{D}$:

$$\begin{aligned}
\mathcal{H}_\infty(x, z) \big|_{(x, z) \xleftarrow{\$} \bar{\mathcal{D}}^{RO}} &= \mathcal{H}_\infty((pk, m, com) \mid RO, sk, st, com, aux(sk, m)) \\
&\quad \begin{array}{l} (sk, pk) \xleftarrow{\$} Gen \\ (sk_{ID}, sk_{Sig}) \leftarrow sk \\ (com, st) \xleftarrow{\$} Com(sk_{ID}) \\ m \xleftarrow{\$} \mathcal{D}^{RO}(sk) \end{array} \\
&\geq \mathcal{H}_\infty(m \mid RO, sk, st, com, aux(sk, m)) \\
&\quad [\dots] \\
&\geq \mathcal{H}_\infty(m \mid RO, sk, aux(sk, m)) \\
&\quad \begin{array}{l} (sk, pk) \xleftarrow{\$} Gen \\ m \xleftarrow{\$} \mathcal{D}^{RO}(sk) \end{array}
\end{aligned}$$

The first inequality holds since the min-entropy can only decrease when considering less random variables and the second inequality holds because the distribution of m is independent of st and com . ■

Game G_2 . This is the same game as the previous one except that it replaces the output of H_1 in the verification of $\mathcal{A}$'s signature by a uniformly random value from H_1 's output space (Line 20).

Claim 23: It holds that

$$\Pr[G_1^A \Rightarrow 1] = \Pr[G_2^A \Rightarrow 1].$$

Proof. Due to the changes in the previous game, $\mathcal{A}$ never queries random oracle RO'_1 with the correct m^* . In contrast, $\mathcal{D}$ could have queried their random oracle RO_1 on the correct values, i.e. the public keys pk_{ID}^*

Adversary $\mathcal{B}_i^{\text{RO}}(y, z)$ <pre> 01 $(\cdot, H_2, \cdot, \cdot) \xleftarrow{\\$} \mathcal{OS}$ 02 $\text{cnt} \leftarrow 0$ 03 $z \rightarrow (\text{sk}, \text{st}, \text{com}, a)$ 04 $\text{sk} \rightarrow (\text{sk}_{\text{ID}}, \text{sk}_{\text{Sig}})$ 05 $\text{pk}_{\text{ID}} \leftarrow \text{derivePK}(\text{sk}_{\text{ID}})$ 06 $\text{pk}_{\text{Sig}} \leftarrow \text{derivePK}(\text{sk}_{\text{Sig}})$ 07 $\sigma_2 \xleftarrow{\\$} \text{Sgn}_2(\text{sk}_{\text{Sig}}, y)$ 08 $\text{chl} \leftarrow H_2(\sigma_2)$ 09 $\text{rsp} \xleftarrow{\\$} \text{Rsp}(\text{sk}_{\text{ID}}, \text{com}, \text{chl}, \text{st})$ 10 $\sigma \leftarrow (\text{rsp}, \sigma_2)$ 11 $(\text{pk}^*, \sigma^*) \xleftarrow{\\$} \mathcal{A}^{\text{RO}'_1(\cdot), \text{RO}_2(\cdot)}(\text{sk}, \sigma, a)$ 12 return $\perp$ </pre> Adversary $\bar{\mathcal{D}}^{\text{RO}}$ <pre> 13 $(\text{sk}_{\text{ID}}, \text{pk}_{\text{ID}}) \xleftarrow{\\$} \text{Gen}_1$ 14 $(\text{sk}_{\text{Sig}}, \text{pk}_{\text{Sig}}) \xleftarrow{\\$} \text{Gen}_2$ 15 $\text{sk} \leftarrow (\text{sk}_{\text{ID}}, \text{sk}_{\text{Sig}})$ 16 $m \xleftarrow{\\$} \mathcal{D}^{\text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{sk})$ 17 $x \leftarrow (\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m \parallel \text{com})$ 18 $z \leftarrow (\text{sk}, \text{st}, \text{com}, \text{aux}(\text{sk}, m))$ 19 return (x, z) </pre>	Oracle $\text{RO}_1(x)$ <pre> 20 return $\bar{\text{RO}}_1(x)$ </pre> Oracle $\text{RO}_2(x)$ <pre> 21 return $H_2(x)$ </pre> Oracle $\text{RO}'_1(x)$ <pre> 22 $\text{cnt} \leftarrow \text{cnt} + 1$ 23 $x \rightarrow \dots \parallel m \parallel \dots$ 24 if $\text{cnt} = i$ 25 return $(\text{pk}_{\text{ID}} \parallel \text{pk}_{\text{Sig}} \parallel m \parallel \text{com})$ 26 return $\bar{\text{RO}}_1(x)$ </pre>
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Figure 30. Adversaries $\mathcal{B}_i$ and $\bar{\mathcal{D}}$ against **HnS** simulating the i -th hybrid between $\mathbf{G}_0/\mathbf{G}_1$ for adversaries $\mathcal{A}$ and $\mathcal{D}$.

and pk_{Sig}^* and the message m^* (and the correct commitment which is not relevant for our argument). However, the information $\mathcal{A}$ receives is independent of the output of such a query because $\mathcal{A}$ obtains sk which is independently generated and not chosen by $\mathcal{D}$, the signature σ which does not involve any additional information from $\mathcal{D}$ except for the message, and the auxiliary information which can only include information about sk and the message m^* itself. Note that the signature that $\mathcal{A}$ receives is based on a public key which must be different from the public key $\mathcal{A}$ outputs which means that the signature cannot contain information of the random oracle query on $\text{pk}_{\text{ID}}^*, \text{pk}_{\text{Sig}}^*, m^*$. Since the query output is independent from $\mathcal{A}$'s view, reprogramming the random oracle is indistinguishable. ■

Final reduction. The final game can be reduced to random-message validity.

Claim 24: There exists an adversary $\mathcal{C}$ against **RMV** of Sig_2 such that

$$\Pr[\mathbf{G}_2^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}_2, \mathcal{C}}^{\{0,1\}^\lambda\text{-RMV}}.$$

Proof. Reduction $\mathcal{C}$ can simulate $\mathbf{G}_2$ for adversary $\mathcal{A}$ as is. When $\mathcal{A}$ outputs a public key and a signature, $\mathcal{C}$ can extract a public key pk_{Sig} and a signature σ_2 for Sig_2 and output these to their own game. Since the message m' is uniform due to the changes in the previous game, $\mathcal{C}$ wins their **RMV** game if $\mathcal{A}$'s signature verifies. ■

The running times of $\mathcal{B}$ and $\mathcal{C}$ are approximately the same as for $\mathcal{A}$ which concludes the proof. ■

E Proofs of Section 6

E.1 Proof of Theorem 10

Theorem 10 ($((\mathbf{SUF}_1 \vee \mathbf{SUF}_2) \wedge \text{Sig}_1 \text{ salt-unique} \Rightarrow \mathbf{SUF})$). If Sig_1 is salt-unique, then for any adversary $\mathcal{A}$, making at most Q_s signing queries and Q_{RO} random oracle queries, against the **SUF-CMA** security of $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda]$ (Figure 7) in the random oracle model, there exist an **SUF-CMA** adversary $\mathcal{B}$ against Sig_2 and an **SUF-CMA** adversary $\mathcal{C}$ against Sig_1 with $t_{\mathcal{A}} \approx t_{\mathcal{B}} \approx t_{\mathcal{C}}$ such that

$$\text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} \leq \min \left\{ \text{Adv}_{\text{Sig}, \mathcal{B}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}, \text{Adv}_{\text{SigS}, \mathcal{C}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}} + Q_{\text{RO}} \cdot (\gamma_{\text{Sig}} 2^{-\kappa} + 2^{-\lambda+1}) \right\}.$$

Proof. We proceed with a sequence of games depicted in Figure 31.

Game $\mathbf{G}_0$. This is the **SUF-CMA** game for construction $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda]$ where the random oracle is instantiated via lazy sampling.

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] = \text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \mathcal{A}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}.$$

Games $\mathbf{G}_0 - \mathbf{G}_3$	Oracle $\text{Sgn}(m)$
01 $\mathcal{Q}, \mathcal{Q}', \mathcal{L}_{\text{H}_1}[], \mathcal{L}_{\text{H}_2}[] \leftarrow \emptyset$	21 $r \xleftarrow{\$} \{0, 1\}^{\kappa}$
02 $(\text{sk}_1, \text{pk}_1) \xleftarrow{\$} \text{SigS.Gen}$	22 $m' \leftarrow \text{RO}_1(\text{pk}_1 \ \text{pk}_2 \ m)$
03 $(\text{sk}_2, \text{pk}_2) \xleftarrow{\$} \text{Sig.Gen}$	23 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(\text{sk}_2, m' \ r)$
04 $\text{pk} \leftarrow (\text{pk}_1, \text{pk}_2)$	24 $\mathcal{Q}' \leftarrow \mathcal{Q}' \cup \{(m' \ r, \sigma_2)\}$
05 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{pk})$	25 if $\mathcal{L}_{\text{H}_2}[m' \ \sigma_2 \ r] \neq \perp$ // $\mathbf{G}_2 - \mathbf{G}_3$
06 if $(m^*, \sigma^*) \in \mathcal{Q}$	26 abort // $\mathbf{G}_2 - \mathbf{G}_3$
07 return 0	27 $h \leftarrow \text{RO}_2(m' \ \sigma_2 \ r)$
08 $\sigma^* \rightarrow (\sigma_1^*, \sigma_2^*)$	28 $\sigma_1 \xleftarrow{\$} \text{Sgn}_{\text{salt}}(\text{sk}_1, h, r)$
09 $r^* \leftarrow \text{Ext}(\text{pk}_1, \sigma_1^*)$	29 $\sigma \leftarrow (\sigma_1, \sigma_2)$
10 $m' \leftarrow \text{RO}_1(\text{pk}_1 \ \text{pk}_2 \ m^*)$	30 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
11 if $\exists x \neq \text{pk}_1 \ \text{pk}_2 \ m^* : \mathcal{L}_{\text{H}_1}[x] = m'$ // $\mathbf{G}_3$	31 return σ
12 abort // $\mathbf{G}_3$	
13 $h^* \leftarrow \text{RO}_2(m' \ \sigma_2^* \ r^*)$	<u>Oracle $\text{RO}_1(\sigma_2 \ r)$</u>
14 if $\exists x \neq m' \ \sigma_2^* \ r^* : \mathcal{L}_{\text{H}_2}[x] = h^*$ // $\mathbf{G}_3$	32 if $\mathcal{L}_{\text{H}_1}[x] = \perp$
15 abort // $\mathbf{G}_3$	33 $\mathcal{L}_{\text{H}_1}[x] \xleftarrow{\$} \{0, 1\}^{\lambda}$
16 if $(m' \ r^*, \sigma_2^*) \in \mathcal{Q}'$ // $\mathbf{G}_1 - \mathbf{G}_3$	34 return $\mathcal{L}_{\text{H}_1}[x]$
17 return 0 // $\mathbf{G}_1 - \mathbf{G}_3$	<u>Oracle $\text{RO}_2(x)$</u>
18 if $\text{Ver}_1(\text{pk}_1, h^*, \sigma_1^*) \wedge \text{Ver}_2(\text{pk}_{\text{Sig}}, m' \ r^*, \sigma_2^*)$	35 if $\mathcal{L}_{\text{H}_2}[x] = \perp$
19 return 1	36 $\mathcal{L}_{\text{H}_2}[x] \xleftarrow{\$} \{0, 1\}^{\lambda}$
20 return 0	37 return $\mathcal{L}_{\text{H}_2}[x]$

Figure 31. Games $\mathbf{G}_0 - \mathbf{G}_3$ for the proof of Theorem 10.

Game $\mathbf{G}_1$. This is the same game as the previous one except that it maintains an additional set $\mathcal{Q}'$ in which the input and output of every Sgn_2 operation is stored. The game also outputs 0 if the tuple $(m' \| r^*, \sigma_2^*)$ from the forgery exists in $\mathcal{Q}'$.

Claim 25: It holds that

$$\Pr[\mathbf{G}_0^{\mathbf{A}} \Rightarrow 1] = \Pr[\mathbf{G}_1^{\mathbf{A}} \Rightarrow 1].$$

Proof. We show that the winning probability does not change. Note that the game terminates and returns 0 in case the message/signature combination corresponds to a signing oracle query. To reach the newly introduced return statement, this cannot be the case. For the new return to trigger, there must be a matching element in $\mathcal{Q}'$. Hence, there was a query to Sgn , with the same values m', r^*, σ_2^* . This implies the same h^* and due to the salt-uniqueness of SigS , σ_1^* is either the same (leading to a 0-return due to the triviality check from above) or invalid (also leading to the game returning 0). ■

Reduction to SUF-CMA of Sig. Claim 26: There exists an adversary $\mathcal{B}$ against **SUF-CMA** such that

$$\Pr[\mathbf{G}_1^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}, \mathcal{B}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}.$$

Proof. The reduction is formalized in Figure 32. The signing oracle can be simulated using $\mathcal{B}$'s signing oracle. If $\mathcal{A}$ wins their game, $\mathcal{B}$'s winning conditions are also fulfilled. The signature is valid and must be fresh due to the check introduced in $\mathbf{G}_1$. ■

<u>Adversary $\mathcal{B}^{\text{Sgn}_{\mathcal{B}}}(\text{pk}_2)$</u>	<u>Oracle $\text{Sgn}(m)$</u>
01 $(H_1, H_2, \cdot, \cdot) \xleftarrow{\$} \mathcal{OS}$	17 $r \xleftarrow{\$} \{0, 1\}^\kappa$
02 $\mathcal{Q}, \mathcal{Q}' \leftarrow \emptyset$	18 $m' \leftarrow H_1(\text{pk}_1 \ \text{pk}_2 \ m)$
03 $(\text{sk}_1, \text{pk}_1) \xleftarrow{\$} \text{SigS.Gen}$	19 $\sigma_2 \xleftarrow{\$} \text{Sgn}_{\mathcal{B}}(m' \ r)$ // sign oracle
04 $\text{pk} \leftarrow (\text{pk}_1, \text{pk}_2)$	20 $\mathcal{Q}' \leftarrow \mathcal{Q}' \cup \{(m' \ r, \sigma_2)\}$
05 $(m^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{pk})$	21 $h \leftarrow H_2(m' \ \sigma_2 \ r)$
06 if $(m^*, \sigma^*) \in \mathcal{Q}$	22 $\sigma_1 \xleftarrow{\$} \text{Sgn}_{\text{salt}}(\text{sk}_1, h, r)$
07 return 0	23 $\sigma \leftarrow (\sigma_1, \sigma_2)$
08 $\sigma^* \rightarrow (\sigma_1^*, \sigma_2^*)$	24 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
09 $r^* \leftarrow \text{Ext}(\text{pk}_1, \sigma_1^*)$	25 return σ
10 $m' \leftarrow H_1(\text{pk}_1 \ \text{pk}_2 \ m^*)$	<u>Oracle $\text{RO}_1(x)$</u>
11 $h^* \leftarrow H_2(m' \ \sigma_2^* \ r^*)$	26 return $H_1(x)$
12 if $(m' \ r^*, \sigma_2^*) \in \mathcal{Q}'$	<u>Oracle $\text{RO}_2(x)$</u>
13 return 0	27 return $H_2(x)$
14 if $\text{Ver}_1(\text{pk}_1, h^*, \sigma_1^*) \wedge \text{Ver}_2(\text{pk}_{\text{Sig}}, m' \ r^*, \sigma_2^*)$	
15 return $(m' \ r^*, \sigma_2^*)$	// win
16 return 0	

Figure 32. Adversary $\mathcal{B}$ against **SUF-CMA** security of **Sig** having access to oracle $\text{Sgn}_{\mathcal{B}}$ simulating $\mathbf{G}_1$ for adversary $\mathcal{A}$.

Game $\mathbf{G}_2$. This is the same game as the previous one except that it aborts in the signing oracle if the random oracle RO_2 was already queried on the input $m' \| \sigma_2 \| r$ before.

Claim 27: It holds that

$$\Pr[\mathbf{G}_1^{\mathcal{A}} \Rightarrow 1] - \Pr[\mathbf{G}_2^{\mathcal{A}} \Rightarrow 1] \leq Q_{\text{RO}} \cdot \gamma_{\text{Sig}} 2^{-\kappa}.$$

Proof. For a fixed element in $\mathcal{L}_H$, the probability that the freshly created signature σ_2 is the same as the second part of the element can be upper bounded by γ_{Sig_2} . The probability that the salt part of the element is the same is at most $2^{-\kappa}$. Since $\mathcal{L}_H$ contains at most Q_{RO} elements, we obtain the claim. ■

Game G_3 . This is the same game as the previous one except that it aborts if there is a collision in one of the random oracles for the output forgery (Line 12 and Line 15).

Claim 28: It holds that

$$\Pr[\mathsf{G}_2^{\mathcal{A}} \Rightarrow 1] - \Pr[\mathsf{G}_3^{\mathcal{A}} \Rightarrow 1] \leq \frac{Q_{\text{RO}}}{2^{\lambda-1}}.$$

Proof. Both RO lists contain at most Q_{RO} many elements and hence each collision probability can be upper bounded by $Q_{\text{RO}}/2^{\lambda}$. ■

*Reduction to **SUF-CMA** of SigS.* Claim 29: There exists an adversary $\mathcal{C}$ against **SUF-CMA** such that

$$\Pr[\mathsf{G}_3^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{SigS}, \mathcal{C}}^{(Q_s, Q_{\text{RO}})\text{-SUF-CMA}}.$$

Proof. The reduction is formalized in Figure 33. The signing oracle can be simulated as follows. First, the reduction samples a uniformly random h and signs it using their own signing oracle. Then, they extract the salt r using algorithm Ext . The distribution of the salt must also be uniform by definition of a salt-based signature.¹⁷ Signature σ_2 can be computed as usual and if the random oracle was already queried on the produced signature σ_2 and the salt, the game aborts due to the changes in G_3 . Otherwise, the reduction can program the random oracle on h . Hence, the reduction simulates the signing oracle perfectly.

If adversary $\mathcal{A}$ wins their game, $\mathcal{C}$'s winning conditions are fulfilled as well: For a contradiction argument, assume that $\mathcal{C}$'s condition is not fulfilled, i.e. h^* was queried to the signing oracle with output σ_1^* . That means there was a signing query for $\mathcal{A}$ that used this h^* and due to the no-collision requirement for RO_2 introduced in G_3 the input to the random oracle query outputting h^* must be the same if $\mathcal{A}$ wins their game. That means, for this signing oracle query we have the same m', σ_2 , and r . Due to the no-collision requirement for RO_1 , the actual message m from the signing query and m^* must also be the same. Finally, since the salt and the message for the salt-based signature scheme are the same, σ_1 from the oracle query and σ_1^* must also be the same due to SigS's salt-uniqueness property. Therefore the signing oracle added the tuple $(m^*, (\sigma_1^*, \sigma_2^*))$ to list $\mathcal{Q}$ which would lead to $\mathcal{A}$ not fulfilling their freshness condition. ■

The running time of $\mathcal{B}$ and $\mathcal{C}$ are approximately the same as for $\mathcal{A}$. Collecting the bounds yields the theorem statement. ■

E.2 Proof of Theorem 13

Theorem 13 (NR). For any adversaries $\mathcal{A}$ and $\mathcal{D}$ against the **NR** security of $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda] := (\text{Gen}, \cdot, \cdot)$ (Figure 7), there exist **HnS** adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$, an **RMV** adversary $\mathcal{C}$ against Sig, and an **RMV** adversary $\mathcal{E}$ against SigS with $t_{\mathcal{A}} = t_{\mathcal{B}} = t_{\mathcal{C}} = t_{\mathcal{E}}$ and $t_{\mathcal{D}} = t_{\bar{\mathcal{D}}}$ such that

$$\text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \text{aux}, \mathcal{A}, \mathcal{D}}^{(Q_{\mathcal{A}}, Q_{\mathcal{D}})\text{-NR}} \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}} + \min \left\{ \text{Adv}_{\text{Sig}, \mathcal{C}}^{\{0, 1\}^\lambda\text{-RMV}}, \text{Adv}_{\text{SigS}, \mathcal{E}}^{\{0, 1\}^\lambda\text{-RMV}} + \frac{Q_{\mathcal{A}}}{2^\lambda} \right\}.$$

and

$$\mathcal{H}_\infty \left(m \mid \text{RO}, \text{sk}, \text{aux}(\text{sk}, m) \right) \leq \mathcal{H}_\infty \left(x \mid \text{RO}, z \right)_{\substack{(\text{sk}, \text{pk}) \leftarrow \text{Gen} \\ m \leftarrow \text{D}^{\text{RO}}(\text{sk})}}_{(x, z) \leftarrow \text{D}^{\text{RO}}}$$

Proof. In Figure 34, we present a sequence of games.

Game G_0 . We start with the **NR** game for $\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda]$:

$$\Pr[\mathsf{G}_0^{\mathcal{A}} \Rightarrow 1] = \text{Adv}_{\text{BoP-3}[\text{SigS}, \text{Sig}, \kappa, \lambda], \mathcal{A}, \mathcal{D}}^{\text{NR}}.$$

¹⁷ This is implied by the distributions of the normal and salt-specific signing to be equal and the extract algorithm to be deterministic.

Adversary $\mathcal{C}^{\text{Sgn}_C}(\text{pk}_1)$	Oracle $\text{Sgn}(m)$
01 $\mathcal{Q}, \mathcal{Q}' \leftarrow \emptyset$	20 $h \leftarrow_{\mathbb{S}} \{0, 1\}^\lambda$
02 $(\text{sk}_2, \text{pk}_2) \leftarrow_{\mathbb{S}} \text{Sig.Gen}$	21 $\sigma_1 \leftarrow_{\mathbb{S}} \text{Sgn}_C(h)$ // sign oracle
03 $\text{pk} \leftarrow (\text{pk}_1, \text{pk}_2)$	22 $r \leftarrow \text{Ext}(\text{pk}_1, \sigma_1)$
04 $(m^*, \sigma^*) \leftarrow_{\mathbb{S}} \mathcal{A}^{\text{Sgn}(\cdot), \text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{pk})$	23 $m' \leftarrow \text{RO}_1(\text{pk}_1 \parallel \text{pk}_2 \parallel m)$
05 if $(m^*, \sigma^*) \in \mathcal{Q}$	24 $\sigma_2 \leftarrow_{\mathbb{S}} \text{Sgn}_2(\text{sk}_2, m' \parallel r)$
06 return 0	25 if $\mathcal{L}_{H_2}[m' \parallel \sigma_2 \parallel r] \neq \perp$
07 $\sigma^* \rightarrow (\sigma_1^*, \sigma_2^*)$	26 abort
08 $r^* \leftarrow \text{Ext}(\sigma_1^*)$	27 $\mathcal{L}_{H_2}[m' \parallel \sigma_2 \parallel r] \leftarrow h$ // program RO
09 $m' \leftarrow \text{RO}_1(\text{pk}_1 \parallel \text{pk}_2 \parallel m^*)$	28 $\mathcal{Q}' \leftarrow \mathcal{Q}' \cup \{(m' \parallel r, \sigma_2)\}$
10 if $\exists x \neq \text{pk}_1 \parallel \text{pk}_2 \parallel m^* : \mathcal{L}_{H_1}[x] = m'$	29 $\sigma \leftarrow (\sigma_1, \sigma_2)$
11 abort	30 $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(m, \sigma)\}$
12 $h^* \leftarrow \text{RO}_2(m' \parallel \sigma_2^* \parallel r^*)$	31 return σ
13 if $\exists x \neq m' \parallel \sigma_2^* \parallel r^* : \mathcal{L}_{H_2}[x] = h^*$	Oracle $\text{RO}_1(x)$
14 abort	32 if $\mathcal{L}_{H_1}[x] = \perp$
15 if $(m' \parallel r^*, \sigma_2^*) \in \mathcal{Q}'$	33 $\mathcal{L}_{H_1}[x] \leftarrow_{\mathbb{S}} \{0, 1\}^\lambda$
16 return 0	34 return $\mathcal{L}_{H_1}[x]$
17 if $\text{Ver}_1(\text{pk}_1, h^*, \sigma_1^*) \wedge \text{Ver}_2(\text{pk}_{\text{Sig}}, m' \parallel r^*, \sigma_2^*)$	Oracle $\text{RO}_2(x)$
18 return (h^*, σ_1^*)	35 if $\mathcal{L}_{H_2}[x] = \perp$
19 return 0	36 $\mathcal{L}_{H_2}[x] \leftarrow_{\mathbb{S}} \{0, 1\}^\lambda$
	37 return $\mathcal{L}_{H_2}[x]$

Figure 33. Adversary $\mathcal{C}$ against **SUF-CMA** security of SigS having access to oracle Sgn_C simulating $\mathbf{G}_3$ for adversary $\mathcal{A}$.

Game $\mathbf{G}_1$. This is the same game as the previous one except that it aborts in the first random oracle if adversary $\mathcal{A}$ queries it on message m^* . To ease the depiction, we denote it using a different oracle but using the same underlying function.

Claim 30: There exist adversaries $\mathcal{B}$ and $\bar{\mathcal{D}}$ against **HnS** such that

$$\mathcal{H}_{\infty}^{(\text{sk}, \text{pk}) \leftarrow_{\mathbb{S}} \text{Gen}} (m \mid \text{RO}, \text{sk}, \text{aux}(\text{sk}, m)) \leq \mathcal{H}_{\infty}^{(x, z) \leftarrow_{\mathbb{S}} \bar{\mathcal{D}}^{\text{RO}}} (x \mid \text{RO}, z)$$

$m \leftarrow_{\mathbb{S}} \mathcal{D}^{\text{RO}}(\text{sk})$

and

$$\Pr [\mathbf{G}_0^{\mathcal{A}} \Rightarrow 1] - \Pr [\mathbf{G}_1^{\mathcal{A}} \Rightarrow 1] \leq Q_{\mathcal{A}} \cdot \text{Adv}_{\mathcal{OS}, \mathcal{B}, \bar{\mathcal{D}}}^{\text{HnS}},$$

with $\mathcal{OS} := \{\{0, 1\}^* \rightarrow \{0, 1\}^\lambda\}$.

Proof. The claim can be proved analogously to Theorem 9. We prove the claim by a sequence of hybrids over the random oracle queries to RO'_1 . The original game $\mathbf{G}_0$ does not abort in the random oracle and the i -th hybrid aborts if there is a random oracle query on m^* within the first i queries to RO'_1 . The i -th reduction is denoted by $\mathcal{B}_i$ and formally constructed in Figure 35. The reduction is an adversary against **HnS** and returns a solution in the i -th query to RO'_1 . We further need to define an appropriate adversary $\bar{\mathcal{D}}$ which is

Games $G_0 - G_3$	Oracle $RO_1(x)$
01 $(H_1, H_2, \cdot, \cdot) \xleftarrow{\$} \mathcal{OS}$	23 return $H_1(x)$
02 $(sk_1, pk_1) \xleftarrow{\$} \text{Gen}_1$	<u>Oracle $RO_2(x)$</u>
03 $(sk_1, pk_2) \xleftarrow{\$} \text{Gen}_2$	24 return $H_2(x)$
04 $sk \leftarrow (sk_1, sk_2)$	<u>Oracle $RO'_1(x)$</u>
05 $m \xleftarrow{\$} \mathcal{D}^{RO_1(\cdot), RO_2(\cdot)}(sk)$	25 $x \rightarrow \dots \ m$
06 $r \xleftarrow{\$} \{0, 1\}^\kappa$	26 if $m = m^*$ // $G_1 - G_3$
07 $m' \leftarrow H_1(pk_1 \ pk_2 \ m)$	27 abort // $G_1 - G_3$
08 $\sigma_2 \xleftarrow{\$} \text{Sgn}_2(sk_2, m' \ r)$	28 return $H_1(x)$
09 $h \leftarrow H_2(m' \ \sigma_2 \ r)$	
10 $\sigma_1 \xleftarrow{\$} \text{Sgn}_{\text{salt}}(sk_1, h, r)$	
11 $\sigma \leftarrow (\sigma_1, \sigma_2)$	
12 $(pk^*, \sigma^*) \xleftarrow{\$} \mathcal{A}^{RO'_1(\cdot), RO_2(\cdot)}(sk, \sigma, \text{aux}(sk, m))$	
13 if $(pk_1, pk_2) \neq pk^*$	
14 return 0	
15 $pk^* \rightarrow (pk_1^*, pk_2^*)$	
16 $\sigma^* \rightarrow (\sigma_1^*, \sigma_2^*)$	
17 $r^* \leftarrow \text{Ext}(pk_1^*, \sigma_1^*)$	
18 $m' \leftarrow H_1(pk_1^* \ pk_2^* \ m^*)$	
19 $m' \xleftarrow{\$} \{0, 1\}^\lambda$ // $G_2 - G_3$	
20 $h^* \leftarrow H_2(m' \ \sigma_2^* \ r^*)$	
21 $h^* \xleftarrow{\$} \{0, 1\}^\lambda$ // G_3	
22 return $\text{Ver}_1(pk_1^*, h^*, \sigma_1^*) \wedge \text{Ver}_2(pk_2^*, m' \ r^*, \sigma_2^*)$	

Figure 34. Games $G_0 - G_3$ for the proof of Theorem 13.

also given in Figure 35. Note that the min-entropy of $\bar{\mathcal{D}}$ is the same as the one of $\mathcal{D}$:

$$\begin{aligned}
\mathcal{H}_\infty_{(x,z) \xleftarrow{\$} \bar{\mathcal{D}}^{RO}}(x \mid \bar{RO}, z) &= \mathcal{H}_\infty_{\substack{(sk, pk) \xleftarrow{\$} \text{Gen} \\ m \xleftarrow{\$} \mathcal{D}^{RO}(sk)}}((pk, m) \mid RO, sk, \text{aux}(sk, m)) \\
&\geq \mathcal{H}_\infty_{\substack{(sk, pk) \xleftarrow{\$} \text{Gen} \\ m \xleftarrow{\$} \mathcal{D}^{RO}(sk)}}(m \mid RO, sk, \text{aux}(sk, m))
\end{aligned}$$

The last inequality holds since the min-entropy can only decrease when considering less random variables. ■

Game G_2 . This is the same game as the previous one except that it replaces the output of H_1 in the verification process after $\mathcal{A}$ output a solution to a uniformly random value from the co-domain of H_1 .

Claim 31: It holds that

$$\Pr[G_1^A \Rightarrow 1] = \Pr[G_2^A \Rightarrow 1].$$

Proof. Due to the changes in the previous game, $\mathcal{A}$ never queries random oracle RO'_1 with the correct m^* . In contrast, $\mathcal{D}$ could have queried their random oracle RO_1 on the correct values, i.e. the public keys pk_1^* and pk_2^* and the message m^* . However, the information $\mathcal{A}$ receives is independent of the output of such a query because $\mathcal{A}$ obtains sk which is independently generated and not chosen by $\mathcal{D}$, the signature σ which does not involve any additional information from $\mathcal{D}$ except for the message, and the auxiliary information

Adversary $\mathcal{B}_i^{\text{RO}}(y, z)$ <pre> 01 $(\cdot, H_2, \cdot, \cdot) \xleftarrow{\\$} \mathcal{OS}$ 02 $\text{cnt} \leftarrow 0$ 03 $z \rightarrow (\text{sk}, a)$ 04 $\text{sk} \rightarrow (\text{sk}_1, \text{sk}_2)$ 05 $\text{pk}_1 \leftarrow \text{derivePK}(\text{sk}_1)$ 06 $\text{pk}_2 \leftarrow \text{derivePK}(\text{sk}_2)$ 07 $r \xleftarrow{\\$} \{0, 1\}^\kappa$ 08 $\sigma_2 \xleftarrow{\\$} \text{Sgn}_2(\text{sk}_2, y \ r)$ 09 $h \leftarrow H_2(y \ \sigma_2 \ r)$ 10 $\sigma_1 \xleftarrow{\\$} \text{Sgn}_{\text{salt}}(\text{sk}_1, h, r)$ 11 $\sigma \leftarrow (\sigma_1, \sigma_2)$ 12 $(\text{pk}^*, \sigma^*) \xleftarrow{\\$} \mathcal{A}^{\text{RO}'_1(\cdot), \text{RO}_2(\cdot)}(\text{sk}, \sigma, a)$ 13 return $\perp$ </pre> Adversary $\bar{\mathcal{D}}^{\text{RO}}$ <pre> 14 $(\text{sk}_1, \text{pk}_1) \xleftarrow{\\$} \text{Gen}_1$ 15 $(\text{sk}_2, \text{pk}_2) \xleftarrow{\\$} \text{Gen}_2$ 16 $\text{sk} \leftarrow (\text{sk}_1, \text{sk}_2)$ 17 $m \xleftarrow{\\$} \mathcal{D}^{\text{RO}_1(\cdot), \text{RO}_2(\cdot)}(\text{sk})$ 18 $x \leftarrow (\text{pk}_1 \ \text{pk}_2 \ m)$ 19 $z \leftarrow (\text{sk}, \text{aux}(\text{sk}, m))$ 20 return (x, z) </pre>	Oracle $\text{RO}_1(x)$ <pre> 21 return $\bar{\text{RO}}_1(x)$ </pre> Oracle $\text{RO}_2(x)$ <pre> 22 return $H_2(x)$ </pre> Oracle $\text{RO}'_1(x)$ <pre> 23 $\text{cnt} \leftarrow \text{cnt} + 1$ 24 $x \rightarrow \dots \ m$ 25 if $\text{cnt} = i$ 26 return $(\text{pk}_1 \ \text{pk}_2 \ m)$ 27 return $\bar{\text{RO}}_1(x)$ </pre>
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Figure 35. Adversaries $\mathcal{B}_i$ and $\bar{\mathcal{D}}$ against **HnS** simulating the i -th hybrid between $\mathcal{G}_0/\mathcal{G}_1$ for adversaries $\mathcal{A}$ and $\mathcal{D}$.

which can only include information about sk and the message m^* itself. Note that the signature that $\mathcal{A}$ receives is based on a public key which must be different from the public key $\mathcal{A}$ outputs which means that the signature cannot contain information of the random oracle query on $\text{pk}_1^*, \text{pk}_2^*, m^*$. Since the query output is independent from $\mathcal{A}$'s view, reprogramming the random oracle is indistinguishable. ■

Reduction to Sig. We can reduce the game to **Sig**'s **RMV**.

Claim 32: There exists an adversary $\mathcal{C}$ against **RMV** such that

$$\Pr[\mathcal{G}_2^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{Sig}, \mathcal{C}}^{\{0,1\}^\lambda\text{-RMV}}.$$

Proof. Reduction $\mathcal{C}$ can simulate $\mathcal{G}_2$ for adversary $\mathcal{A}$ as is. When $\mathcal{A}$ outputs a public key a signature, $\mathcal{C}$ can extract a public key pk_2 and a signature σ_2 for **Sig**₂. Further they can extract the salt r for signature scheme **SigS** via the extraction oracle **Ext**. Then, they can output $(\text{pk}_2, \sigma_2, r)$ which is a valid solution for their game in case $\mathcal{A}$ wins because $\mathcal{A}$ produces a valid signature which verifies for a randomly chosen message with an appendix r . ■

Game $\mathcal{G}_3$. This is the same game as the previous one except that it replaces the output of H_2 in the verification process after $\mathcal{A}$ output a solution to a uniformly random value from the co-domain of H_2 .

Claim 33: It holds that

$$\Pr[\mathcal{G}_2^{\mathcal{A}} \Rightarrow 1] - \Pr[\mathcal{G}_3^{\mathcal{A}} \Rightarrow 1] \leq \frac{Q_{\mathcal{A}}}{2^\lambda}.$$

Proof. Due to the changes in the last game, m' is uniformly random. Random oracle $\mathbf{RO}_2$ was queried on that value before with probability at most $\frac{1}{2^\lambda}$. The distributions are only distinguishable if $\mathcal{A}$ queries the random oracle on that value. Hence, the claim follows by taking at most $Q_{\mathcal{A}}$ queries from $\mathcal{A}$ into account. ■

Reduction to SigS. We can reduce the game to SigS's **RMV**.

Claim 34: There exists an adversary $\mathcal{E}$ against **RMV** such that

$$\Pr[\mathbf{G}_3^{\mathcal{A}} \Rightarrow 1] \leq \text{Adv}_{\text{SigS}, \mathcal{E}}^{\{0,1\}^\lambda\text{-RMV}}.$$

Proof. The reduction works similar to the reduction $\mathcal{C}$ with the exception that $\mathcal{E}$ outputs the empty string compared to the salt r . ■

Combining all bounds concludes the proof. ■